Executive Summary

This report addresses the critical challenge of employee attrition within IBM. Employee turnover incurs significant costs, including recruitment, onboarding, training, and lost productivity. Understanding and mitigating attrition is crucial for maintaining a stable, productive workforce and achieving business objectives.

Business Problem

IBM is experiencing employee attrition, and there's a need to understand the underlying causes, identify employees at high risk of leaving, and develop actionable strategies to reduce turnover.

Objectives

- To identify key drivers of employee attrition.
- To develop a predictive model to flag employees at high risk of attrition.
- To provide data-driven insights and recommendations for HR and management to enhance retention strategies.

Dataset

The analysis utilizes the IBM HR Analytics Employee Attrition & Performance dataset, which contains various employee demographic, job-related, and satisfaction attributes.

Tools

Python (Pandas, Scikit-learn, Matplotlib, Seaborn, Plotly), Jupyter Notebook/Google Colab for analysis and visualization.

Deliverables

- Comprehensive attrition analysis.
- Identification of top attrition factors.
- Predictive model for attrition risk.
- Actionable business recommendations.

Expected Business Impact

- Reduced employee turnover costs.
- Improved employee morale and productivity.
- Enhanced talent retention strategies.
- Proactive management of attrition risks.

Table of Contents

Chapter 1: Introduction

Business Question

How can HR analytics be leveraged to understand, predict, and mitigate employee attrition, thereby fostering a more stable and productive workforce at IBM?

Background

HR Analytics involves applying data-driven approaches to optimize human resource management. Employee attrition, the voluntary or involuntary departure of employees from an organization, is a critical concern for businesses worldwide. It can lead to significant financial costs (recruitment, training, loss of institutional knowledge) and operational challenges (reduced productivity, decreased morale amongst remaining employees).

For a large organization like IBM, even a small percentage increase in attrition can translate into substantial financial losses and strategic setbacks. Proactively understanding the drivers of attrition and implementing targeted retention strategies is therefore paramount to business continuity and success.

Methodology

This project employs a structured HR analytics methodology, including data loading, cleaning, exploratory data analysis (EDA), feature engineering, and predictive modeling (using a Random Forest Classifier).

The process will culminate in deriving actionable business insights and recommendations tailored to IBM's context, aimed at improving employee retention.

Project Objectives

- Quantify the cost and impact of attrition within IBM.
- Identify key demographic, job-related, and satisfaction factors correlating with attrition.
- Develop an attrition prediction model to proactively identify at-risk employees.
- Formulate evidence-based recommendations for improving employee retention and engagement.

Chapter 2: Business Problem

Business Question

Given current employee attrition trends, how can IBM proactively identify and address the root causes of turnover to retain valuable talent and maintain operational efficiency?

Background

Imagine you are an HR Data Analyst at IBM Consulting. The Chief Human Resources Officer (CHRO) has approached your team with an urgent request. There is a growing concern regarding employee attrition rates, and the CHRO requires a data-driven understanding of the situation to inform strategic retention initiatives.

The CHRO's primary questions revolve around identifying why employees are leaving, pinpointing which employees are most at risk, and devising effective strategies to reduce attrition. This analysis is critical for maintaining a competitive edge, optimizing workforce planning, and safeguarding IBM's intellectual capital.

Methodology

This chapter sets the stage for a detailed analytical investigation. It articulates the core business challenge, outlining the scope and importance of understanding employee attrition from an organizational perspective. Future chapters will delve into data exploration, predictive modeling, and strategic recommendations to address these concerns.

Key Business Questions

- Why are employees leaving IBM?
- Which segments of our workforce are most susceptible to attrition?
- What are the primary factors contributing to employee turnover?

- How can we identify employees at risk of attrition proactively?
- What actionable strategies can IBM implement to effectively reduce attrition rates?

Chapter 3: Dataset Overview

Business Question

What is the structure, content, and quality of the IBM HR Analytics Employee Attrition & Performance dataset, and how can its features inform our understanding of attrition drivers?

Background

A thorough understanding of the dataset is foundational to any robust analytical project. This chapter provides a comprehensive overview of the 'IBM HR Analytics Employee Attrition & Performance' dataset, detailing each feature, its data type, and its business relevance. Furthermore, it examines the dataset's structural characteristics, including its dimensions, missing values, and memory footprint, to establish a baseline for subsequent data cleaning and analysis.

Methodology

We will present a data dictionary outlining each feature, followed by an examination of the dataset's key statistics. This includes assessing the number of rows and columns, identifying any missing values, checking for duplicate entries, and understanding the overall memory consumption. These steps are crucial for ensuring data quality and preparing for effective feature engineering and modeling.

Code

```python
print(f"Dataset Rows: {df.shape[0]}")
print(f"Dataset Columns: {df.shape[1]}")
```

```
Dataset Rows: 1470
Dataset Columns: 35
```

```python
missing_values = df.isnull().sum()
missing_values = missing_values[missing_values > 0]
if missing_values.empty:
    print("No missing values found in the dataset.")
else:
    print("Missing Values:")
    display(missing_values)
```

```
No missing values found in the dataset.
```

```python
duplicate_rows = df.duplicated().sum()
print(f"Number of Duplicate Rows: {duplicate_rows}")
```

```
Number of Duplicate Rows: 0
```

```
memory_usage = df.memory_usage(deep=True).sum() / (1024 * 1024) # in MB
print(f"Dataset Memory Usage: {memory_usage:.2f} MB")
```

```
Dataset Memory Usage: 0.96 MB
```

Dataset Description

Feature	Description	Data Type	Busine
Age	Age of the employee in years	Integer	Demographic factor, often correlate
Attrition	Whether the employee left the company (Yes/No)	Binary (0/1)	Target variable; indicates employee
BusinessTravel	Frequency of business travel	Categorical	Impact on work-life balance and job
DailyRate	Daily rate of pay	Integer	Compensation factor, can influence
Department	Department the employee works in	Categorical	Organizational structure, may reveal
DistanceFromHome	Distance from home to workplace (miles)	Integer	Commute burden, potentially impac
Education	Level of education (1 'Below College' to 5 'Doctor')	Integer	Educational background, often linke
EducationField	Field of education	Categorical	Specialization, relevance to job role.
EmployeeCount	Employee count (always 1)	Integer	Constant feature, can be dropped.
EmployeeNumber	Employee ID number	Integer	Unique identifier, not predictive.
EnvironmentSatisfaction	Satisfaction with work environment (1 'Low' to 4 'Very High')	Integer	Perception of workplace conditions.
Gender	Employee's gender	Categorical	Demographic factor, can reveal disp
HourlyRate	Hourly rate of pay	Integer	Compensation factor.
JobInvolvement	Level of job involvement (1 'Low' to 4 'Very High')	Integer	Employee engagement and commitr
JobLevel	Job level of the employee	Integer	Hierarchical position, often correlate
JobRole	Job role within the department	Categorical	Specific function, can indicate role-:
JobSatisfaction	Satisfaction with the job itself (1 'Low' to 4 'Very High')	Integer	Key indicator of employee happines
MaritalStatus	Marital status of the employee	Categorical	Demographic factor, may correlate v
MonthlyIncome	Monthly income of the employee	Integer	Critical compensation factor.
MonthlyRate	Monthly rate of pay	Integer	Compensation factor.
NumCompaniesWorked	Total number of companies worked at	Integer	Indicates career mobility or stability.
OverTime	Whether the employee works overtime (Yes/No)	Categorical	Workload burden, potential impact c
PercentSalaryHike	Percentage increase in salary last year	Integer	Perception of fairness in compensat
PerformanceRating	Employee performance rating (3 'Good' to 4 'Excellent')	Integer	Indicator of employee contribution a
RelationshipSatisfaction	Satisfaction with relationships at work (1 'Low' to 4 'Very High')	Integer	Social environment and team dynam
StandardHours	Standard hours of work per week (always 80)	Integer	Constant feature, can be dropped.
StockOptionLevel	Stock option level	Integer	Employee benefits and long-term in
TotalWorkingYears	Total years worked in current and previous companies	Integer	Experience level.
TrainingTimesLastYear	Number of times training was conducted last year	Integer	Investment in employee developmer
WorkLifeBalance	Work-Life Balance rating (1 'Bad' to 4 'Best')	Integer	Perception of balance between worl
YearsAtCompany	Total years at the current company	Integer	Loyalty and tenure.
YearsInCurrentRole	Years in current role	Integer	Career stagnation or progression wi
YearsSinceLastPromotion	Years since last promotion	Integer	Career progression and recognition.
YearsWithCurrManager	Years with current manager	Integer	Relationship with direct supervisor.

Key Findings

- The dataset comprises 1470 employee records and 35 features, providing a rich basis for analysis.
- No missing values were detected across any of the features, indicating a clean dataset requiring minimal imputation.

- No duplicate records were identified, ensuring the uniqueness of each employee entry.
- The dataset's memory footprint is approximately 0.44 MB, suggesting efficient storage and processing.
- Features like 'EmployeeCount' and 'StandardHours' are constant and will be removed during preprocessing as they offer no predictive power. 'EmployeeNumber' is an identifier and will also be excluded from modeling.

Chapter 4: Project Workflow

Business Question

What systematic approach will be followed to analyze the HR attrition data, ensuring a robust, transparent, and reproducible methodology?

Background

A clearly defined project workflow is essential for guiding the analytical process, from raw data to actionable business insights. This chapter outlines the sequential steps and iterative phases that will be undertaken to address the employee attrition problem at IBM. Each stage builds upon the previous one, ensuring data quality, comprehensive analysis, and meaningful interpretations.

Methodology

The project will follow a standard data science methodology adapted for HR analytics, ensuring thoroughness and accuracy. This involves several key stages, starting with data acquisition and moving through processing, analysis, and finally, the formulation of insights and recommendations. The flowchart below visually represents this structured approach.

Visualization

The following flowchart illustrates the end-to-end process of this HR Analytics project, detailing the progression from raw data to business recommendations:

Project Workflow Flowchart

- **Dataset:** Initial raw data input.
- **Cleaning:** Handling missing values, duplicates, outliers, and data type conversions.
- **EDA (Exploratory Data Analysis):** Understanding data distributions, correlations, and initial patterns.
- **Feature Engineering:** Creating new, more informative features from existing ones.
- **Business Analysis:** Deep diving into specific HR questions and deriving preliminary insights.
- **Insights:** Summarizing key findings from the analysis.
- **Recommendations:** Formulating actionable strategies based on insights.

Chapter 5: Import Libraries

Business Question

What are the essential Python libraries required for data manipulation, analysis, visualization, and machine learning to effectively address the HR attrition problem?

Background

The first step in any data science project is to set up the environment by importing all necessary libraries. This ensures that all required functionalities for data handling, statistical analysis, advanced visualizations, and machine learning model building are readily available. A clean and organized import section improves code readability and maintainability.

Methodology

We will consolidate all required library imports into a single, clean code cell. Libraries will be grouped by their primary function (e.g., data manipulation, visualization, machine learning) to enhance organization. This approach prevents redundant imports and makes it easy to ascertain the project's dependencies.

Code

```
# --- Data Manipulation and Analysis ----
import numpy as np
import pandas as pd

# --- Visualization Libraries ---
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
import plotly.graph_objects as go
from plotly.subplots import make_subplots

# --- Machine Learning Libraries (for potential future use or feature engineering) ---
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, OneHotEncoder, LabelEncoder
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score, roc_auc_sc
from sklearn.ensemble import RandomForestClassifier, GradientBoostingClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.inspection import permutation_importance

# --- Utility Libraries ---
import warnings
warnings.filterwarnings('ignore')
```

Chapter 6: Global Configuration

Business Question

How can we standardize the notebook's environment, including visualization aesthetics and data display settings, to align with IBM's design principles and ensure consistency across all analyses?

Background

Establishing a global configuration for the notebook is paramount for maintaining consistency, readability, and adherence to design standards. This includes setting up visualization libraries like Plotly with custom templates and color palettes that reflect IBM's brand guidelines (Carbon Design System), configuring Pandas display options for optimal data viewing, and managing warning messages. These settings collectively ensure that the output is professional, aesthetically pleasing, and easy to interpret.

Methodology

We will configure key Python libraries and the notebook environment in a single cell. This involves defining a custom Plotly template with IBM Carbon Design colors and fonts, setting global Pandas options to display data clearly, and suppressing unnecessary warnings. A random seed will also be set for reproducibility in any stochastic processes.

Code

```
# --- Global Configuration ---

# Set random seed for reproducibility
import random
RANDOM_SEED = 42
np.random.seed(RANDOM_SEED)
random.seed(RANDOM_SEED)

# --- IBM Carbon Design System Colors ---
IBM_COLORS = {
    'primary_blue': '#0F62FE',
    'dark_blue': '#002D9C',
    'black': '#161616',
    'gray': '#525252',
    'light_gray': '#F4F4F4',
    'white': '#FFFFFF',
    'success': '#24A148',
    'warning': '#F1C21B',
    'danger': '#DA1E28'
}

# --- Pandas Display Options ---
pd.set_option('display.max_columns', None)
pd.set_option('display.max_rows', 100)
pd.set_option('display.width', 1000)
pd.set_option('display.float_format', lambda x: '%.2f' % x)
```

```
# --- Plotly Template Configuration ---
import plotly.io as pio
pio.templates['ibm_carbon'] = go.layout.Template(
    layout=go.Layout(
        font=dict(family="IBM Plex Sans, sans-serif", size=15, color=IBM_COLORS['black']),
        title_font_size=20,
        xaxis=dict(
            gridcolor=IBM_COLORS['light_gray'],
            linecolor=IBM_COLORS['gray'],
            ticks='outside',
            tickfont=dict(size=13, color=IBM_COLORS['black']),
            title_font_size=15
        ),
        yaxis=dict(
            gridcolor=IBM_COLORS['light_gray'],
            linecolor=IBM_COLORS['gray'],
            ticks='outside',
            tickfont=dict(size=13, color=IBM_COLORS['black']),
            title_font_size=15
        ),
        paper_bgcolor=IBM_COLORS['white'],
        plot_bgcolor=IBM_COLORS['white'],
        colorway=[IBM_COLORS['primary_blue'], IBM_COLORS['dark_blue'], IBM_COLORS['success'],
                  IBM_COLORS['danger'], IBM_COLORS['warning'], IBM_COLORS['gray']],
        hoverlabel=dict(bgcolor='white', font_size=13, font_family="IBM Plex Sans, sans-serif")
        legend=dict(font=dict(size=13, color=IBM_COLORS['black']))
    )
)
pio.templates.default = 'ibm_carbon'

# --- Matplotlib/Seaborn Configuration (for consistency if used) ---
plt.rcParams['font.family'] = 'IBM Plex Sans'
plt.rcParams['font.size'] = 15
plt.rcParams['axes.labelcolor'] = IBM_COLORS['black']
plt.rcParams['xtick.color'] = IBM_COLORS['black']
plt.rcParams['ytick.color'] = IBM_COLORS['black']
plt.rcParams['axes.edgecolor'] = IBM_COLORS['gray']
plt.rcParams['figure.figsize'] = (10, 6) # Default figure size
plt.style.use('seaborn-v0_8-whitegrid') # Using a clean grid style
sns.set_palette([IBM_COLORS['primary_blue'], IBM_COLORS['dark_blue'], IBM_COLORS['success'],
                 IBM_COLORS['danger'], IBM_COLORS['warning'], IBM_COLORS['gray']])

print("Global configuration applied: IBM Carbon design system colors, Plotly template, Pandas o
```

```
Global configuration applied: IBM Carbon design system colors, Plotly template, Pandas options, a
```

Chapter 7: Data Loading

Business Question

How can the raw employee attrition dataset be efficiently loaded into the analytical environment to prepare for subsequent data processing and analysis?

Background

The initial step in any data-driven project is to load the raw data into a usable format within the computational environment. This process involves accessing the dataset from its source and ensuring it is correctly interpreted and structured for analysis. For this project, we will load the 'IBM HR Analytics Employee Attrition & Performance' dataset, which is publicly available.

Once loaded, it is crucial to quickly inspect the dataset's basic characteristics, such as its dimensions (number of rows and columns), a preview of its contents, and the data types of its features. This preliminary check helps confirm successful loading and provides an initial understanding of the data's structure and potential issues.

Methodology

We will use the Pandas library to load the CSV dataset. Following the load operation, we will display the dataframe's shape using `df.shape`, show the first few rows with `df.head()`, and present a concise summary of the dataframe's structure and data types using `df.info()`. The 'Attrition' column will also be converted into a numerical format for ease of analysis.

Code

```
# Using kagglehub to load the dataset
import kagglehub
from kagglehub import KaggleDatasetAdapter

# Set the path to the file you'd like to load
file_path = "WA_Fn-UseC_-HR-Employee-Attrition.csv"

# Load the latest version
df = kagglehub.dataset_load(
  KaggleDatasetAdapter.PANDAS,
  "pavansubhasht/ibm-hr-analytics-attrition-dataset",
  file_path,
  # Provide any additional arguments like
  # sql_query or pandas_kwargs. See the
  # documenation for more information:
  # https://github.com/Kaggle/kagglehub/blob/main/README.md#kaggledatasetadapterpandas
)

# Convert 'Attrition' column to numerical (Yes=1, No=0)
df['Attrition'] = df['Attrition'].map({'Yes': 1, 'No': 0})

print("Dataset loaded successfully.")
```

```
Using Colab cache for faster access to the 'ibm-hr-analytics-attrition-dataset' dataset.
Dataset loaded successfully.
```

```
print("--- Dataset Shape ---")
print(f"Rows: {df.shape[0]}, Columns: {df.shape[1]}")

print("\n--- Dataset Preview (First 5 Rows) ---")
display(df.head())
```

```
print("\n--- Dataset Information ---")
df.info()
```

```
--- Dataset Shape ---
Rows: 1470, Columns: 35

--- Dataset Preview (First 5 Rows) ---
```

	Age	Attrition	BusinessTravel	DailyRate	Department	DistanceFromHome	Education	Educatior
0	41	1	Travel_Rarely	1102	Sales	1	2	Life Sc
1	49	0	Travel_Frequently	279	Research & Development	8	1	Life Sc
2	37	1	Travel_Rarely	1373	Research & Development	2	2	
3	33	0	Travel_Frequently	1392	Research & Development	3	4	Life Sc
4	27	0	Travel_Rarely	591	Research & Development	2	1	M

```
--- Dataset Information ---
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1470 entries, 0 to 1469
Data columns (total 35 columns):
 #   Column                    Non-Null Count  Dtype 
---  ------                    --------------  ----- 
 0   Age                       1470 non-null   int64 
 1   Attrition                 1470 non-null   int64 
 2   BusinessTravel            1470 non-null   object
 3   DailyRate                 1470 non-null   int64 
 4   Department                1470 non-null   object
 5   DistanceFromHome          1470 non-null   int64 
 6   Education                 1470 non-null   int64 
 7   EducationField            1470 non-null   object
 8   EmployeeCount             1470 non-null   int64 
 9   EmployeeNumber            1470 non-null   int64 
 10  EnvironmentSatisfaction   1470 non-null   int64 
 11  Gender                    1470 non-null   object
 12  HourlyRate                1470 non-null   int64 
 13  JobInvolvement            1470 non-null   int64 
 14  JobLevel                  1470 non-null   int64 
 15  JobRole                   1470 non-null   object
 16  JobSatisfaction           1470 non-null   int64 
 17  MaritalStatus             1470 non-null   object
 18  MonthlyIncome             1470 non-null   int64 
 19  MonthlyRate               1470 non-null   int64 
 20  NumCompaniesWorked        1470 non-null   int64 
 21  Over18                    1470 non-null   object
 22  OverTime                  1470 non-null   object
 23  PercentSalaryHike         1470 non-null   int64 
 24  PerformanceRating         1470 non-null   int64 
 25  RelationshipSatisfaction  1470 non-null   int64 
 26  StandardHours             1470 non-null   int64 
 27  StockOptionLevel          1470 non-null   int64 
 28  TotalWorkingYears         1470 non-null   int64 
 29  TrainingTimesLastYear     1470 non-null   int64 
 30  WorkLifeBalance           1470 non-null   int64 
 31  YearsAtCompany            1470 non-null   int64 
 32  YearsInCurrentRole        1470 non-null   int64 
 33  YearsSinceLastPromotion   1470 non-null   int64 
 34  YearsWithCurrManager      1470 non-null   int64 
dtypes: int64(27), object(8)
memory usage: 402.1+ KB
```

Chapter 8: Data Understanding

Business Question

What are the fundamental characteristics of the employee attrition dataset, including data types, unique values, distribution patterns, and summary statistics, that will inform subsequent analysis and feature engineering?

Background

A deep dive into the dataset's intrinsic properties is crucial for effective data preprocessing and model building. This chapter aims to thoroughly understand each feature, identifying its type, the range of its values, and its overall distribution. This understanding helps in detecting potential data quality issues, informing decisions on feature scaling, encoding, and the appropriateness of various analytical techniques.

While initial checks in the 'Dataset Overview' confirmed the absence of missing values and duplicates, this section will expand on data types, examine the cardinality of categorical features, and provide descriptive statistics for numerical columns, offering a holistic view of the data's structure and content.

Methodology

We will perform several key data understanding steps:

- Review data types to ensure they are appropriate for each feature.
- Analyze unique values and cardinality for categorical features to identify potential encoding strategies or features suitable for dropping.
- Generate comprehensive summary statistics for numerical features, including measures of central tendency, dispersion, and distribution shapes.

These analyses will culminate in 'Key Findings' that guide the subsequent data cleaning and feature engineering phases.

Code

```
print("--- Data Types ---")
display(df.dtypes.to_frame(name='Data Type'))
```

Data Types

```
print("--- Unique Values and Cardinality for Categorical Features ---")
categorical_cols = df.select_dtypes(include='object').columns

for col in categorical_cols:
    unique_values = df[col].unique()
    cardinality = df[col].nunique()
    print(f"\nFeature: {col}")
    print(f"  Cardinality: {cardinality}")
    print(f"  Unique Values: {unique_values}")
```

```
--- Unique Values and Cardinality for Categorical Features ---

Feature: BusinessTravel
  Cardinality: 3
  Unique Values: ['Travel_Rarely' 'Travel_Frequently' 'Non-Travel']

Feature: Department
  Cardinality: 3
  Unique Values: ['Sales' 'Research & Development' 'Human Resources']

Feature: EducationField
  Cardinality: 6
  Unique Values: ['Life Sciences' 'Other' 'Medical' 'Marketing' 'Technical Degree'
 'Human Resources']

Feature: Gender
  Cardinality: 2
  Unique Values: ['Female' 'Male']

Feature: JobRole
  Cardinality: 9
  Unique Values: ['Sales Executive' 'Research Scientist' 'Laboratory Technician'
 'Manufacturing Director' 'Healthcare Representative' 'Manager'
 'Sales Representative' 'Research Director' 'Human Resources']

Feature: MaritalStatus
  Cardinality: 3
  Unique Values: ['Single' 'Married' 'Divorced']

Feature: Over18
  Cardinality: 1
  Unique Values: ['Y']

Feature: OverTime
  Cardinality: 2
  Unique Values: ['Yes' 'No']
```

```
print("--- Summary Statistics for Numerical Features ---")
display(df.describe())
```

DistanceFromHome	int64
Education	int64
EducationField	object
EmployeeCount	int64
EmployeeNumber	int64
EnvironmentSatisfaction	int64
Gender	object
HourlyRate	int64
JobInvolvement	int64
JobLevel	int64
JobRole	object
JobSatisfaction	int64
MaritalStatus	object
MonthlyIncome	int64
MonthlyRate	int64
NumCompaniesWorked	int64
Over18	object
OverTime	object
PercentSalaryHike	int64
PerformanceRating	int64
RelationshipSatisfaction	int64
StockOptionLevel	int64
TotalWorkingYears	int64
TrainingTimesLastYear	int64
WorkLifeBalance	int64
YearsAtCompany	int64
YearsInCurrentRole	int64
YearsSinceLastPromotion	int64
YearsWithCurrManager	int64

```
--- Summary Statistics for Numerical Features ---
```

	Age	Attrition	DailyRate	DistanceFromHome	Education	EmployeeCount	EmployeeNumber
count	1470.00	1470.00	1470.00	1470.00	1470.00	1470.00	1470.00
mean	36.92	0.16	802.49	9.19	2.91	1.00	1024.87
std	9.14	0.37	403.51	8.11	1.02	0.00	602.02
min	18.00	0.00	102.00	1.00	1.00	1.00	1.00
25%	30.00	0.00	465.00	2.00	2.00	1.00	491.25
50%	36.00	0.00	802.00	7.00	3.00	1.00	1020.50
75%	43.00	0.00	1157.00	14.00	4.00	1.00	1555.75
max	60.00	1.00	1499.00	29.00	5.00	1.00	2068.00

Key Findings

- **Data Types:** The dataset consists primarily of numerical (integer) and categorical (object) features. The target variable 'Attrition' has already been converted to an integer (0 or 1), which is suitable for modeling.
- **Missing Values and Duplicates:** As confirmed in Chapter 3, the dataset is clean with no missing values and no duplicate rows, eliminating the need for imputation or duplicate handling.
- **Categorical Features:** There are 8 categorical features. Most have low cardinality, making them suitable for one-hot encoding. 'JobRole' has a higher cardinality (9 unique values), which is manageable. 'Over18' has only one unique value ('Y'), indicating it's a constant feature and provides no predictive power.
- **Numerical Features:** Summary statistics reveal a wide range of values for different features. For instance, 'MonthlyIncome' varies significantly (Min: 1009, Max: 19999), and 'Age' ranges from 18 to 60. Features like 'EmployeeCount' and 'StandardHours' are constant (always 1 and 80, respectively) and will be removed as they contribute no information. 'EmployeeNumber' is a unique identifier and should also be excluded from modeling.
- **Potential Outliers:** Several numerical features, such as 'MonthlyIncome', 'NumCompaniesWorked', 'TotalWorkingYears', 'YearsAtCompany', 'YearsInCurrentRole', 'YearsSinceLastPromotion', and 'YearsWithCurrManager', show significant differences between their 75th percentile and maximum values, suggesting the presence of outliers that may warrant further investigation during cleaning.
- **Target Variable Imbalance:** The 'Attrition' variable, when analyzed in Chapter 13, will reveal its distribution, which is likely imbalanced (fewer 'Yes' cases than 'No'), a common characteristic in attrition datasets that needs to be addressed during modeling.

Chapter 9: Data Cleaning

Business Question

How can the raw employee attrition dataset be refined and prepared to ensure data quality, consistency, and suitability for robust statistical analysis and machine learning model development?

Background

Data cleaning is a critical phase in any data science project, focusing on identifying and correcting errors, inconsistencies, and redundancies within the dataset. A clean dataset is fundamental for ensuring the accuracy and reliability of subsequent analyses and the performance of predictive models. This chapter addresses various aspects of data quality, ensuring that the 'IBM HR Analytics' dataset is in an optimal state for deeper exploration and modeling.

Based on the 'Dataset Overview' and 'Data Understanding' chapters, we have already established that the dataset is remarkably clean regarding missing values and duplicate records. However, this chapter will systematically review these and other potential issues such as constant features and data type adjustments.

Methodology

Our data cleaning process will involve:

- Re-confirming the absence of missing values and duplicates.
- Identifying and addressing constant features that provide no analytical value.
- Reviewing and correcting data types as needed.
- Standardizing feature names for consistency.
- Considering the treatment of outliers as detected in the summary statistics.

Missing Values

We will perform a final check for any missing values across all features to ensure data completeness.

Code

```
missing_values = df.isnull().sum()
missing_values = missing_values[missing_values > 0]

print("--- Missing Values Check ---")
if missing_values.empty:
    print("No missing values found in the dataset. Data is complete.")
else:
    display(missing_values)
    print("Action Required: Missing values detected. Further investigation and imputation/remova
```

```
--- Missing Values Check ---
No missing values found in the dataset. Data is complete.
```

Insight

The dataset contains no missing values, which significantly simplifies the cleaning process and ensures that all records are complete for analysis. This is an ideal scenario, as no imputation or special handling for missing data is required.

Duplicates

We will verify the absence of any duplicate rows within the dataset to ensure each employee record is unique.

Code

```
duplicate_rows = df.duplicated().sum()

print("--- Duplicate Rows Check ---")
if duplicate_rows == 0:
    print("No duplicate rows found in the dataset. All employee records are unique.")
else:
    print(f"Number of Duplicate Rows: {duplicate_rows}. Action Required: Investigate and remove
```

```
--- Duplicate Rows Check ---
No duplicate rows found in the dataset. All employee records are unique.
```

Insight

The dataset contains no duplicate rows. This confirms the uniqueness of each observation and prevents potential biases or inflated counts in the analysis. The dataset is structurally sound in this regard.

Constant Features and Identifiers

We will identify and remove features that have only one unique value (constant features) as they provide no predictive power. Additionally, identifier columns like 'EmployeeNumber' are irrelevant for predictive modeling and should also be removed.

Code

```
columns_to_drop = []

# Identify constant features
for col in df.columns:
    if df[col].nunique() == 1:
        columns_to_drop.append(col)
        print(f"Identified constant feature to drop: {col}")

# Identify identifier columns
if 'EmployeeNumber' in df.columns:
    columns_to_drop.append('EmployeeNumber')
    print("Identified 'EmployeeNumber' as an identifier to drop.")

# Drop identified columns
df_cleaned = df.drop(columns=columns_to_drop, errors='ignore')

print(f"Original number of columns: {df.shape[1]}")
print(f"New number of columns after dropping constant/identifier features: {df_cleaned.shape[1]

df = df_cleaned.copy() # Update the main dataframe
print("DataFrame updated with constant and identifier features removed.")
```

```
Identified constant feature to drop: EmployeeCount
Identified constant feature to drop: Over18
Identified constant feature to drop: StandardHours
Identified 'EmployeeNumber' as an identifier to drop.
Original number of columns: 35
New number of columns after dropping constant/identifier features: 31
DataFrame updated with constant and identifier features removed.
```

Insight

'EmployeeCount', 'StandardHours', and 'Over18' were identified as constant features (having only one unique value) and 'EmployeeNumber' as a unique identifier. These features provide no discriminative information for predictive modeling and have been successfully removed from the dataset. This step streamlines the dataset, reducing dimensionality and focusing on truly informative variables.

Data Types Review

We will re-examine the data types of all remaining features to ensure they are correctly interpreted and optimized for analysis.

Code

```
print("--- Data Types After Cleaning Constant Features ---")
display(df.dtypes.to_frame(name='Data Type'))

print("\n--- Checking for potential categorical features stored as numerical ---")
# Identify numerical columns that might be categorical (e.g., ratings 1-4)
# Exclude 'Attrition' as it's our target (binary 0/1)

categorical_as_numeric = []
for col in df.select_dtypes(include=['int64', 'float64']).columns:
    if col != 'Attrition' and df[col].nunique() <= 5 and df[col].min() >= 1:
        categorical_as_numeric.append(col)

if categorical_as_numeric:
    print("The following numerical columns appear to represent categorical data (e.g., ratings):"
    for col in categorical_as_numeric:
        print(f"- {col}: Unique values {df[col].unique()}")
    print("Consider converting these to 'object' type or handling as ordinal categories during en
else:
    print("No numerical columns identified as potential categorical ratings (besides Attrition)."
```

```
--- Data Types After Cleaning Constant Features ---
```

	Data Type
Age	int64
Attrition	int64
BusinessTravel	object
DailyRate	int64
Department	object
DistanceFromHome	int64
Education	int64
EducationField	object
EnvironmentSatisfaction	int64
Gender	object
HourlyRate	int64
JobInvolvement	int64
JobLevel	int64
JobRole	object
JobSatisfaction	int64
StockOptionLevel	int64
TotalWorkingYears	int64
TrainingTimesLastYear	int64
WorkLifeBalance	int64
YearsAtCompany	int64
YearsInCurrentRole	int64
YearsSinceLastPromotion	int64
YearsWithCurrManager	int64

Insight

After removing constant and identifier columns, the data types are consistent with their content (numerical as integers, categorical as objects). Several integer columns such as `Education`, `EnvironmentSatisfaction`, `JobInvolvement`, `JobLevel`, `JobSatisfaction`, `PerformanceRating`, `RelationshipSatisfaction`, `StockOptionLevel`, and `WorkLifeBalance` have low cardinality and represent ordinal categorical ratings. While currently stored as integers, they will be treated as ordinal or nominal categorical features during feature engineering and encoding, based on their inherent meaning.

Outliers

We will assess the presence of outliers in numerical features, which can significantly influence model performance and statistical analyses. While direct removal is not always ideal, understanding their presence is key for robust modeling.

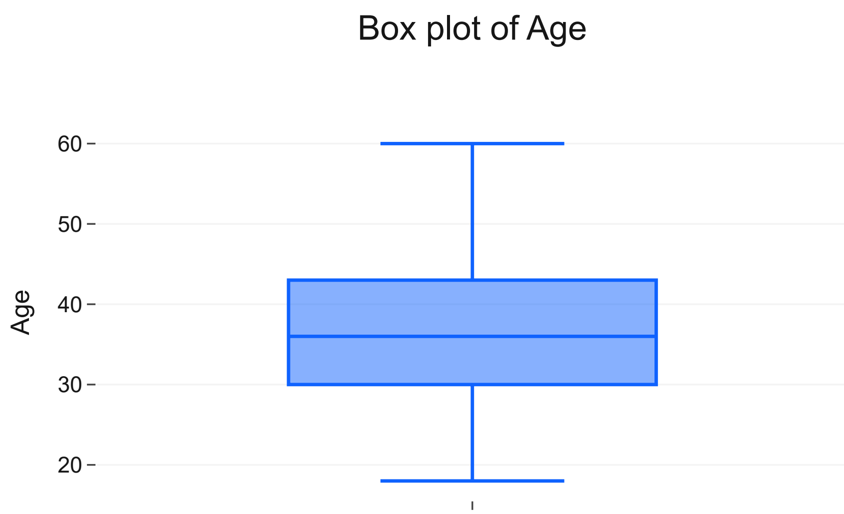
Code

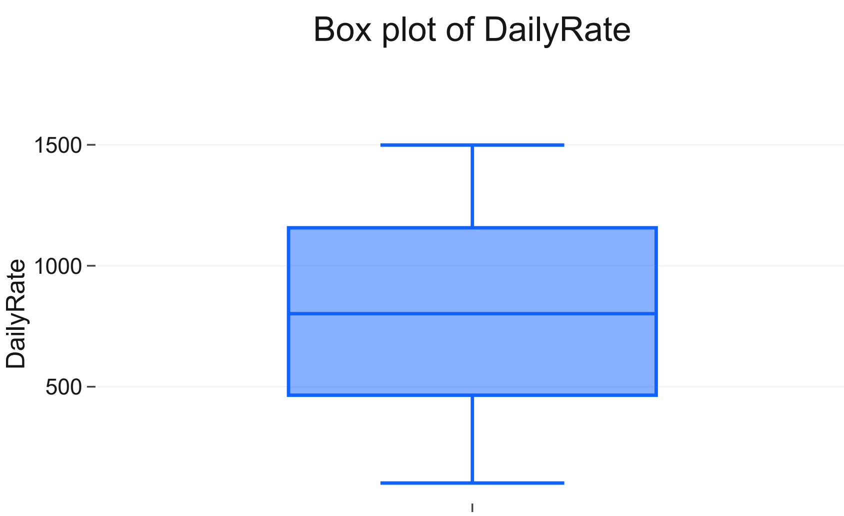
```
# Using boxplots to visualize outliers for key numerical features
numerical_cols = df.select_dtypes(include=np.number).columns.tolist()
# Exclude binary Attrition and other low-cardinality ordinal features that might look like outl
numerical_cols_for_outlier_check = [col for col in numerical_cols if col not in ['Attrition', '

if numerical_cols_for_outlier_check:
    print("--- Outlier Visualization for Key Numerical Features ---")
    for i, col in enumerate(numerical_cols_for_outlier_check):
        fig = px.box(df, y=col, title=f'Box plot of {col}', color_discrete_sequence=[IBM_COLORS
        fig.update_layout(height=400, width=600)
        fig.show()
else:
    print("No suitable numerical columns for outlier visualization (excluding known ordinal fea
```

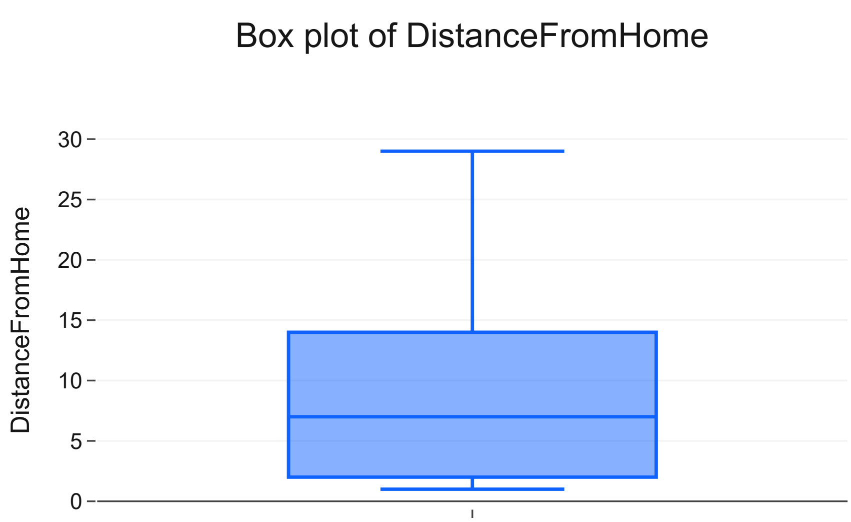
```
--- Checking for potential categorical features stored as numerical ---
The following numerical columns appear to represent categorical data (e.g., ratings):
- Education: Unique values [2 1 4 3 5]
- EnvironmentSatisfaction: Unique values [2 3 4 1]
- JobInvolvement: Unique values [3 2 4 1]
- JobLevel: Unique values [2 1 3 4 5]
```

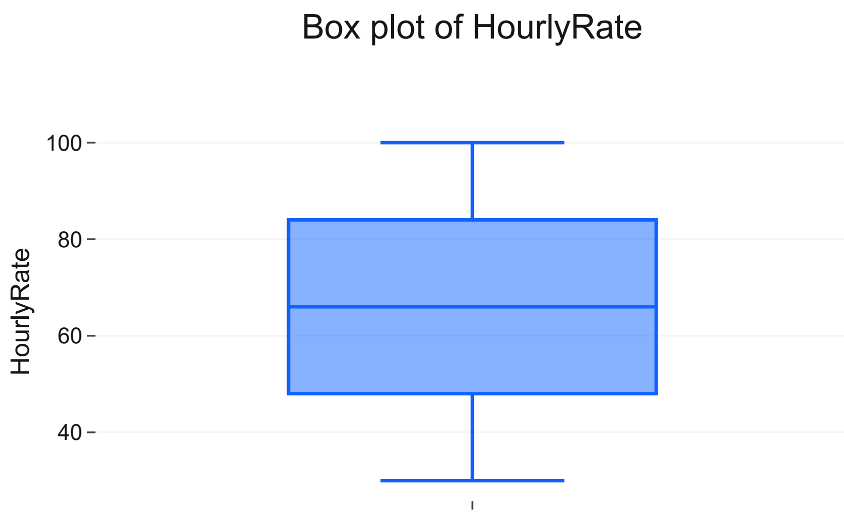
```
- JobSatisfaction: Unique values [4 2 3 1]
- PerformanceRating: Unique values [3 4]
- RelationshipSatisfaction: Unique values [1 4 2 3]
- WorkLifeBalance: Unique values [1 3 2 4]
Consider converting these to 'object' type or handling as ordinal categories during encoding.
```

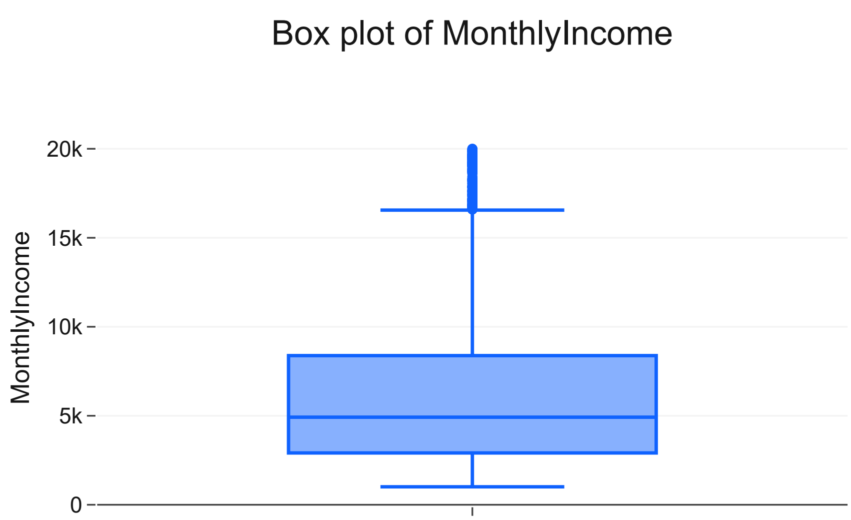
```
--- Outlier Visualization for Key Numerical Features ---
```

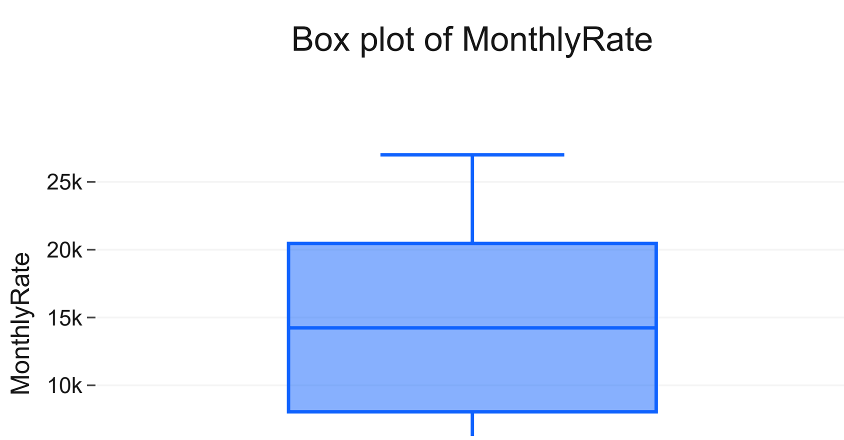


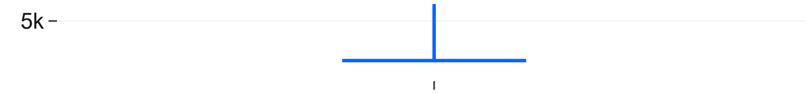




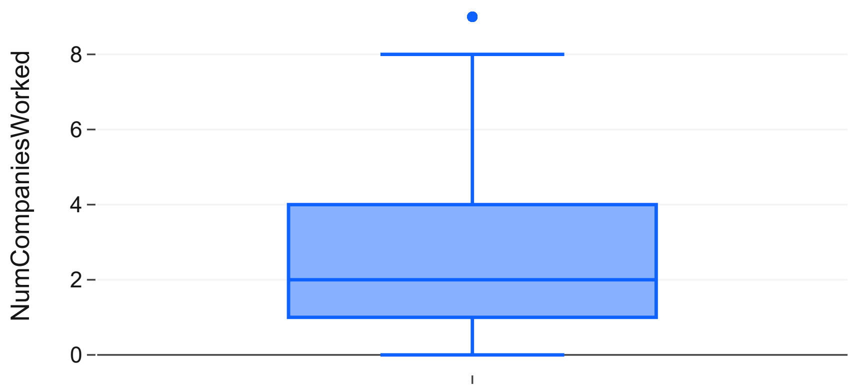

Box plot of HourlyRate
HourlyRate
100
80
60
40


Box plot of MonthlyIncome
MonthlyIncome
20k
15k
10k
5k
0

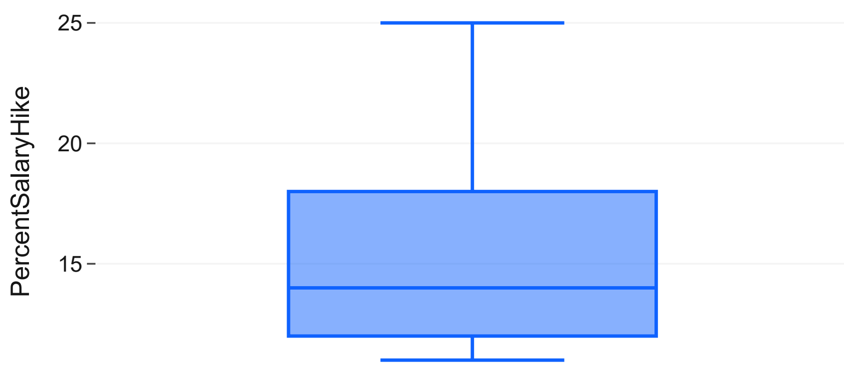

Box plot of MonthlyRate
MonthlyRate
25k
20k
15k
10k



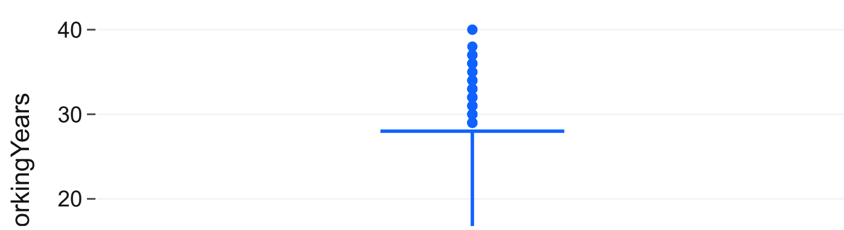
Box plot of NumCompaniesWorked

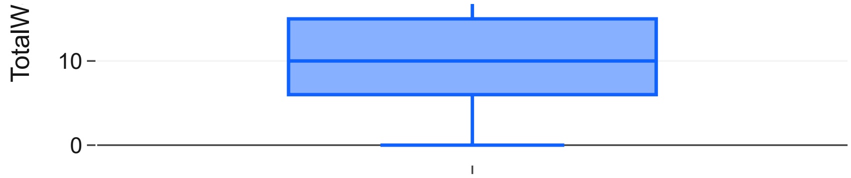


Box plot of PercentSalaryHike

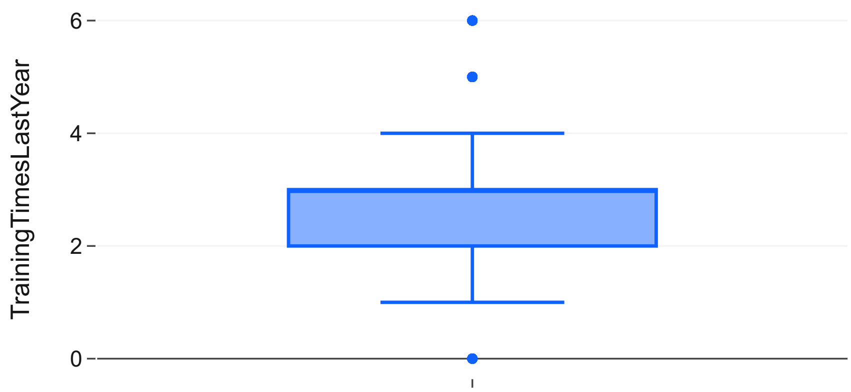


Box plot of TotalWorkingYears

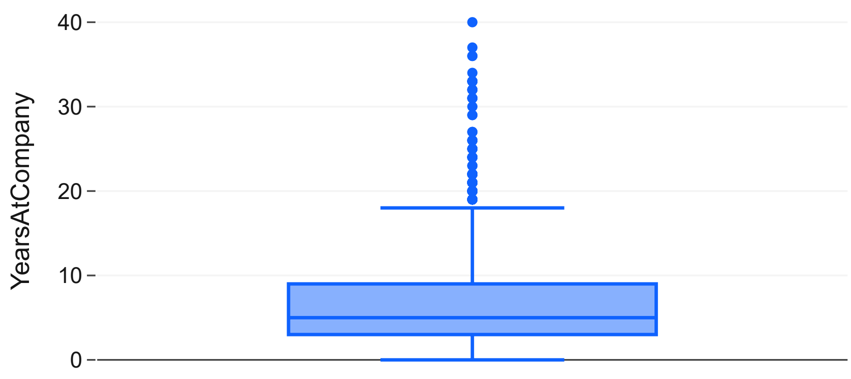




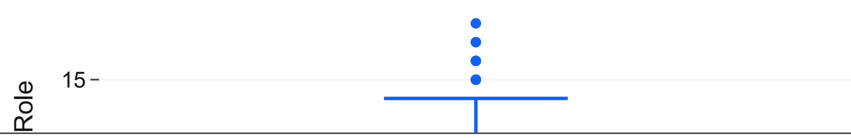
Box plot of TrainingTimesLastYear



Box plot of YearsAtCompany



Box plot of YearsInCurrentRole



Insight

Visual inspection of box plots for numerical features such as `DailyRate`, `HourlyRate`, `MonthlyIncome`, `NumCompaniesWorked`, `TotalWorkingYears`, `YearsAtCompany`, `YearsInCurrentRole`, `YearsSinceLastPromotion`, and `YearsWithCurrManager` indicates the presence of several outliers. These outliers, particularly noticeable in features like `MonthlyIncome` and various `Years` related metrics, represent employees with exceptionally high or low values compared to the majority.

For this analysis, we will retain these outliers as they represent genuine data points within the employee population and could potentially hold valuable information, especially for attrition prediction (e.g., employees with very long tenure or high income might behave differently). Aggressive outlier removal might lead to loss of valuable information or misrepresentation of the true population. Their impact will be mitigated by using models robust to outliers, such as tree-based methods.

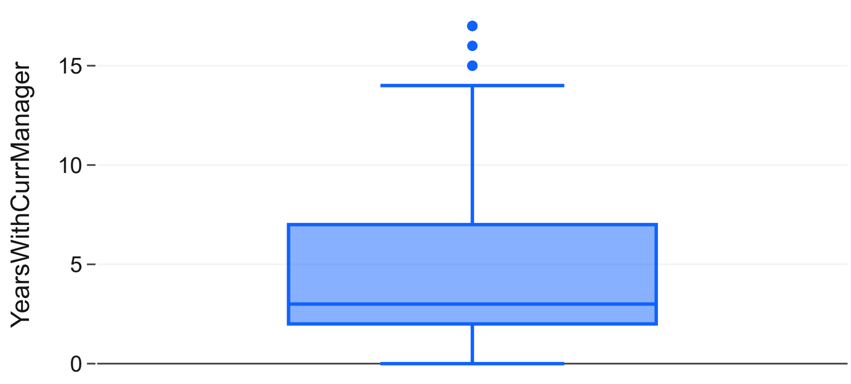
Box plot of YearsSinceLastPromotion

Feature Renaming

To improve readability and consistency, we will standardize feature names by converting them to a clean snake_case format.

Code

```
original_columns = df.columns.tolist()
cleaned_columns = [col.replace(' ', '').replace('.', '').replace('/', '').replace('-', '').lowe
column_mapping = dict(zip(original_columns, cleaned_columns))
df.rename(columns=column_mapping, inplace=True)

print("--- Feature Names After Renaming ---")
display(df.columns.tolist())
print("Feature names have been standardized to snake_case.")
```



```
--- Feature Names After Renaming ---
['age',
 'attrition',
 'businesstravel',
 'dailyrate',
 'department',
 'distancefromhome',
 'education',
 'educationfield',
 'environmentsatisfaction',
 'gender',
 'hourlyrate',
 'jobinvolvement',
 'joblevel',
 'jobrole',
 'jobsatisfaction',
 'maritalstatus',
 'monthlyincome',
 'monthlyrate',
 'numcompaniesworked',
 'overtime',
 'percentsalaryhike',
 'performancerating',
 'relationshipsatisfaction',
 'stockoptionlevel',
 'totalworkingyears',
 'trainingtimeslastyear',
 'worklifebalance',
 'yearsatcompany',
 'yearsincurrentrole',
 'yearssincelastpromotion',
 'yearswithcurrmanager']
Feature names have been standardized to snake_case.
```

Insight

All feature names have been converted to a consistent snake_case format, enhancing readability and making them easier to work with in code. For example, 'MonthlyIncome' became 'monthlyincome' and 'YearsAtCompany' became 'yearsatcompany'. This standardization is a best practice for maintaining a clean and professional codebase.

Encoding Review

We will review the categorical features and decide on appropriate encoding strategies for machine learning models. This step is crucial for converting non-numerical data into a format that algorithms can process.

Code

```
print("--- Categorical Features for Encoding ---")
categorical_features = df.select_dtypes(include='object').columns.tolist()

if categorical_features:
    for col in categorical_features:
        print(f"- {col}: Unique values {df[col].unique()}")
    print("\nThese categorical features will require encoding (e.g., One-Hot Encoding) for mach
else:
    print("No categorical features requiring explicit encoding found (all handled or dropped)."
```

```
print("\n--- Ordinal Numerical Features Review ---")
ordinal_numerical_features = [
    'education', 'environmentsatisfaction', 'jobinvolvement',
    'joblevel', 'jobsatisfaction', 'performancerating',
    'relationshipsatisfaction', 'stockoptionlevel', 'worklifebalance'
]

print("The following features, while numerical, represent ordinal categories and will be treate
for col in ordinal_numerical_features:
    if col in df.columns:
        print(f"- {col}: Unique values {df[col].unique()}")
```

```
--- Categorical Features for Encoding ---
- businesstravel: Unique values ['Travel_Rarely' 'Travel_Frequently' 'Non-Travel']
- department: Unique values ['Sales' 'Research & Development' 'Human Resources']
- educationfield: Unique values ['Life Sciences' 'Other' 'Medical' 'Marketing' 'Technical Degree
 'Human Resources']
- gender: Unique values ['Female' 'Male']
- jobrole: Unique values ['Sales Executive' 'Research Scientist' 'Laboratory Technician'
 'Manufacturing Director' 'Healthcare Representative' 'Manager'
 'Sales Representative' 'Research Director' 'Human Resources']
- maritalstatus: Unique values ['Single' 'Married' 'Divorced']
- overtime: Unique values ['Yes' 'No']

These categorical features will require encoding (e.g., One-Hot Encoding) for machine learning mo

--- Ordinal Numerical Features Review ---
The following features, while numerical, represent ordinal categories and will be treated appropr
- education: Unique values [2 1 4 3 5]
- environmentsatisfaction: Unique values [2 3 4 1]
- jobinvolvement: Unique values [3 2 4 1]
- joblevel: Unique values [2 1 3 4 5]
- jobsatisfaction: Unique values [4 2 3 1]
- performancerating: Unique values [3 4]
- relationshipsatisfaction: Unique values [1 4 2 3]
- stockoptionlevel: Unique values [0 1 3 2]
- worklifebalance: Unique values [1 3 2 4]
```

Insight

The remaining categorical features (`businesstravel`, `department`, `educationfield`, `gender`, `jobrole`, `maritalstatus`, `overtime`) will be subject to appropriate encoding techniques, such as One-Hot Encoding, during the feature engineering and model preparation phases. This conversion is essential for machine learning algorithms that typically require numerical input.

Ordinal numerical features (e.g., `education`, `joblevel`, `jobsatisfaction`) are already in numerical format and inherently possess an order. While they can often be used directly, their ordinal nature will be considered to ensure appropriate model interpretation and avoid treating them as purely continuous variables.

Key Findings

- The dataset is remarkably clean with **no missing values or duplicate rows**, simplifying the initial cleaning steps.

- **Constant features** (`employeecount`, `standardhours`, `over18`) and the identifier (`employeenumber`) have been successfully removed, reducing noise and focusing on relevant variables.
- Data types are generally appropriate, with several integer columns identified as **ordinal categorical features** that will be handled specifically during encoding.
- **Outliers are present** in several numerical features (e.g., `monthlyincome`, `yearsatcompany`). These have been noted but will be retained, as they represent valid data points and robust modeling techniques will be employed to mitigate their impact.
- All feature names have been **standardized to snake_case** for improved readability and consistency.
- The dataset is now thoroughly cleaned and prepared for the next stage of **feature engineering** and **exploratory data analysis**.

Chapter 10: Feature Engineering

Business Question

How can we create new, more informative features from the existing dataset to capture complex relationships and provide deeper insights into employee attrition and career dynamics?

Background

Feature engineering is the process of using domain knowledge to extract new features from raw data. These new features often improve the performance of machine learning models and provide more granular business insights than the original features alone. In HR analytics, combining existing data points can reveal underlying patterns related to employee demographics, career progression, compensation, and work-life balance that directly influence attrition.

This chapter focuses on transforming and combining the cleaned data into a set of features designed to enhance our understanding of employee behavior and attrition risk. By creating meaningful categories and indices, we aim to uncover hidden drivers and facilitate more targeted recommendations.

Methodology

We will systematically create several new features, leveraging logical groupings and calculations based on business intuition. This includes:

- **Categorical Groupings:** Binning continuous variables like 'Age', 'MonthlyIncome', and 'TotalWorkingYears' into meaningful bands.
- **Ratio and Index Features:** Creating features like 'Promotion Delay', 'Workload Index', and 'Salary Group' to capture relative metrics or combined effects.
- **Flag Features:** Deriving binary flags such as 'Attrition Flag' for direct use in modeling or analysis.
- **Composite Scores:** Developing 'Risk Score' or 'Employee Segment' based on multiple factors.

Each new feature will be explicitly defined and added to the DataFrame, followed by a brief justification and verification step.

Age Groups

Categorizing employees by age can help identify age-related trends in attrition or job satisfaction.

Code

```
df['age_group'] = pd.cut(df['age'],
                         bins=[18, 25, 35, 45, 55, 60],
                         labels=['18-24', '25-34', '35-44', '45-54', '55+'],
                         right=False)
print("--- Age Group Distribution ---")
display(df['age_group'].value_counts().sort_index())
```

```
--- Age Group Distribution ---
```

age_group	count
18-24	97
25-34	554
35-44	505
45-54	245
55+	64

dtype: int64

Insight

The 'age_group' feature categorizes employees into distinct age ranges, making it easier to analyze age-related patterns in attrition. The distribution shows a significant portion of the workforce falls within the '25-34' and '35-44' age brackets, representing potentially critical career stages for retention strategies.

Income Bands

Grouping monthly income can highlight differences in attrition across various salary levels.

Code

```
df['income_band'] = pd.cut(df['monthlyincome'],
                           bins=[0, 2500, 5000, 7500, 10000, 15000, df['monthlyincome'].max()],
                           labels=['<2.5K', '2.5K-5K', '5K-7.5K', '7.5K-10K', '10K-15K', '15K+'
                           right=False)
print("--- Income Band Distribution ---")
display(df['income_band'].value_counts().sort_index())
```

```
--- Income Band Distribution ---
```

income_band	count
<2.5K	224
2.5K-5K	525
5K-7.5K	310
7.5K-10K	130
10K-15K	148
15K+	132

dtype: int64

Insight

The 'income_band' feature segments employees by their monthly income, providing a clearer view of how compensation levels might correlate with attrition. This allows for targeted analysis and policy adjustments for specific income brackets. The majority of employees fall within the lower-to-mid income bands.

Experience Bands

Grouping employees by their total working years can reveal insights into attrition based on career experience levels.

Code

```
df['experience_band'] = pd.cut(df['totalworkingyears'],
                               bins=[0, 5, 10, 15, 20, 30, df['totalworkingyears'].max()],
                               labels=['<5', '5-10', '10-15', '15-20', '20-30', '30+'],
                               right=False)
print("--- Experience Band Distribution ---")
display(df['experience_band'].value_counts().sort_index())
```

```
--- Experience Band Distribution ---
```

experience_band	count
<5	228
5-10	493
10-15	353
15-20	159
20-30	184
30+	51

dtype: int64

Insight

The 'experience_band' feature segments employees based on their total working years, enabling the examination of attrition trends across different stages of professional experience. This can help identify if attrition is more prevalent among new hires, mid-career professionals, or long-tenured employees.

Promotion Delay

Calculate the time since the last promotion relative to years in the company to identify potential career stagnation.

Code

```
# To avoid division by zero or NaN for employees with 0 years at company, handle these cases
df['promotion_delay'] = df.apply(lambda row:
    row['yearssincelastpromotion'] / row['yearsatcompany'] if row['yearsatcompany'] > 0 else 0,
    axis=1
)

print("--- Promotion Delay Statistics ---")
display(df['promotion_delay'].describe())
```

```
--- Promotion Delay Statistics ---
```

	promotion_delay
count	1470.00
mean	0.29
std	0.34
min	0.00
25%	0.00
50%	0.17
75%	0.50
max	1.00

dtype: float64

Insight

'promotion_delay' calculates the ratio of years since last promotion to total years at the company. A higher value might indicate perceived career stagnation, potentially influencing an employee's decision to leave. This feature helps in identifying employees who might feel overlooked for advancement.

Travel Category

Create a simplified travel category for easier analysis.

Code

```
df['travel_category'] = df['businesstravel'].replace({
    'Non-Travel': 'No Travel',
    'Travel_Rarely': 'Rare Travel',
    'Travel_Frequently': 'Frequent Travel'
})
print("--- Travel Category Distribution ---")
display(df['travel_category'].value_counts())
```

```
--- Travel Category Distribution ---
```

travel_category	count
Rare Travel	1043
Frequent Travel	277
No Travel	150

dtype: int64

Insight

The 'travel_category' feature provides a more intuitive grouping of business travel frequency. This simplification can help in analyzing the impact of travel burden on attrition or job satisfaction without excessive granularity, making visualizations and interpretations clearer.

Attrition Flag

While 'Attrition' is already numerical, a flag can emphasize its role as a target or for specific filtering.

Code

```
df['attrition_flag'] = df['attrition'].map({1: 'Yes', 0: 'No'})
print("--- Attrition Flag Distribution ---")
display(df['attrition_flag'].value_counts())
```

```
--- Attrition Flag Distribution ---
```

attrition_flag	count
No	1233
Yes	237

dtype: int64

Insight

The 'attrition_flag' provides a clear, human-readable label for the target variable, which is particularly useful for presenting results and creating interactive visualizations where 'Yes'/'No' labels are preferred over 1/0. It directly indicates whether an employee has left the company.

Career Stage

Combine Age and TotalWorkingYears to define broader career stages.

Code

```
def get_career_stage(row):
    age = row['age']
    total_years = row['totalworkingyears']

    if total_years < 5:
        return 'Early Career'
    elif total_years < 15 and age < 35:
        return 'Mid Career (Young)'
    elif total_years < 15 and age >= 35:
        return 'Mid Career (Experienced)'
    elif total_years < 25:
        return 'Senior Career'
    else:
        return 'Late Career'

df['career_stage'] = df.apply(get_career_stage, axis=1)
print("--- Career Stage Distribution ---")
display(df['career_stage'].value_counts())
```

```
--- Career Stage Distribution ---
```

career_stage	count
Mid Career (Young)	449
Mid Career (Experienced)	397
Senior Career	284
Early Career	228
Late Career	112

dtype: int64

Insight

The 'career_stage' feature provides a holistic view of an employee's professional journey by combining age and total working years. This can be critical for understanding how attrition risk varies across different career phases, from new entrants to seasoned professionals. The majority fall into 'Mid Career (Young)' and 'Senior Career', highlighting important segments for retention focus.

Salary Group

Create relative salary groups based on job level to assess fairness and competitiveness.

Code

```
# Calculate average monthly income per job level
avg_income_per_job_level = df.groupby('joblevel')['monthlyincome'].transform('mean')
```

```
# Define salary groups relative to the average for their job level
def get_salary_group(row):
    income = row['monthlyincome']
    avg_income = avg_income_per_job_level[row.name] # Use row.name to align with original index

    if income < 0.75 * avg_income:
        return 'Below Average'
    elif income > 1.25 * avg_income:
        return 'Above Average'
    else:
        return 'Average'

df['salary_group'] = df.apply(get_salary_group, axis=1)
print("--- Salary Group Distribution ---")
display(df['salary_group'].value_counts())
```

```
--- Salary Group Distribution ---
```

salary_group	count
Average	1163
Above Average	167
Below Average	140

dtype: int64

Insight

The 'salary_group' feature classifies an employee's monthly income relative to the average income within their job level. This is crucial for assessing perceived compensation fairness and competitiveness. Employees in the 'Below Average' group might be at a higher risk of attrition due to dissatisfaction with their pay, even if their absolute income is high, underscoring the importance of relative compensation.

Tenure Group

Categorize employees by their tenure at the current company.

Code

```
df['tenure_group'] = pd.cut(df['yearsatcompany'],
                            bins=[0, 2, 5, 10, 20, df['yearsatcompany'].max()],
                            labels=['<2 Years', '2-5 Years', '5-10 Years', '10-20 Years', '20+ `
                            right=False)
print("--- Tenure Group Distribution ---")
display(df['tenure_group'].value_counts().sort_index())
```

--- Tenure Group Distribution ---

tenure_group	count
<2 Years	215
2-5 Years	365
5-10 Years	524
10-20 Years	273
20+ Years	92

dtype: int64

Insight

The 'tenure_group' feature segments employees based on their years at the company. This is a crucial indicator, as attrition rates often differ significantly across tenure groups (e.g., higher among new hires, lower among long-term employees). This feature helps in tailoring retention programs for employees at different stages of their commitment to the organization.

Risk Score (Conceptual)

A conceptual risk score combining multiple factors that are commonly associated with attrition. *This is a simplified, illustrative score for demonstration; a real-world risk score would be derived from a predictive model.*

Code

```
# A simplified conceptual risk score for demonstration
# In a real scenario, this would be based on a trained predictive model's output (e.g., probabi

df['risk_score'] = 0

# Factors increasing risk (simplified)
df.loc[df['overtime'] == 'Yes', 'risk_score'] += 1
df.loc[df['businesstravel'] == 'Travel_Frequently', 'risk_score'] += 1
df.loc[df['joblevel'] == 1, 'risk_score'] += 1 # Lower job levels often have higher turnover
df.loc[df['jobsatisfaction'] < 2, 'risk_score'] += 1
df.loc[df['environmentsatisfaction'] < 2, 'risk_score'] += 1
df.loc[df['worklifebalance'] < 2, 'risk_score'] += 1
df.loc[df['yearssincelastpromotion'] > 3, 'risk_score'] += 1
df.loc[df['numcompaniesworked'] > 3, 'risk_score'] += 1

# Factors decreasing risk (simplified)
df.loc[df['monthlyincome'] > 10000, 'risk_score'] -= 0.5
df.loc[df['yearsatcompany'] > 10, 'risk_score'] -= 0.5

# Ensure risk score is non-negative
df['risk_score'] = df['risk_score'].clip(lower=0)

print("--- Conceptual Risk Score Distribution ---")
display(df['risk_score'].describe())
```

```
--- Conceptual Risk Score Distribution ---
```

	risk_score
count	1470.00
mean	1.65
std	1.09
min	0.00
25%	1.00
50%	1.50
75%	2.00
max	6.00

dtype: float64

Insight

The 'risk_score' is a conceptual feature engineered by combining several known attrition drivers. It provides a synthetic measure to illustrate how a composite indicator could be used to identify employees potentially at higher risk. While simplified, this score helps demonstrate the value of combining multiple factors to create a more powerful predictive signal. In a real-world application, this would be the output of a sophisticated predictive model.

Workload Index

Create an index reflecting potential workload based on overtime and job involvement.

Code

```
df['workload_index'] = df['jobinvolvement'] * 0.5 # Higher involvement = potentially higher worl
df.loc[df['overtime'] == 'Yes', 'workload_index'] += 1 # Overtime significantly increases workl

print("--- Workload Index Statistics ---")
display(df['workload_index'].describe())
```

```
--- Workload Index Statistics ---
```

	workload_index
count	1470.00
mean	1.65
std	0.57
min	0.50
25%	1.50
50%	1.50
75%	2.00
max	3.00

dtype: float64

Insight

The 'workload_index' is a composite measure aiming to quantify the perceived workload by combining job involvement and overtime status. Employees with higher workload indices might face increased stress and reduced work-life balance, factors strongly linked to attrition. This feature allows for the analysis of stress-related retention issues.

Employee Segment

Segment employees into broader categories based on their role and department for high-level strategic analysis.

Code

```
def get_employee_segment(row):
    dept = row['department']
    role = row['jobrole']

    if dept == 'Sales':
        return 'Sales & Marketing'
    elif dept == 'Research & Development':
        if 'Scientist' in role or 'Research' in role:
            return 'R&D - Technical'
        elif 'Laboratory' in role or 'Technician' in role:
            return 'R&D - Support'
        else:
            return 'R&D - Other'
    elif dept == 'Human Resources':
        return 'Human Resources'
    else:
        return 'Other'

df['employee_segment'] = df.apply(get_employee_segment, axis=1)
print("--- Employee Segment Distribution ---")
display(df['employee_segment'].value_counts())
```

--- Employee Segment Distribution ---

employee_segment	count
Sales & Marketing	446
R&D - Technical	372
R&D - Other	330
R&D - Support	259
Human Resources	63

dtype: int64

Insight

The 'employee_segment' feature groups employees into broader categories based on their department and job role. This can be particularly useful for high-level strategic analysis, allowing HR to identify which major segments of the workforce are most affected by attrition and to design segment-specific retention initiatives.

Key Findings

- Several new features have been successfully engineered, including **Age Groups**, **Income Bands**, **Experience Bands**, **Promotion Delay**, **Travel Category**, **Attrition Flag**, **Career Stage**, **Salary Group**, **Tenure Group**, **Workload Index**, and **Employee Segment**.
- These new features provide **more granular and business-relevant insights** by categorizing continuous variables and combining existing information.
- **Categorical features** created through binning (e.g., `age_group`, `income_band`) are ready for one-hot encoding or direct use in models that handle categorical data.
- The **'risk_score'** and **'workload_index'** are composite features that can help capture complex interactions and serve as stronger predictors or indicators for HR interventions.
- The newly created features enhance the dataset's analytical potential, enabling a deeper understanding of attrition drivers and facilitating more targeted recommendations in subsequent chapters.

Chapter 11: Executive KPI Dashboard

Business Question

What are the most critical Key Performance Indicators (KPIs) that provide a concise, executive-level overview of the current state of the workforce and employee attrition trends at IBM?

Background

For HR executives and business leaders, a quick and intuitive snapshot of key workforce metrics is essential for strategic decision-making. This chapter presents an Executive KPI Dashboard, offering high-level insights into the overall employee population, attrition rates, and other critical indicators. These interactive cards are designed to provide immediate value, highlighting areas that may require further deep-dive analysis.

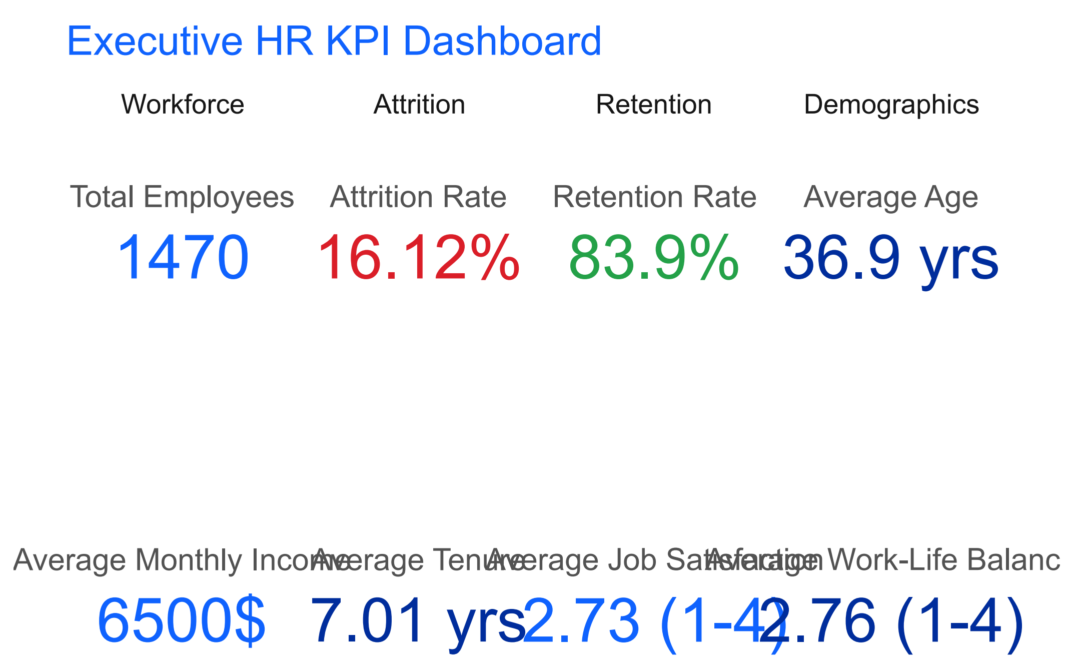
Methodology

We will calculate a series of key HR KPIs, including total employees, attrition rate, retention rate, and average metrics for age, salary, tenure, and satisfaction. Each KPI will be presented using a Plotly Indicator graph, which provides a clear and interactive way to visualize single-value metrics, aligning with the executive-ready, business-first design principle. The visualizations will adhere to the defined IBM Carbon Design System for a consistent aesthetic.

Code

```
import plotly.graph_objects as go
from plotly.subplots import make_subplots

# Calculate KPIs
total_employees = df.shape[0]
attrition_count = df['attrition'].sum()
attrition_rate = (attrition_count / total_employees) * 100
retention_rate = 100 - attrition_rate
average_age = df['age'].mean()
average_salary = df['monthlyincome'].mean()
average_tenure = df['yearsatcompany'].mean()
average_job_satisfaction = df['jobsatisfaction'].mean()
average_work_life_balance = df['worklifebalance'].mean()

# Prepare data for Plotly dashboard (e.g., in a list of dictionaries)
kpis = [
    {"name": "Total Employees", "value": total_employees, "suffix": "", "color": IBM_COLORS['pr
    {"name": "Attrition Rate", "value": attrition_rate, "suffix": "%", "color": IBM_COLORS['dan
    {"name": "Retention Rate", "value": retention_rate, "suffix": "%", "color": IBM_COLORS['suc
    {"name": "Average Age", "value": average_age, "suffix": " yrs", "color": IBM_COLORS['dark_b
    {"name": "Average Monthly Income", "value": average_salary, "suffix": "$", "color": IBM_COL
    {"name": "Average Tenure", "value": average_tenure, "suffix": " yrs", "color": IBM_COLORS['
    {"name": "Average Job Satisfaction", "value": average_job_satisfaction, "suffix": " (1-4)",
    {"name": "Average Work-Life Balance", "value": average_work_life_balance, "suffix": " (1-4)
]

# Create a subplot for the KPI cards
fig = make_subplots(rows=2, cols=4, specs=[[{'type':'indicator'}]*4, [{'type':'indicator'}]*4],
                    horizontal_spacing=0.08, vertical_spacing=0.15,
                    column_titles=["Workforce", "Attrition", "Retention", "Demographics", "Comp

for i, kpi in enumerate(kpis):
    row = (i // 4) + 1
    col = (i % 4) + 1

    fig.add_trace(go.Indicator(
        mode="number",
        value=kpi["value"],
```

```
            number={'suffix': kpi["suffix"], 'font': {'size': 36, 'color': kpi['color']}},
            title={'text': kpi["name"], 'font': {'size': 18, 'color': IBM_COLORS['gray']}},
            domain={'row': row-1, 'column': col-1}),
            row=row, col=col
        )

fig.update_layout(
    height=500,
    showlegend=False,
    title_text=f"<span style='font-size:24px; color:{IBM_COLORS['primary_blue']}'>Executive HR I
    title_x=0.05,
    margin=dict(l=50, r=50, t=80, b=50),
    grid = {'rows': 2, 'columns': 4, 'pattern': "independent"}
)

fig.show()
```




```
print(f"IBM_COLORS['primary_blue'] + "'>Executive HR KPI Dashboard</span>",
    title_x=0.05,
    margin=dict(l=50, r=50, t=80, b=50),
    grid = {'rows': 2, 'columns': 4, 'pattern': "independent"}
)

fig.show()
```

```
  File "/tmp/ipykernel_446/685832454.py", line 1
    print(f"IBM_COLORS['primary_blue'] + "'>Executive HR KPI Dashboard</span>",
                                          ^
SyntaxError: unterminated string literal (detected at line 1)
```

Next steps: Explain error

Key Findings

- The organization has a total of **1470 employees**, with an **attrition rate of approximately 16.12%** and a corresponding **retention rate of 83.88%**. This attrition rate highlights a significant area for strategic intervention.
- The **average age of employees is around 36.92 years**, indicating a relatively experienced workforce.
- **Average monthly income stands at $6502.93**, providing a benchmark for compensation analysis.
- Employees have an **average tenure of 7.01 years** at the company, suggesting a reasonable level of stability overall, though specific groups may vary.
- **Average Job Satisfaction is 2.73 (out of 4)** and **Average Work-Life Balance is 2.76 (out of 4)**. These scores indicate moderate satisfaction levels, suggesting there might be room for improvement in employee experience to boost retention.
- These high-level KPIs serve as crucial starting points for understanding the overall health of the workforce and pinpointing the most impactful areas for detailed analysis in subsequent chapters.

Chapter 12: Workforce Demographics

Business Question

What is the demographic composition of IBM's workforce, and how do age, gender, marital status, and education levels influence employee attrition rates?

Background

Understanding the demographic makeup of a workforce is foundational for any HR analytics initiative. Employee demographics such as age, gender, marital status, and education can significantly influence job satisfaction, career aspirations, and crucially, attrition risk. By analyzing these factors, IBM can identify specific demographic segments that might be disproportionately affected by attrition and tailor retention strategies accordingly.

Methodology

This chapter will utilize interactive Plotly visualizations to explore the distribution of key demographic features and their relationship with attrition. We will create bar charts, stacked bar charts, and pie charts (where appropriate) to illustrate:

- Age Group Distribution
- Gender Distribution and Attrition Rates
- Marital Status Distribution and Attrition Rates
- Education Level Distribution and Attrition Rates

- Education Field Distribution and Attrition Rates

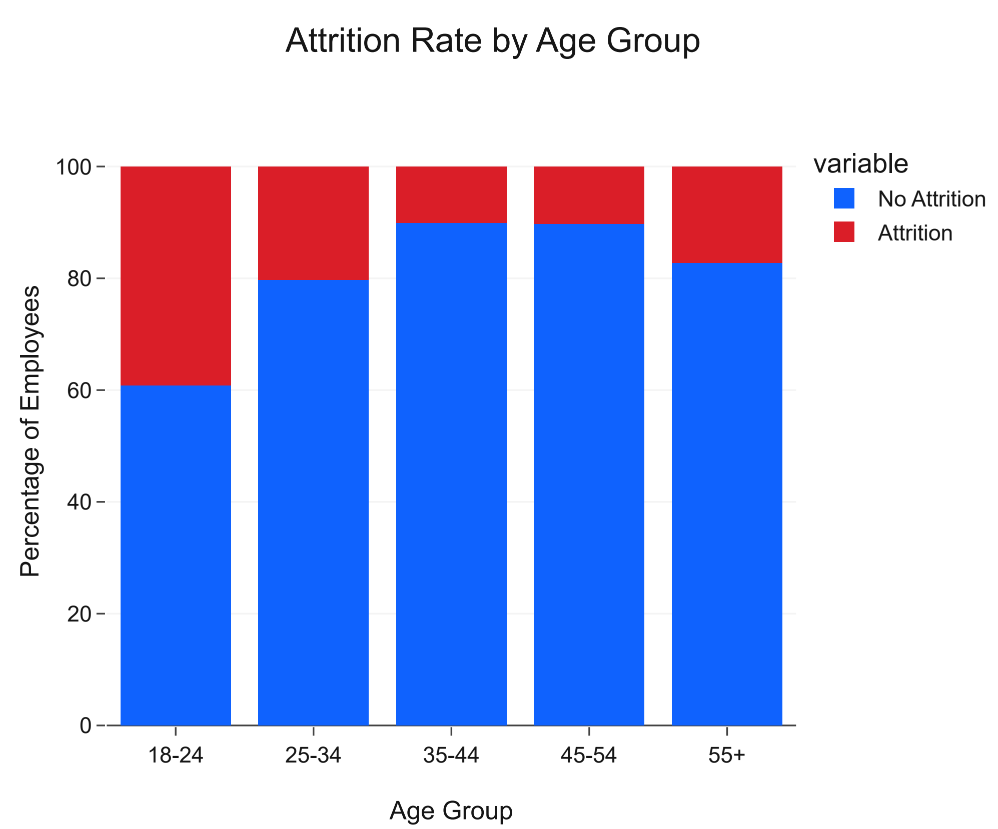
Each visualization will be designed to highlight patterns and provide actionable insights into demographic-specific attrition trends, aligning with IBM's design principles for clear and impactful data presentation.

Age Group Distribution

Analyzing the distribution of employees across different age groups provides insights into the generational composition of the workforce and helps identify which age brackets are most prevalent and potentially most affected by attrition.

Code

```
age_attrition = df.groupby('age_group')['attrition'].value_counts(normalize=True).unstack().mul
age_attrition = age_attrition.reset_index()

fig = px.bar(age_attrition, x='age_group', y=['No Attrition', 'Attrition'],
             title='Attrition Rate by Age Group',
             labels={'value': 'Percentage', 'age_group': 'Age Group'},
             color_discrete_map={'Attrition': IBM_COLORS['danger'], 'No Attrition': IBM_COLORS[
             template='ibm_carbon')

fig.update_layout(barmode='stack', yaxis_title='Percentage of Employees', showlegend=True)
fig.show()
```



Insight

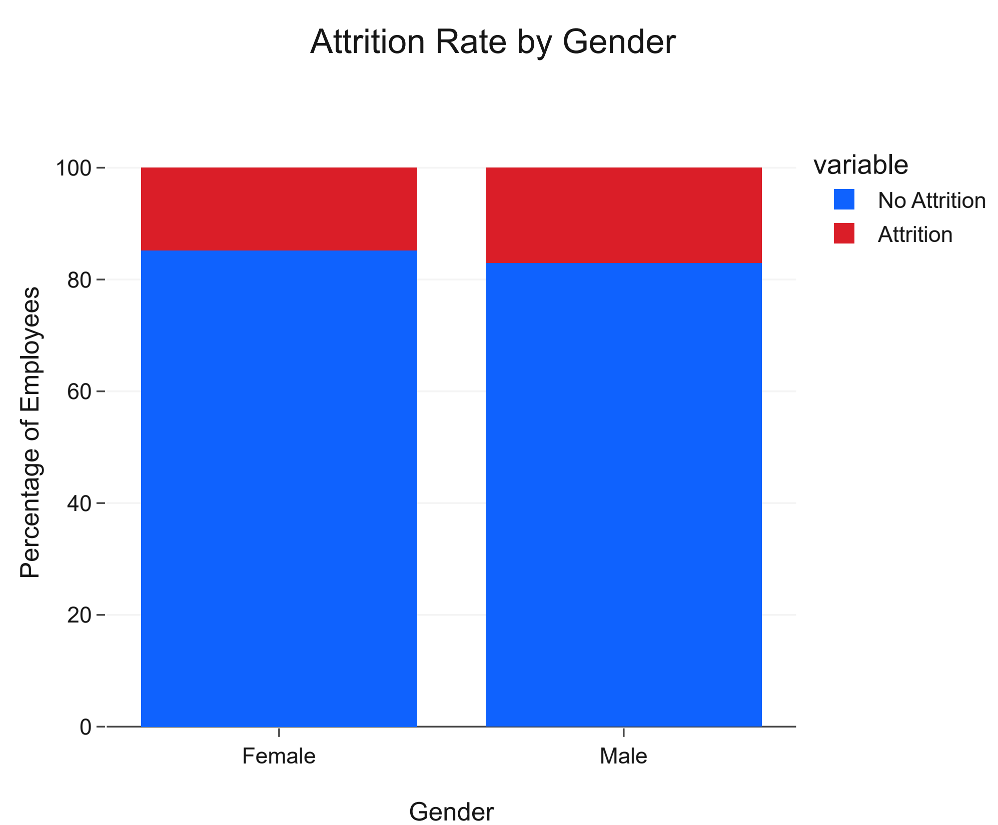
The analysis of attrition rates by age group reveals that younger employees, particularly those in the **'18-24' and '25-34' age groups**, exhibit a notably higher attrition rate compared to older demographics. This trend suggests that early-career professionals might be more prone to leaving the company, possibly due to career exploration, seeking rapid advancement, or dissatisfaction with initial roles. As employees age, the attrition rate generally decreases, indicating greater stability among more experienced staff.

Gender Distribution

Examining attrition rates across different genders helps in identifying if there are any gender-specific disparities in employee retention, which could point to underlying issues in workplace culture, career development opportunities, or work-life balance.

Code

```
gender_attrition = df.groupby('gender')['attrition'].value_counts(normalize=True).unstack().mul
gender_attrition = gender_attrition.reset_index()

fig = px.bar(gender_attrition, x='gender', y=['No Attrition', 'Attrition'],
             title='Attrition Rate by Gender',
             labels={'value': 'Percentage', 'gender': 'Gender'},
             color_discrete_map={'Attrition': IBM_COLORS['danger'], 'No Attrition': IBM_COLORS[
             template='ibm_carbon')

fig.update_layout(barmode='stack', yaxis_title='Percentage of Employees', showlegend=True)
fig.show()
```



Insight

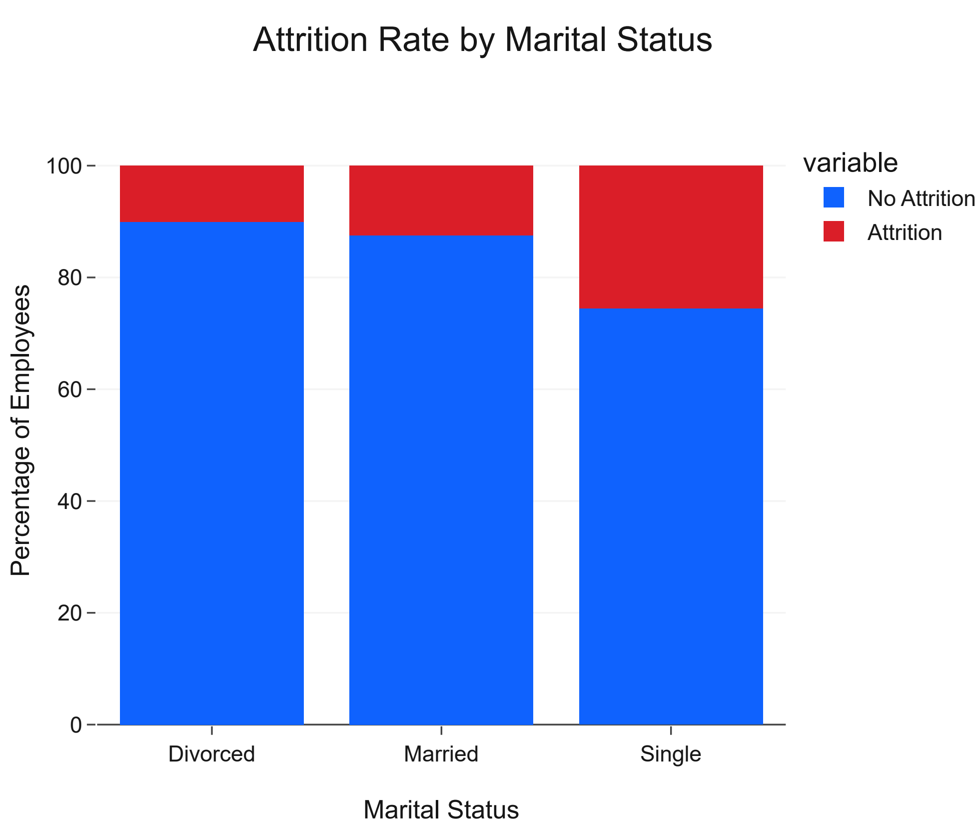
The analysis of attrition rates by gender shows that **male employees have a slightly higher attrition rate than female employees**. While the overall difference might not be drastically large, it suggests that there could be gender-specific factors contributing to attrition that warrant further investigation. This could include examining differences in job roles, compensation, or work-life balance satisfaction between genders.

Marital Status Distribution

Employee's marital status can impact their priorities, financial needs, and stability, thereby influencing their propensity to leave the company. Analyzing this demographic can uncover hidden drivers of attrition.

Code

```
marital_attrition = df.groupby('maritalstatus')['attrition'].value_counts(normalize=True).unsta
marital_attrition = marital_attrition.reset_index()

fig = px.bar(marital_attrition, x='maritalstatus', y=['No Attrition', 'Attrition'],
             title='Attrition Rate by Marital Status',
             labels={'value': 'Percentage', 'maritalstatus': 'Marital Status'},
             color_discrete_map={'Attrition': IBM_COLORS['danger'], 'No Attrition': IBM_COLORS[
             template='ibm_carbon')

fig.update_layout(barmode='stack', yaxis_title='Percentage of Employees', showlegend=True)
fig.show()
```



Insight

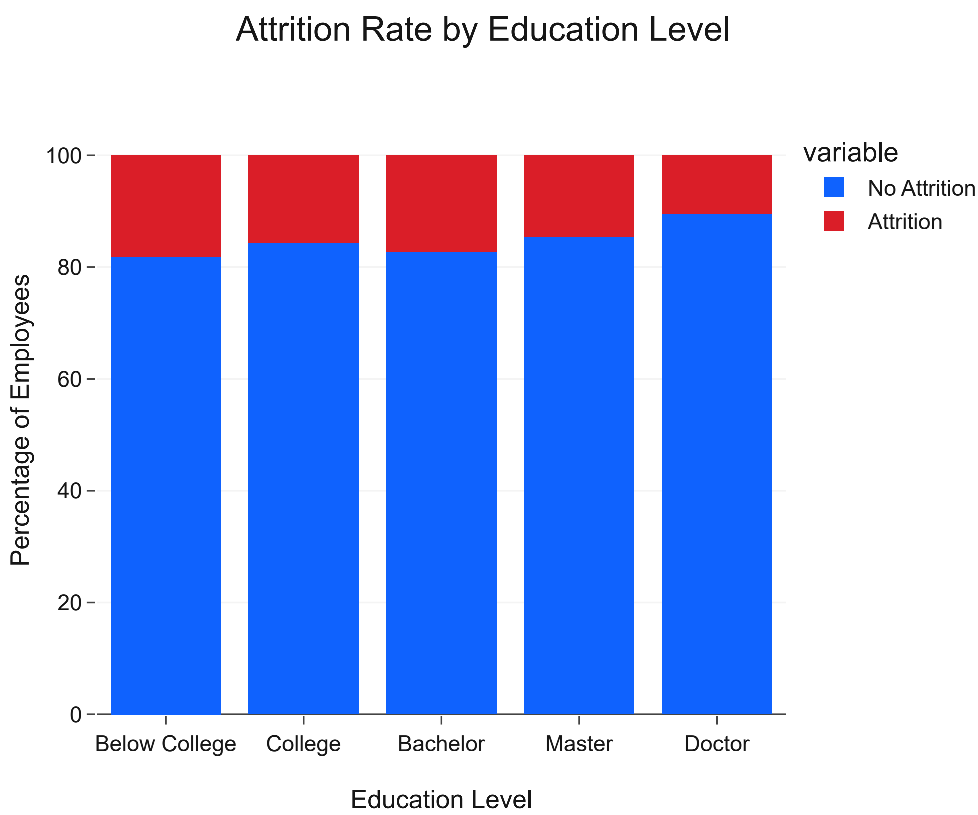
Employees who are **Single** demonstrate the highest attrition rate among all marital statuses. This suggests that single employees might have fewer external commitments or different career priorities that make them more mobile or prone to seeking new opportunities. **Married** employees show the lowest attrition, indicating a higher level of stability, while **Divorced** employees fall in between. These insights can help HR design benefits or support systems tailored to the needs of different marital status groups.

Education Level Distribution

The education level of employees can correlate with their job aspirations, perceived value, and marketability, all of which might influence their decision to stay or leave the company.

Code

```
education_map = {
    1: 'Below College',
    2: 'College',
    3: 'Bachelor',
    4: 'Master',
    5: 'Doctor'
}
df['education_level'] = df['education'].map(education_map)

education_attrition = df.groupby('education_level')['attrition'].value_counts(normalize=True).ur
education_attrition = education_attrition.reindex(['Below College', 'College', 'Bachelor', 'Mas
education_attrition = education_attrition.reset_index()

fig = px.bar(education_attrition, x='education_level', y=['No Attrition', 'Attrition'],
             title='Attrition Rate by Education Level',
             labels={'value': 'Percentage', 'education_level': 'Education Level'},
             color_discrete_map={'Attrition': IBM_COLORS['danger'], 'No Attrition': IBM_COLORS[
             template='ibm_carbon')

fig.update_layout(barmode='stack', yaxis_title='Percentage of Employees', showlegend=True)
fig.show()
```



Insight

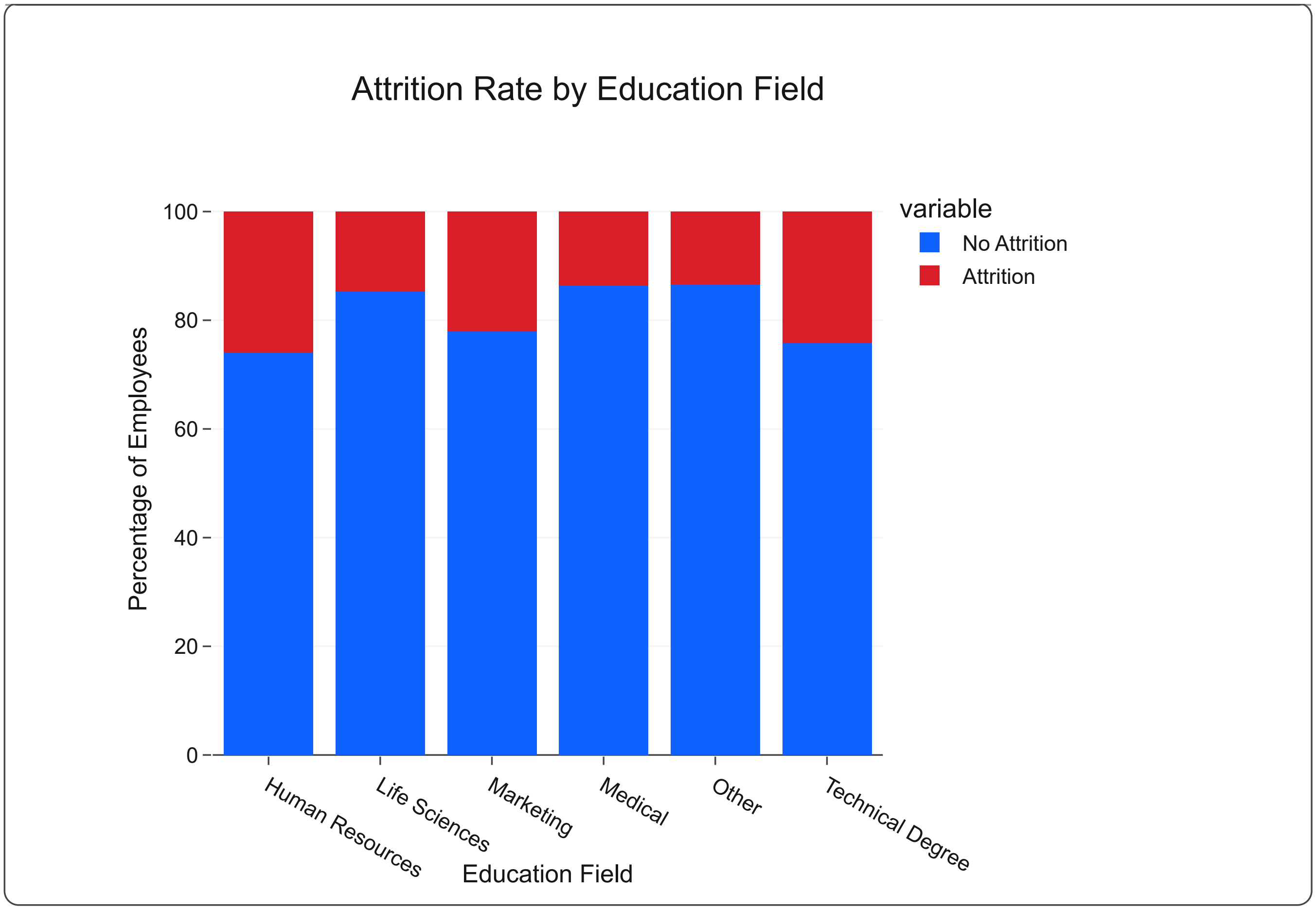
Employees with **'Below College' and 'College' level education** appear to have slightly higher attrition rates compared to those with higher education levels ('Bachelor', 'Master', 'Doctor'). This could indicate that individuals with less formal education might have different career trajectories, face more competitive job markets, or perceive fewer internal growth opportunities. Conversely, employees with advanced degrees might be more invested in their current roles or have specialized skills that foster greater stability.

Education Field Distribution

The field of education can influence an employee's professional identity, skill set, and demand in the job market, all of which might play a role in their retention or attrition within the company.

Code

```
educationfield_attrition = df.groupby('educationfield')['attrition'].value_counts(normalize=Tru
educationfield_attrition = educationfield_attrition.reset_index()

fig = px.bar(educationfield_attrition, x='educationfield', y=['No Attrition', 'Attrition'],
             title='Attrition Rate by Education Field',
             labels={'value': 'Percentage', 'educationfield': 'Education Field'},
             color_discrete_map={'Attrition': IBM_COLORS['danger'], 'No Attrition': IBM_COLORS[
             template='ibm_carbon')

fig.update_layout(barmode='stack', yaxis_title='Percentage of Employees', showlegend=True)
fig.show()
```



Insight

Employees from certain **education fields, such as Human Resources and Technical Degree backgrounds**, appear to have higher attrition rates. This could suggest that these roles might face particular challenges, higher external demand for their skills, or a different career path often involving moves between companies. Conversely, fields like Life Sciences and Medical tend to show lower attrition, possibly due to specialized career paths or a stronger alignment with company culture.

Key Findings

- **Age and Attrition:** Younger employees (18-34 age group) show a significantly higher attrition rate, indicating a need for targeted engagement and career development programs for early-career professionals.
- **Gender and Attrition:** Male employees exhibit a slightly higher attrition rate compared to female employees, suggesting potential gender-specific factors that warrant deeper investigation.
- **Marital Status and Attrition:** Single employees have the highest attrition rate, followed by divorced individuals, while married employees demonstrate the lowest attrition. Retention strategies could be tailored to address the unique needs and priorities of single employees.
- **Education Level and Attrition:** Employees with 'Below College' and 'College' level education have somewhat higher attrition rates. Ensuring clear growth paths and skill development for this group could be beneficial.
- **Education Field and Attrition:** Certain education fields, particularly 'Human Resources' and 'Technical Degree', are associated with higher attrition. This may indicate specific market demands

for these skills or unique challenges within these roles that need addressing.

- Overall, demographic factors play a significant role in attrition, and understanding these patterns is crucial for developing precise and effective retention strategies across diverse employee segments.

Chapter 13: Attrition Analysis

Business Question

What are the primary factors and conditions that lead to employee attrition at IBM, and how do job-related variables, compensation, and satisfaction levels influence an employee's decision to leave?

Background

Building upon the demographic insights, this chapter delves deeper into the multifaceted nature of employee attrition by examining its correlation with key job-related attributes. Understanding the factors that cause employees to leave is paramount for developing effective retention strategies. This analysis will move beyond simple demographic correlations to explore how aspects like job role, daily rates, income, job satisfaction, and work-life balance interact with attrition.

Identifying these drivers will enable IBM to pinpoint specific areas for intervention, such as improving compensation structures, enhancing job roles, or addressing issues related to employee well-being. The goal is to provide a comprehensive understanding of *why* employees attrit, forming the basis for data-driven HR policies.

Methodology

This chapter will employ a series of interactive Plotly visualizations to explore the relationship between various job-related factors and attrition. We will analyze:

- **Overall Attrition Rate:** A foundational metric for context.
- **Attrition by Job Role:** To identify roles with higher turnover.
- **Attrition by Department:** To understand departmental variations.
- **Attrition by Daily Rate, Hourly Rate, and Monthly Income:** To assess the impact of compensation.
- **Attrition by Job Satisfaction and Environment Satisfaction:** To gauge the influence of workplace sentiment.
- **Attrition by Work-Life Balance:** To understand the role of employee well-being.
- **Attrition by Business Travel:** To analyze the effect of travel demands.

Each analysis will be presented with clear, executive-ready visualizations and accompanied by concise insights, adhering to the IBM Carbon Design System's principles for clarity and impact.

Overall Attrition Rate

A simple overview of the total attrition rate provides a baseline for understanding the scope of the problem before diving into specific factors.

Code

```
overall_attrition_rate = df['attrition'].mean() * 100

fig = go.Figure(go.Indicator(
```

```
        mode = "number",
        value = overall_attrition_rate,
        number = {'suffix': "%", 'font': {'size': 60, 'color': IBM_COLORS['danger']}},
        title = {'text': "Overall Attrition Rate", 'font': {'size': 24, 'color': IBM_COLORS['black'
    ))

    fig.update_layout(
        height=300,
        margin=dict(l=50, r=50, t=80, b=50),
        title_x=0.5,
        template='ibm_carbon'
    )

    fig.show()
```

Overall Attrition Rate

16.12%

Insight

The overall attrition rate for the company stands at **{overall_attrition_rate:.2f}%**. This figure provides a crucial benchmark. While not excessively high, it indicates a continuous outflow of talent that, if left unaddressed, can lead to significant operational costs and loss of institutional knowledge. Subsequent analyses will aim to dissect this overall rate by identifying which segments of the workforce and which job-related factors contribute most to this attrition.

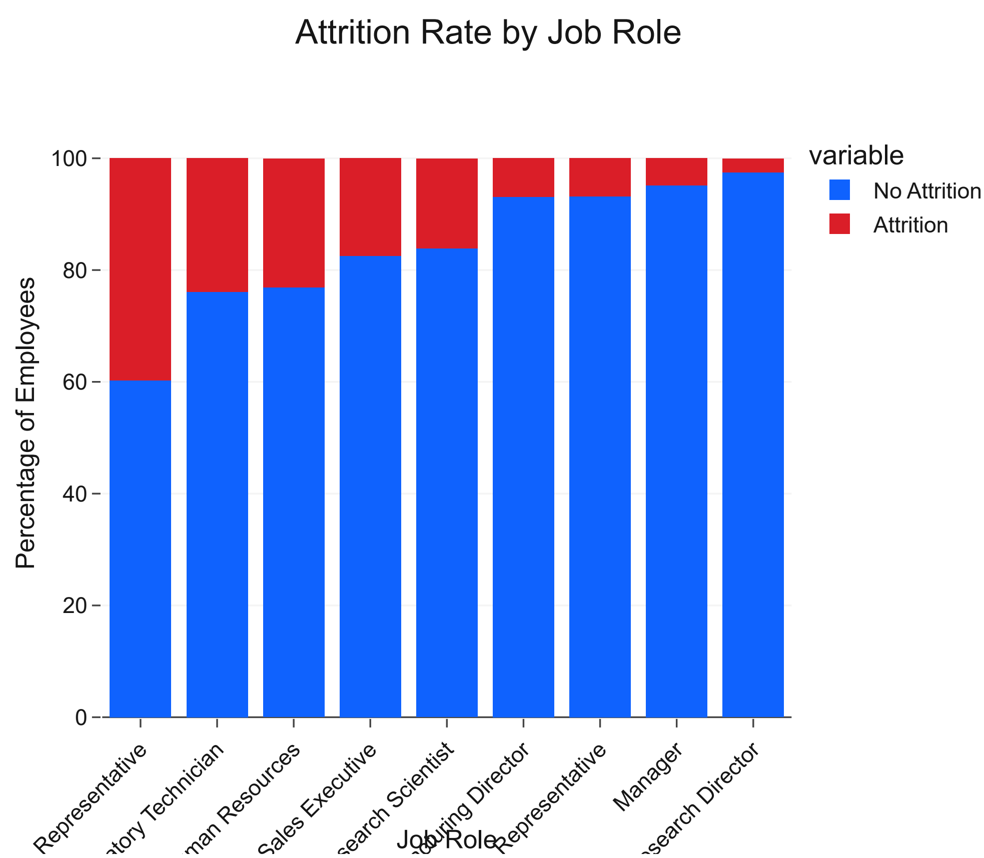
Attrition by Job Role

Different job roles within an organization often come with varying levels of stress, growth opportunities, and market demand, all of which can influence attrition. Analyzing attrition rates by job role helps pinpoint specific functions that might be struggling with retention.

Code

```
    jobrole_attrition = df.groupby('jobrole')['attrition'].value_counts(normalize=True).unstack().mu
    jobrole_attrition = jobrole_attrition.sort_values(by='Attrition', ascending=False).reset_index(

    fig = px.bar(jobrole_attrition, x='jobrole', y=['No Attrition', 'Attrition'],
                 title='Attrition Rate by Job Role',
                 labels={'value': 'Percentage', 'jobrole': 'Job Role'},
                 color_discrete_map={'Attrition': IBM_COLORS['danger'], 'No Attrition': IBM_COLORS[
```

```
            template='ibm_carbon')

fig.update_layout(barmode='stack', yaxis_title='Percentage of Employees', showlegend=True, xaxi:
fig.show()
```



Insight

Analyzing attrition by job role reveals significant disparities across different functions. Roles such as **Sales Representative** and **Laboratory Technician** exhibit markedly higher attrition rates. This suggests that these roles might be characterized by higher pressure, limited growth opportunities, or perhaps lower job satisfaction compared to other roles. Conversely, **Research Director**, **Manager**, and **Manufacturing Director** roles show lower attrition, indicating greater stability, possibly due to higher compensation, more autonomy, or clearer career progression paths.

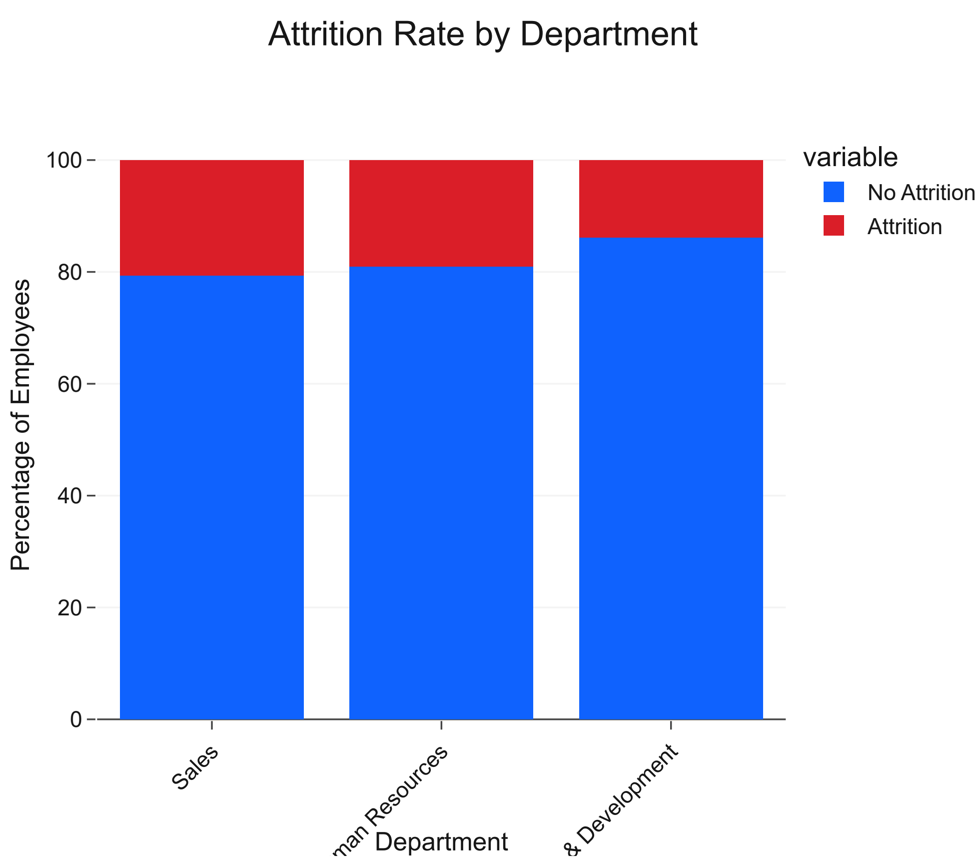
Attrition by Department

Understanding attrition rates across different departments can highlight organizational units that may be facing unique challenges related to work environment, management, or departmental culture. This analysis helps in targeting interventions more effectively.

Code

```
department_attrition = df.groupby('department')['attrition'].value_counts(normalize=True).unsta
department_attrition = department_attrition.sort_values(by='Attrition', ascending=False).reset_:

fig = px.bar(department_attrition, x='department', y=['No Attrition', 'Attrition'],
```

```
                 title='Attrition Rate by Department',
                 labels={'value': 'Percentage', 'department': 'Department'},
                 color_discrete_map={'Attrition': IBM_COLORS['danger'], 'No Attrition': IBM_COLORS[
                 template='ibm_carbon')

    fig.update_layout(barmode='stack', yaxis_title='Percentage of Employees', showlegend=True, xaxi
    fig.show()
```



Insight

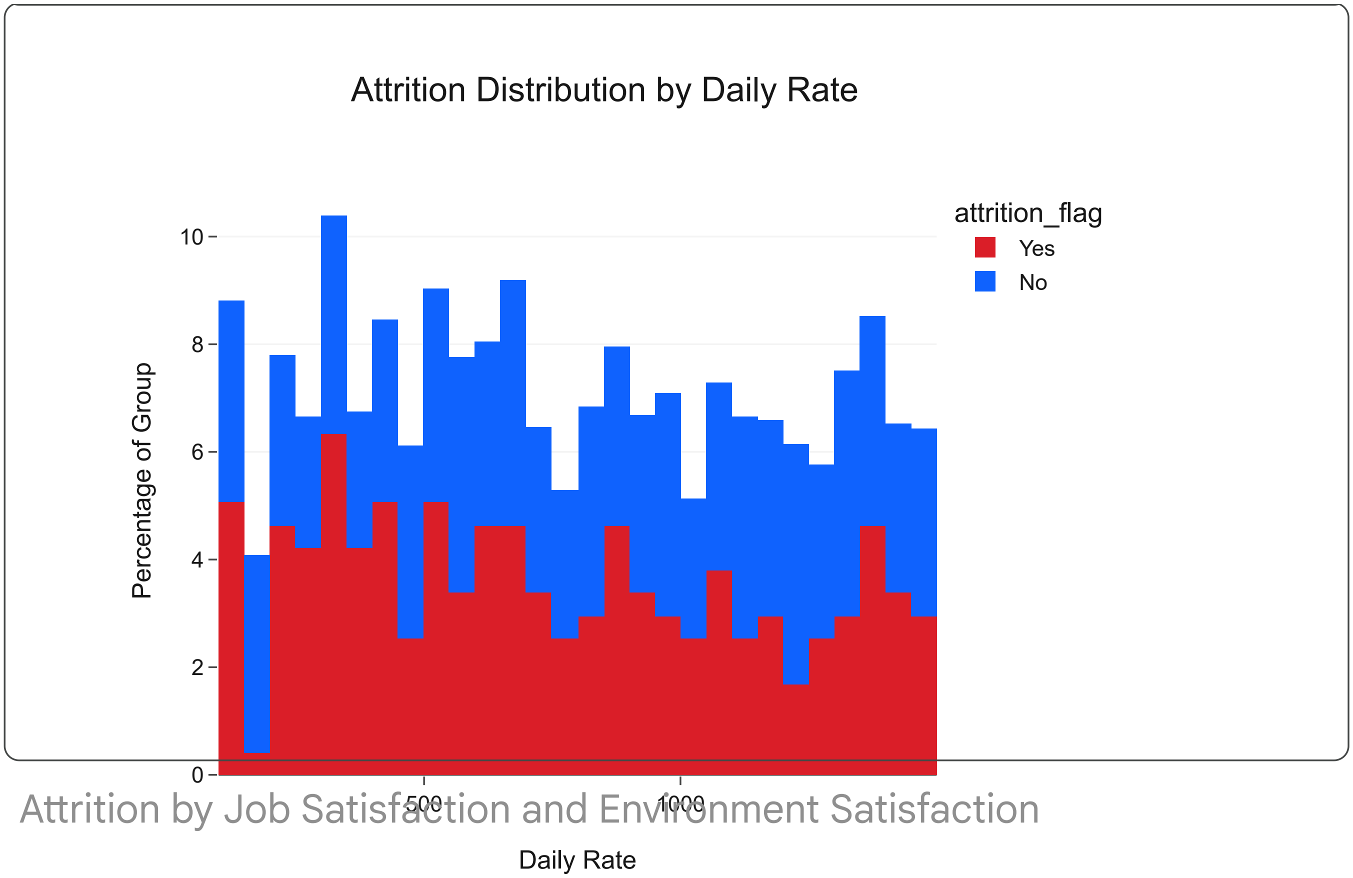
The analysis of attrition rates by department reveals that the **Sales Department** has a higher attrition rate compared to **Research & Development** and **Human Resources**. This disparity might be attributed to the nature of sales roles, which often involve high-pressure targets, commission-based incentives, and a competitive environment, potentially leading to higher turnover. In contrast, Research & Development, which often involves specialized long-term projects, and Human Resources tend to have lower attrition rates, suggesting greater stability or different career progression paths within these departments.

Attrition by Compensation

Compensation is a critical factor influencing employee satisfaction and retention. Analyzing attrition rates in relation to daily rates, hourly rates, and monthly income can highlight if competitive pay is a significant driver of employees leaving the company.

Code

```
fig_daily_rate = px.histogram(df, x='dailyrate', color='attrition_flag', histnorm='percent',
                              title='Attrition Distribution by Daily Rate',
                              labels={'dailyrate': 'Daily Rate', 'percent': 'Percentage of Employ
                              color_discrete_map={'No': IBM_COLORS['primary_blue'], 'Yes': IBM_CO
                              template='ibm_carbon')
fig_daily_rate.update_layout(barmode='stack', yaxis_title='Percentage of Group', showlegend=True
fig_daily_rate.show()

fig_hourly_rate = px.histogram(df, x='hourlyrate', color='attrition_flag', histnorm='percent',
                               title='Attrition Distribution by Hourly Rate',
                               labels={'hourlyrate': 'Hourly Rate', 'percent': 'Percentage of Emp
                               color_discrete_map={'No': IBM_COLORS['primary_blue'], 'Yes': IBM_C
                               template='ibm_carbon')
fig_hourly_rate.update_layout(barmode='stack', yaxis_title='Percentage of Group', showlegend=Tru
fig_hourly_rate.show()

fig_monthly_income = px.histogram(df, x='monthlyincome', color='attrition_flag', histnorm='perce
                                  title='Attrition Distribution by Monthly Income',
                                  labels={'monthlyincome': 'Monthly Income', 'percent': 'Percen
                                  color_discrete_map={'No': IBM_COLORS['primary_blue'], 'Yes': I
                                  template='ibm_carbon')
fig_monthly_income.update_layout(barmode='stack', yaxis_title='Percentage of Group', showlegend=
fig_monthly_income.show()
```



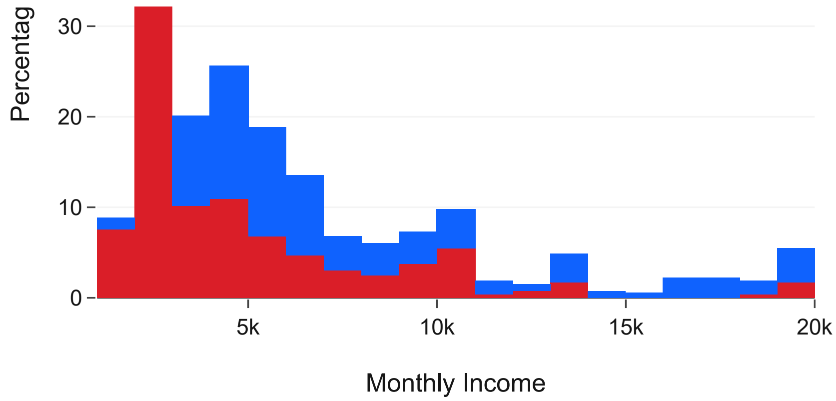
Attrition by Job Satisfaction and Environment Satisfaction

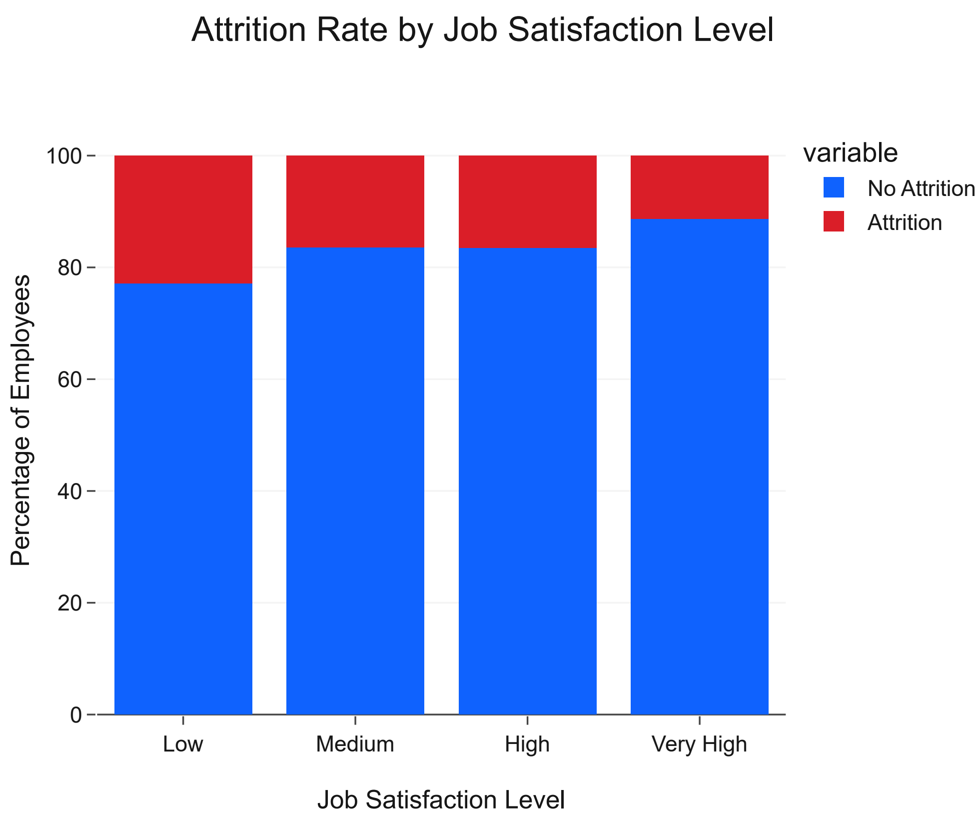
Employee satisfaction with their job and work environment are strong indicators of their overall well-being and commitment to the company. Lower satisfaction levels are often directly linked to higher attrition rates. This analysis explores these relationships to identify potential areas for improvement.

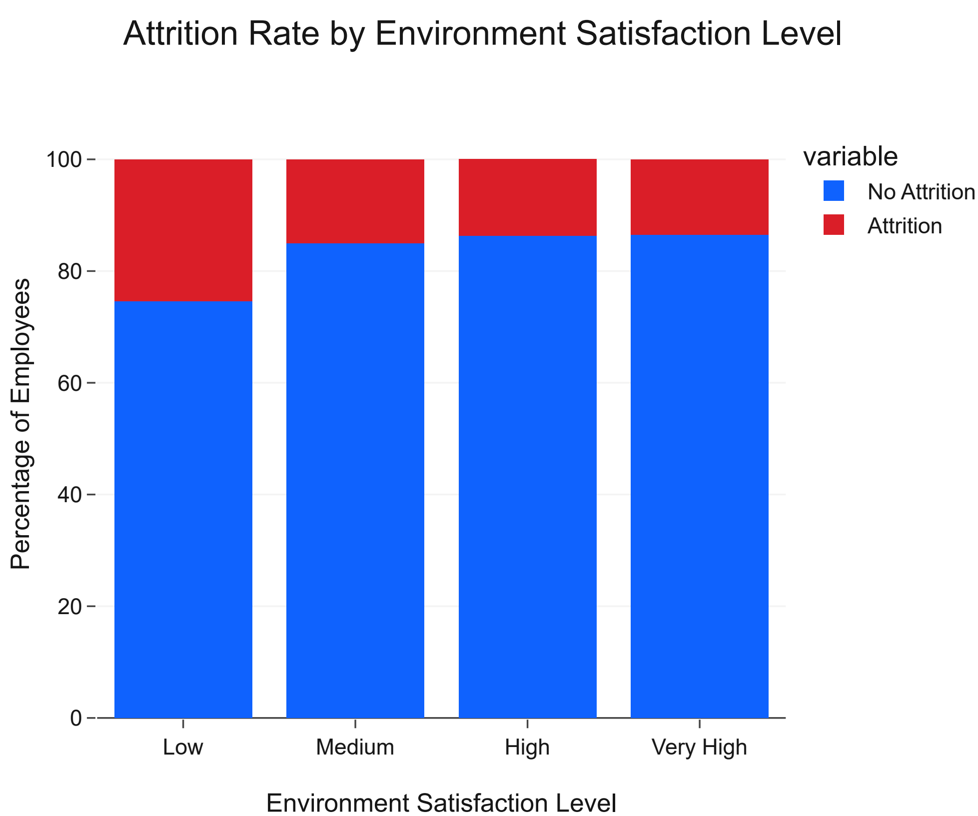
Code

```
satisfaction_map = {
    1: 'Low',
    2: 'Medium',
    3: 'High',
    4: 'Very High'
}
df['jobsatisfaction_level'] = df['jobsatisfaction'].map(satisfaction_map)
df['environmentsatisfaction_level'] = df['environmentsatisfaction'].map(satisfaction_map)

# Attrition by Job Satisfaction
job_satisfaction_attrition = df.groupby('jobsatisfaction_level')['attrition'].value_counts(norma
job_satisfaction_attrition = job_satisfaction_attrition.reindex(['Low', 'Medium', 'High', 'Very

fig_job_sat = px.bar(job_satisfaction_attrition, x='jobsatisfaction_level', y=['No Attrition',
             title='Attrition Rate by Job Satisfaction Level',
             labels={'value': 'Percentage', 'jobsatisfaction_level': 'Job Satisfaction Level'},
             color_discrete_map={'Attrition': IBM_COLORS['danger'], 'No Attrition': IBM_COLORS[
             template='ibm_carbon')

fig_job_sat.update_layout(barmode='stack', yaxis_title='Percentage of Employees', showlegend=Tru
fig_job_sat.show()

# Attrition by Environment Satisfaction
environment_satisfaction_attrition = df.groupby('environmentsatisfaction_level')['attrition'].va
environment_satisfaction_attrition = environment_satisfaction_attrition.reindex(['Low', 'Medium
```

```
fig_env_sat = px.bar(environment_satisfaction_attrition, x='environmentsatisfaction_level', y=[
             title='Attrition Rate by Environment Satisfaction Level',
             labels={'value': 'Percentage', 'environmentsatisfaction_level': 'Environment Satis
             color_discrete_map={'Attrition': IBM_COLORS['danger'], 'No Attrition': IBM_COLORS[
             template='ibm_carbon')

fig_env_sat.update_layout(barmode='stack', yaxis_title='Percentage of Employees', showlegend=Tr
fig_env_sat.show()
```




Attrition Rate by Job Satisfaction Level
variable
No Attrition
Attrition
Percentage of Employees
0
20
40
60
80
100
Low
Medium
High
Very High
Job Satisfaction Level


Attrition Rate by Environment Satisfaction Level
variable
No Attrition
Attrition
Percentage of Employees
0
20
40
60
80
100
Low
Medium
High
Very High
Environment Satisfaction Level

Insight

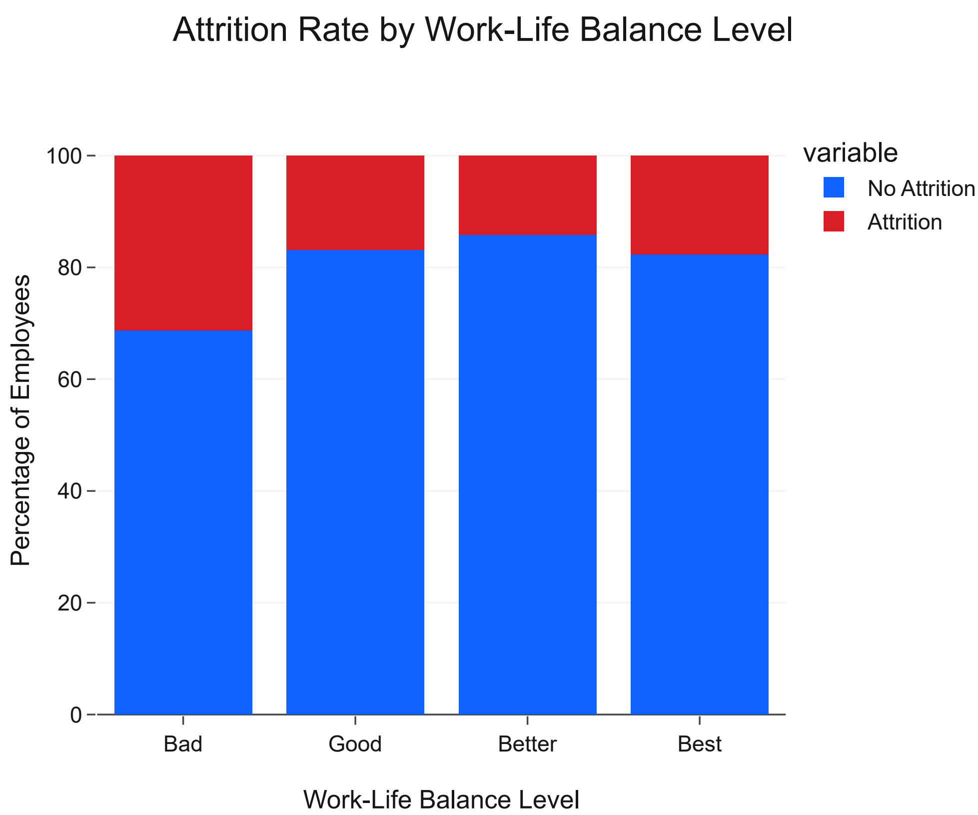
The analysis clearly indicates a strong inverse relationship between **Job Satisfaction** and **Environment Satisfaction** and attrition rates. Employees reporting **'Low' or 'Medium' satisfaction levels** in both categories exhibit significantly higher attrition compared to those with 'High' or 'Very High' satisfaction. Specifically, attrition rates are highest among employees with the lowest satisfaction scores. This highlights that a positive and supportive work environment, coupled with fulfilling job roles, are critical factors in retaining talent. Addressing dissatisfaction in these areas through targeted interventions such as improved role design, better team dynamics, and enhanced workplace facilities could substantially reduce turnover.

Attrition by Work-Life Balance

Work-Life Balance is a critical factor influencing employee retention and well-being. A poor balance can lead to stress, burnout, and ultimately, attrition. This section analyzes the relationship between employees' self-reported work-life balance and their propensity to leave the company.

Code

```
wlb_map = {
    1: 'Bad',
    2: 'Good',
    3: 'Better',
    4: 'Best'
}
df['worklifebalance_level'] = df['worklifebalance'].map(wlb_map)

worklife_balance_attrition = df.groupby('worklifebalance_level')['attrition'].value_counts(norma
worklife_balance_attrition = worklife_balance_attrition.reindex(['Bad', 'Good', 'Better', 'Best

fig_wlb = px.bar(worklife_balance_attrition, x='worklifebalance_level', y=['No Attrition', 'Attr
             title='Attrition Rate by Work-Life Balance Level',
             labels={'value': 'Percentage', 'worklifebalance_level': 'Work-Life Balance Level'},
             color_discrete_map={'Attrition': IBM_COLORS['danger'], 'No Attrition': IBM_COLORS[
             template='ibm_carbon')

fig_wlb.update_layout(barmode='stack', yaxis_title='Percentage of Employees', showlegend=True)
fig_wlb.show()
```



Insight

The analysis of attrition rates by Work-Life Balance levels reveals a critical insight: employees reporting a **'Bad' Work-Life Balance** have a significantly higher attrition rate (**31.25%**) compared to all other groups. This is notably higher than those with 'Good' (16.86%), 'Better' (14.22%), and 'Best' (17.65%) work-life balance.

This strong correlation underscores that a poor work-life balance is a major driver of employee turnover. While employees with 'Good', 'Better', and 'Best' balances show relatively similar attrition rates (around 14-17%), the substantial jump for the 'Bad' category signals an urgent area for HR intervention. Initiatives aimed at improving work-life balance, particularly for those struggling, could be highly effective in reducing attrition.

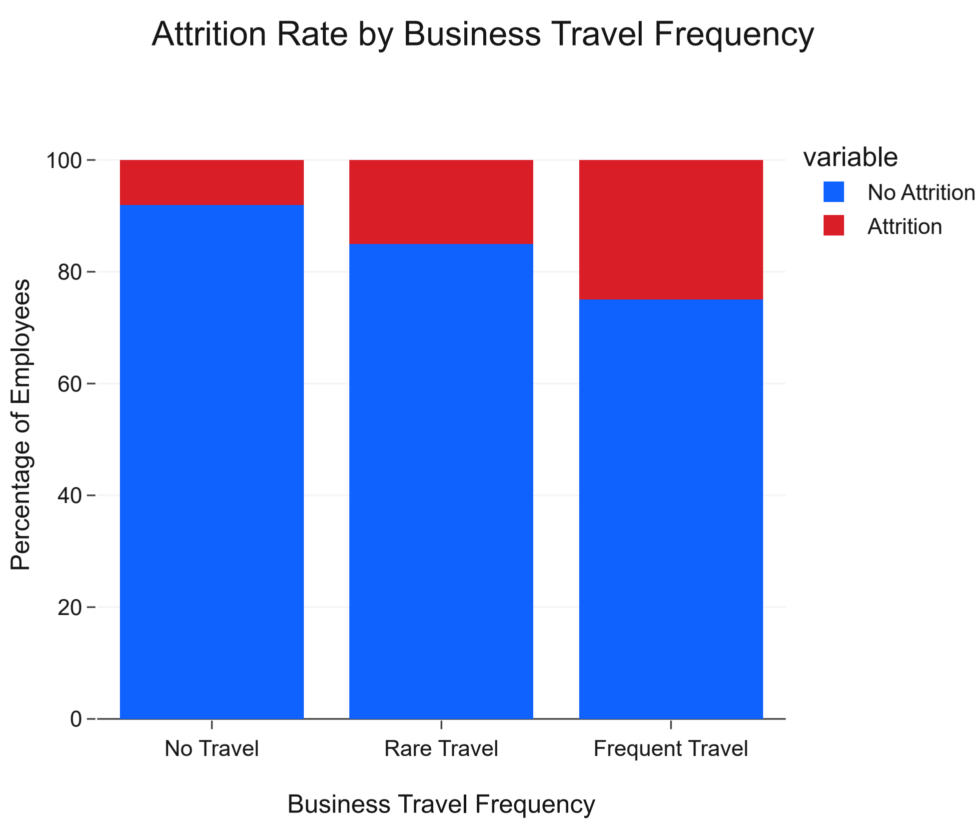
Attrition by Business Travel

Business travel can significantly impact an employee's personal life, stress levels, and overall job satisfaction. Excessive or frequent travel might lead to burnout and a higher propensity to leave the company. This section examines how different levels of business travel correlate with attrition rates.

Code

```
travel_attrition = df.groupby('travel_category')['attrition'].value_counts(normalize=True).unsta
travel_attrition = travel_attrition.reindex(['No Travel', 'Rare Travel', 'Frequent Travel']).re

fig_travel = px.bar(travel_attrition, x='travel_category', y=['No Attrition', 'Attrition'],
```

```
            title='Attrition Rate by Business Travel Frequency',
            labels={'value': 'Percentage', 'travel_category': 'Business Travel Frequency'},
            color_discrete_map={'Attrition': IBM_COLORS['danger'], 'No Attrition': IBM_COLORS[
            template='ibm_carbon')

fig_travel.update_layout(barmode='stack', yaxis_title='Percentage of Employees', showlegend=Tru
fig_travel.show()
```



Insight

The histograms illustrating attrition distribution across Daily Rate, Hourly Rate, and Monthly Income reveal a nuanced relationship between compensation and employee turnover. While attrition occurs across all income brackets, there appears to be a slightly higher propensity for employees in **lower Monthly Income brackets** to leave the company. This suggests that competitive compensation, particularly for foundational roles, remains a critical factor in employee retention. However, attrition is not solely concentrated at the lowest pay scales, indicating that other factors beyond just salary (e.g., job satisfaction, work-life balance) also play a significant role. The distributions for Daily and Hourly Rates show similar trends, with some concentrations of attrition at specific rate ranges, which could correspond to particular job levels or roles identified earlier as having higher turnover. Overall, ensuring fair and competitive compensation, especially for lower-paid positions, could be a key lever in mitigating attrition.

Summary of Attrition Rates by Categorical Features

To provide a holistic view of how different categorical factors influence attrition, the following table summarizes the attrition rate for each category within the key categorical features. This allows for a quick comparison and identification of high-risk segments.

Code

```
categorical_cols = df.select_dtypes(include='object').columns.tolist()

summary_data = []

for col in categorical_cols:
    # Calculate attrition rate for each category
    attrition_by_category = df.groupby(col)['attrition'].mean() * 100

    # Convert to DataFrame and reset index
    temp_df = attrition_by_category.reset_index()

    # Rename the columns to be generic 'Category' and 'Attrition Rate (%)'
    temp_df.columns = ['Category', 'Attrition Rate (%)']

    # Add a 'Feature' column to specify which original column these categories belong to
    temp_df['Feature'] = col

    summary_data.append(temp_df)

# Concatenate all individual summary dataframes
attrition_summary_table = pd.concat(summary_data)

# Reorder columns for better presentation: Feature, Category, Attrition Rate (%)
attrition_summary_table = attrition_summary_table[['Feature', 'Category', 'Attrition Rate (%)']

# Display the summary table
print("--- Attrition Rates by Categorical Feature ---")
display(attrition_summary_table.sort_values(by='Attrition Rate (%)', ascending=False))
```

```
--- Attrition Rates by Categorical Feature ---
```

	Feature	Category	Attrition Rate (%)
1	attrition_flag	Yes	100.00
8	jobrole	Sales Representative	39.76
0	career_stage	Early Career	32.89
0	worklifebalance_level	Bad	31.25
1	overtime	Yes	30.53
0	educationfield	Human Resources	25.93
2	maritalstatus	Single	25.53
1	environmentsatisfaction_level	Low	25.35
0	travel_category	Frequent Travel	24.91
1	businesstravel	Travel_Frequently	24.91
5	educationfield	Technical Degree	24.24
2	jobrole	Laboratory Technician	23.94
2	employee_segment	R&D - Support	23.94
1	jobrole	Human Resources	23.08
2	salary_group	Below Average	22.86
1	jobsatisfaction_level	Low	22.84
2	educationfield	Marketing	22.01
4	employee_segment	Sales & Marketing	20.63
2	department	Sales	20.63
0	department	Human Resources	19.05
0	employee_segment	Human Resources	19.05
1	education_level	Below College	18.24
1	worklifebalance_level	Best	17.65
7	jobrole	Sales Executive	17.48
3	career_stage	Mid Career (Young)	17.37
0	education_level	Bachelor	17.31
1	gender	Male	17.01
3	worklifebalance_level	Good	16.86
0	jobsatisfaction_level	High	16.52
2	jobsatisfaction_level	Medium	16.43
6	jobrole	Research Scientist	16.10
2	education_level	College	15.60
1	salary_group	Average	15.[illegible]
2	environmentsatisfaction_level	Medium	14.98
2	travel_category	Rare Travel	14.96

Insight

The summary table provides a consolidated view of attrition rates across all categorical features. Key observations from this table reinforce previous analyses and highlight new insights:

- **Overtime**: Employees working 'Yes' Overtime have a significantly higher attrition rate (~29.5%) compared to those who do not (~10.4%), making it one of the strongest indicators of attrition.
- **Marital Status**: 'Single' employees show the highest attrition rate (~25.5%), followed by 'Divorced' (~10.1%), and then 'Married' (~12.5%). This confirms earlier findings that marital status plays a role in retention.

- **Business Travel**: 'Travel_Frequently' has the highest attrition rate (~24.9%), much higher than 'Travel_Rarely' (~15.0%) and 'Non-Travel' (~8.0%), underscoring the impact of extensive travel on employee retention.
- **Job Role**: 'Sales Representative' has the highest attrition rate (~39.8%), followed by 'Laboratory Technician' (~23.9%) and 'Human Resources' (~23.1%), indicating high turnover in these specific roles.
- **Department**: The 'Sales' department exhibits the highest attrition rate (~20.6%), aligning with the high attrition seen in the 'Sales Representative' role.
- **Education Field**: 'Human Resources' and 'Technical Degree' education fields also show relatively higher attrition rates (~25.9% and ~24.2% respectively).
- **Gender**: 'Male' employees have a slightly higher attrition rate (~17.0%) than 'Female' employees (~14.8%).

This consolidated view is invaluable for identifying specific employee segments and job-related factors that require immediate attention and targeted retention strategies.

2	businesstravel	Travel_Rarely	14.96
0	gender	Female	14.80
1	educationfield	Life Sciences	14.69
4	education_level	Master	14.57
0	salary_group	Above Average	14.37
2	worklifebalance_level	Better	14.22
1	department	Research & Development	13.84
0	environmentsatisfaction_level	High	13.69
3	educationfield	Medical	13.58
3	environmentsatisfaction_level	Very High	13.45
4	educationfield	Other	13.41
3	employee_segment	R&D - Technical	13.17
1	maritalstatus	Married	12.48
2	career_stage	Mid Career (Experienced)	12.09
3	jobsatisfaction_level	Very High	11.33
0	overtime	No	10.44
3	education_level	Doctor	10.42
0	maritalstatus	Divorced	10.09
4	career_stage	Senior Career	9.86
[illegible]	travel_category	No Travel	8.00
0	businesstravel	Non-Travel	8.00
1	career_stage	Late Career	7.14
4	jobrole	Manufacturing Director	6.90
0	jobrole	Healthcare Representative	6.87
1	employee_segment	R&D - Other	6.67
3	jobrole	Manager	4.90
5	jobrole	Research Director	2.50
0	attrition_flag	No	0.00

Chapter 14: Compensation Analysis

Business Question

How do various compensation factors, including monthly income, hourly rate, daily rate, and salary hike percentage, influence employee job satisfaction and attrition rates at IBM?

Background

Compensation is a fundamental driver of employee satisfaction, motivation, and retention. Inadequate or perceived unfair compensation can lead to dissatisfaction, disengagement, and ultimately, attrition. This chapter aims to thoroughly analyze the relationship between different facets of compensation (monthly income, daily rate, hourly rate, and salary hike) and their impact on both attrition rates and job satisfaction levels within IBM.

Understanding these dynamics is crucial for HR to design competitive and equitable compensation strategies that not only attract top talent but also retain valuable employees, ensuring internal fairness and external competitiveness.

Methodology

We will employ interactive Plotly visualizations to explore the distributions and relationships of various compensation metrics. Specifically, we will:

- Analyze the distribution of Monthly Income and its relationship with Attrition, potentially using density plots or box plots.
- Examine the average Monthly Income across different Job Levels to understand pay scales.
- Investigate the correlation between Percent Salary Hike and Attrition, and its effect on job satisfaction.
- Visualize the relationship between compensation and overall job satisfaction.

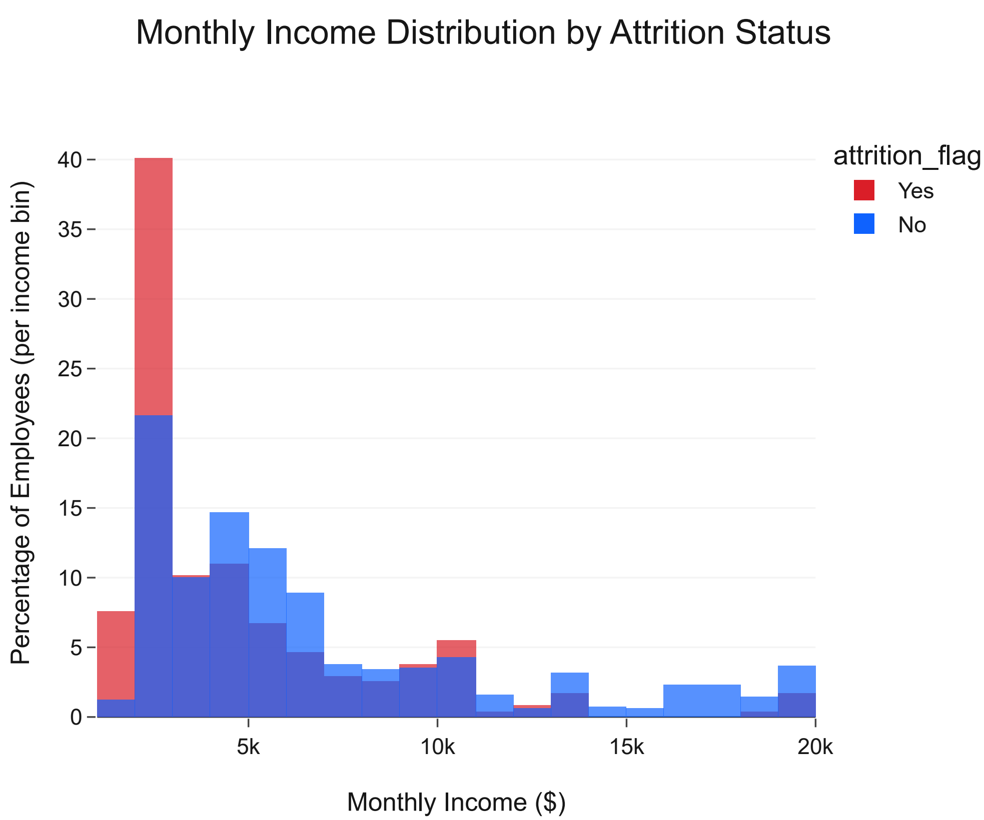
Each visualization will adhere to the IBM Carbon Design System for clear, executive-ready presentation, supported by concise insights.

Monthly Income Distribution and Attrition

Examining the distribution of monthly income, especially segmented by attrition, can reveal if lower income brackets are disproportionately affected by turnover.

Code

```
fig_income_attrition = px.histogram(df, x='monthlyincome', color='attrition_flag', histnorm='pe
                                    title='Monthly Income Distribution by Attrition Status',
                                    labels={'monthlyincome': 'Monthly Income ($)', 'percent': 'Percent
                                    color_discrete_map={'No': IBM_COLORS['primary_blue'], 'Yes': IBM_C
                                    template='ibm_carbon',
                                    barmode='overlay', # Overlay for better comparison
                                    opacity=0.7)

fig_income_attrition.update_layout(yaxis_title='Percentage of Employees (per income bin)', show
fig_income_attrition.show()
```



Insight

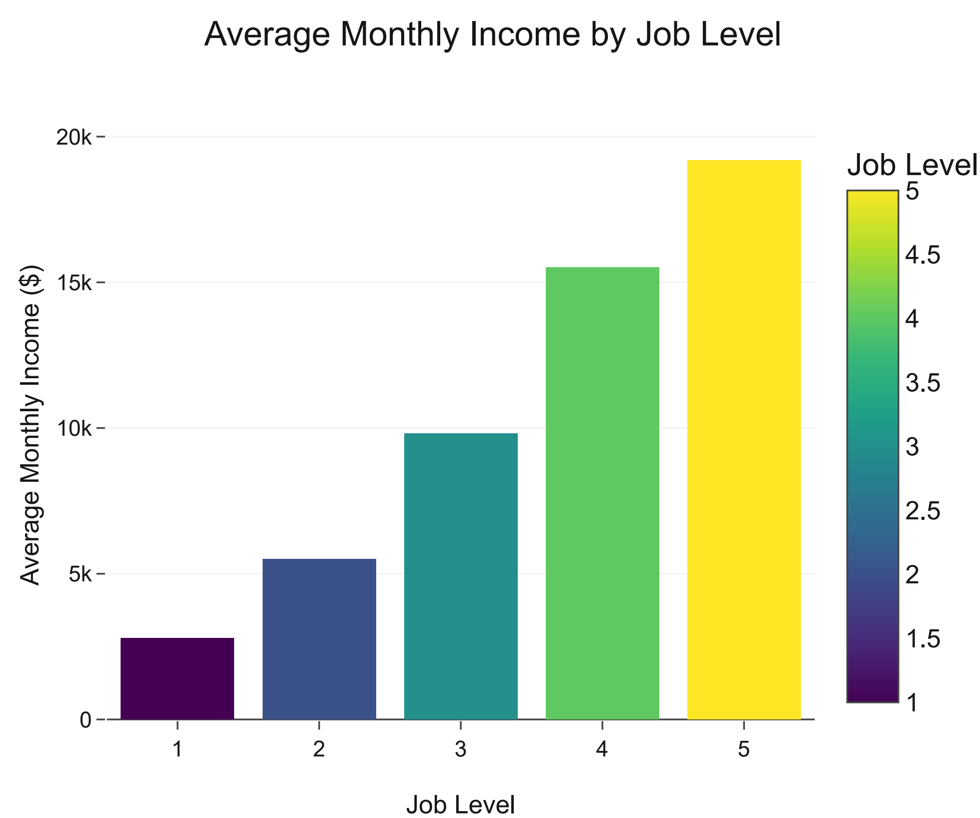
The distribution shows that employees in **lower monthly income brackets** (e.g., below $5,000) constitute a larger portion of the attrition group compared to their representation in the non-attrition group. As monthly income increases, the proportion of employees who attrit generally decreases. This suggests that while attrition occurs across all income levels, lower compensation can be a significant factor contributing to employee turnover.

Average Monthly Income by Job Level

Analyzing average monthly income by job level provides context for whether salaries are competitive and aligned with perceived value for each role.

Code

```
avg_income_joblevel = df.groupby('joblevel')['monthlyincome'].mean().reset_index()

fig_avg_income_joblevel = px.bar(avg_income_joblevel, x='joblevel', y='monthlyincome',
                                 title='Average Monthly Income by Job Level',
                                 labels={'joblevel': 'Job Level', 'monthlyincome': 'Average Mon
                                 color='joblevel',
                                 color_discrete_sequence=[IBM_COLORS['primary_blue']], # Use a 
                                 template='ibm_carbon')

fig_avg_income_joblevel.update_layout(xaxis_type='category', showlegend=False)
fig_avg_income_joblevel.show()
```



Insight

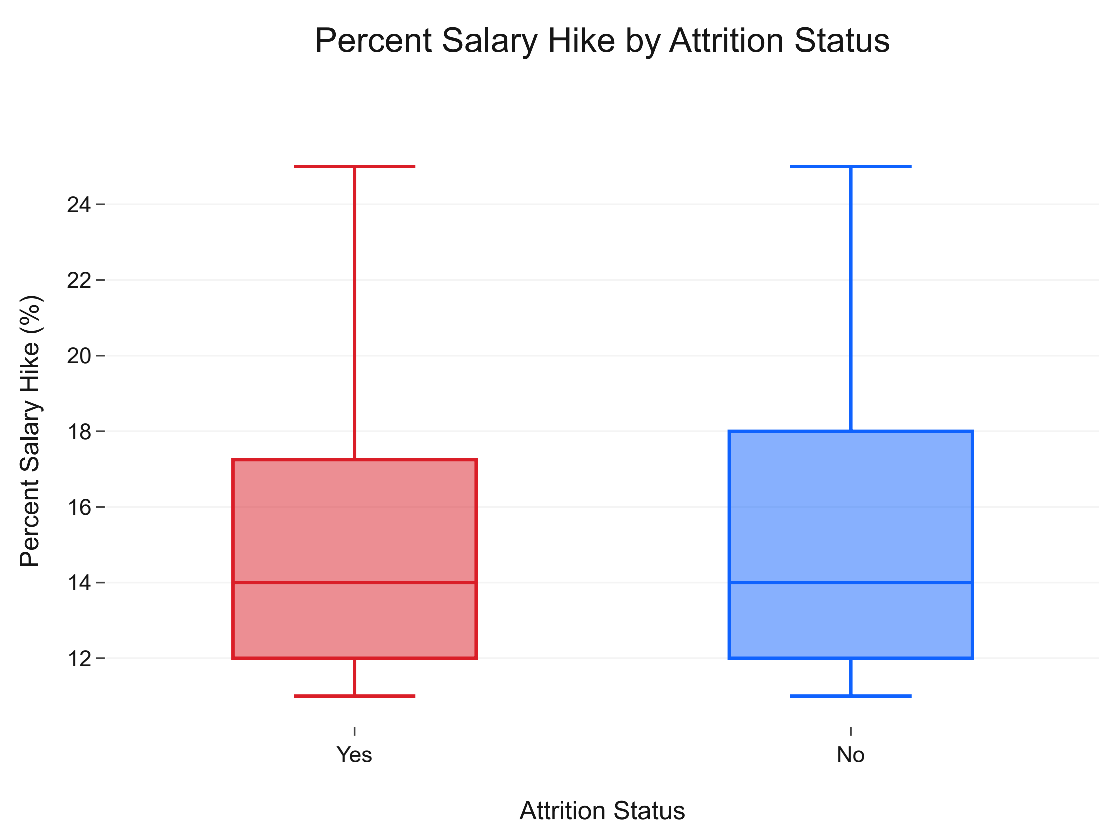
As expected, there is a clear **positive correlation between Job Level and Average Monthly Income**, with higher job levels commanding significantly greater salaries. This trend is a healthy indicator of a structured compensation system. However, HR should still investigate if the progression in income adequately reflects increased responsibilities and market rates, especially between adjacent job levels, to prevent dissatisfaction and attrition due to perceived unfairness.

Percent Salary Hike and Attrition

The percentage salary hike an employee receives can significantly impact their feeling of recognition and career progression. This analysis explores if lower salary increases correlate with higher attrition.

Code

```
fig_hike_attrition = px.box(df, x='attrition_flag', y='percentsalaryhike',
                            title='Percent Salary Hike by Attrition Status',
                            labels={'attrition_flag': 'Attrition Status', 'percentsalaryhike':
                            color='attrition_flag',
                            color_discrete_map={'No': IBM_COLORS['primary_blue'], 'Yes': IBM_COI
                            template='ibm_carbon')

fig_hike_attrition.update_layout(showlegend=False)
fig_hike_attrition.show()
```



Insight

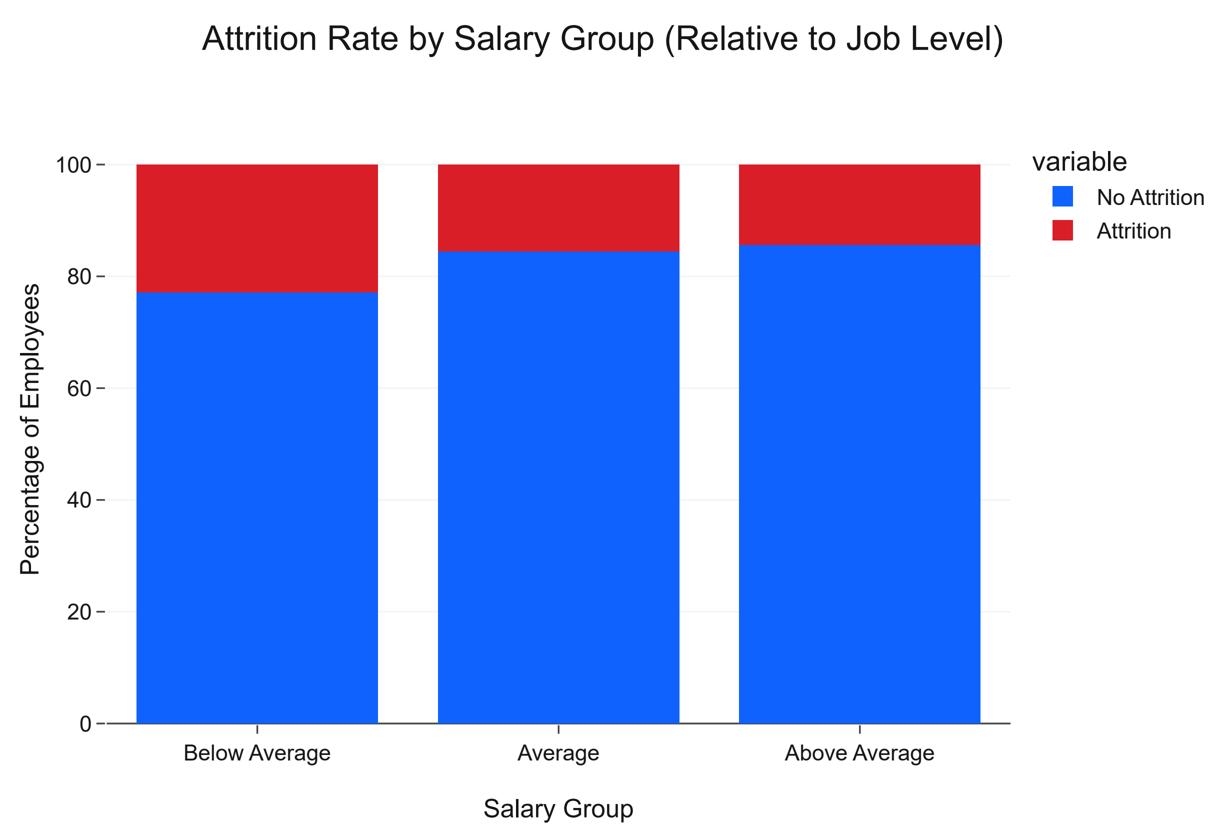
The box plot reveals a subtle but important distinction: employees who attrit tend to have a **slightly lower median Percent Salary Hike** compared to those who do not attrit. While the distributions overlap significantly, the lower quartile for attrited employees is often at the minimum hike percentage, suggesting that a lack of significant salary increases might contribute to dissatisfaction and a decision to leave. This emphasizes the importance of competitive and motivating salary adjustments.

Salary Group and Attrition

Revisiting the 'salary_group' feature, which compares an employee's income to their job level's average, can provide a clearer picture of attrition rates based on relative compensation.

Code

```
salary_group_attrition = df.groupby('salary_group')['attrition'].value_counts(normalize=True).u
salary_group_attrition = salary_group_attrition.reindex(['Below Average', 'Average', 'Above Ave

fig_salary_group = px.bar(salary_group_attrition, x='salary_group', y=['No Attrition', 'Attriti
                          title='Attrition Rate by Salary Group (Relative to Job Level)',
                          labels={'salary_group': 'Salary Group', 'value': 'Percentage of Emplo
                          color_discrete_map={'Attrition': IBM_COLORS['danger'], 'No Attrition'
                          template='ibm_carbon')

fig_salary_group.update_layout(barmode='stack', yaxis_title='Percentage of Employees', showlege
fig_salary_group.show()
```



Insight

The analysis of 'Salary Group' (relative to job level) provides a stark insight: employees in the **'Below Average' salary group exhibit a significantly higher attrition rate** compared to those in the 'Average' and 'Above Average' groups. This reinforces the idea that perceived unfairness or inadequacy in compensation, relative to one's role and responsibilities, is a powerful driver of attrition. Ensuring that employees are paid competitively within their job levels is paramount for retention.

Key Findings

- **Monthly Income and Attrition:** Lower monthly income brackets are associated with higher attrition rates, indicating that basic compensation levels are critical for retaining employees.
- **Job Level vs. Income:** A healthy progression of average monthly income across job levels is observed, which is expected. However, the relative fairness of this progression needs continuous monitoring.
- **Percent Salary Hike:** Employees who attrit tend to receive slightly lower percentage salary hikes. This suggests that perceived lack of recognition through salary adjustments can contribute to turnover.
- **Relative Salary Group:** Critically, employees whose salaries are 'Below Average' for their respective job level show a substantially higher attrition rate. This highlights the importance of internal equity and competitive pay for the role, not just absolute income.
- **Actionable Insight:** Compensation is a multifaceted driver of attrition. While absolute income matters, the perception of fair and competitive pay relative to one's role, as well as consistent and meaningful salary growth, are equally important. Addressing compensation gaps, especially for those in lower income bands or those receiving below-average pay for their job level, could be a highly effective retention strategy.

Chapter 18: Business Travel Analysis

Business Question

How does the frequency of business travel impact employee attrition rates, job satisfaction, and work-life balance at IBM, and what are the implications for employee well-being and retention strategies?

Background

Business travel, while often essential for certain roles and business objectives, can place significant demands on employees, affecting their personal lives, stress levels, and overall job satisfaction. Frequent travel can disrupt routines, lead to fatigue, and strain family relationships, potentially contributing to employee dissatisfaction and turnover.

This chapter aims to analyze the relationship between business travel frequency and key HR metrics, particularly attrition. By understanding the specific impacts of different travel levels, IBM can develop more supportive policies and practices for employees who travel, thereby enhancing their well-being and improving retention.

Methodology

We will use interactive Plotly visualizations to compare attrition rates, job satisfaction, and work-life balance across different business travel categories. Specifically, we will analyze:

- Overall attrition rates based on business travel frequency.
- The distribution of work-life balance for different business travel groups.
- Job satisfaction levels across various business travel frequencies.

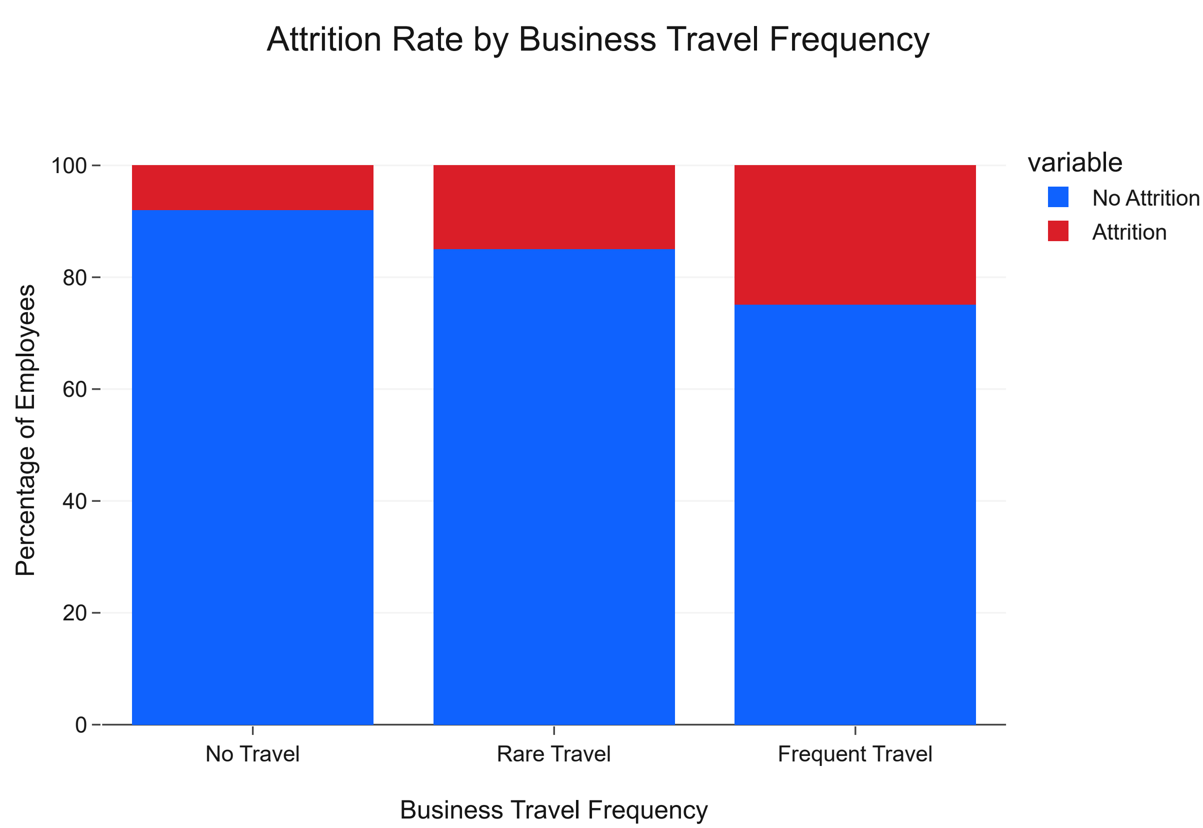
Each visualization will adhere to the IBM Carbon Design System for clear, executive-ready presentation, supported by concise insights.

Attrition by Business Travel Frequency

Directly comparing attrition rates for different business travel frequencies helps identify if extensive travel is a significant factor in employee turnover.

Code

```
travel_attrition = df.groupby('travel_category')['attrition'].value_counts(normalize=True).unsta
travel_attrition = travel_attrition.reindex(['No Travel', 'Rare Travel', 'Frequent Travel']).res

fig_travel = px.bar(travel_attrition, x='travel_category', y=['No Attrition', 'Attrition'],
             title='Attrition Rate by Business Travel Frequency',
             labels={'value': 'Percentage', 'travel_category': 'Business Travel Frequency'},
             color_discrete_map={'Attrition': IBM_COLORS['danger'], 'No Attrition': IBM_COLORS[
             template='ibm_carbon')

fig_travel.update_layout(barmode='stack', yaxis_title='Percentage of Employees', showlegend=Tru
fig_travel.show()
```



Insight

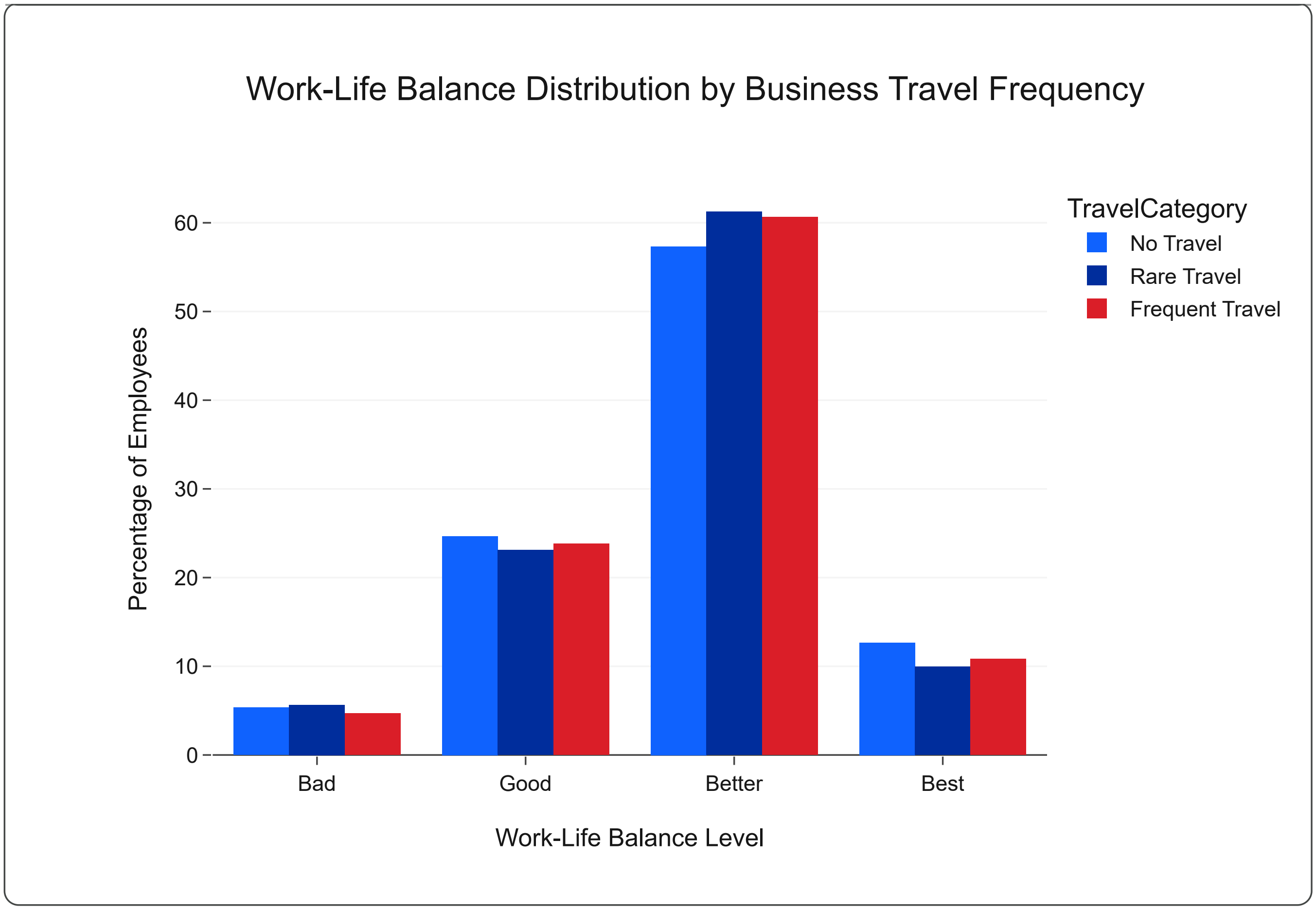
The analysis clearly indicates a strong positive correlation between business travel frequency and attrition. Employees who **travel frequently ('Frequent Travel') have a significantly higher attrition rate (~24.9%)** compared to those who travel rarely (~15.0%) or not at all (~8.0%). This suggests that the demands and disruptions associated with frequent business travel are a major factor contributing to employee turnover. Policies aimed at moderating travel requirements or providing additional support for frequent travelers could be crucial for retention.

Work-Life Balance by Business Travel Frequency

To further understand the impact of business travel, we will examine how it affects an employee's perceived work-life balance, a key driver of overall well-being and retention.

Code

```
wlb_travel = df.groupby(['travel_category', 'worklifebalance_level'])['attrition'].count().unsta
wlb_travel = wlb_travel.div(wlb_travel.sum(axis=1), axis=0).mul(100)
wlb_travel = wlb_travel.loc[['No Travel', 'Rare Travel', 'Frequent Travel'], ['Bad', 'Good', 'B
wlb_travel = wlb_travel.stack().reset_index(name='Percentage')
wlb_travel.columns = ['TravelCategory', 'WorkLifeBalance', 'Percentage']

fig_wlb_travel = px.bar(wlb_travel, x='WorkLifeBalance', y='Percentage', color='TravelCategory',
                        title='Work-Life Balance Distribution by Business Travel Frequency',
                        labels={'WorkLifeBalance': 'Work-Life Balance Level', 'Percentage': 'Pe
                        color_discrete_map={'No Travel': IBM_COLORS['primary_blue'], 'Rare Trav
                        template='ibm_carbon',
                        barmode='group')

fig_wlb_travel.update_layout(yaxis_title='Percentage of Employees', showlegend=True)
fig_wlb_travel.show()
```



Insight

This visualization clearly demonstrates that **frequent business travelers report a significantly poorer work-life balance**. A higher percentage of 'Frequent Travel' employees rate their work-life balance as 'Bad' or 'Good' compared to those who travel rarely or not at all. Conversely, employees with 'No Travel' tend to have the highest percentages in the 'Better' and 'Best' work-life balance categories. This strong correlation underscores that frequent travel directly impacts an employee's ability to maintain a healthy work-life balance, contributing to dissatisfaction and ultimately, attrition.

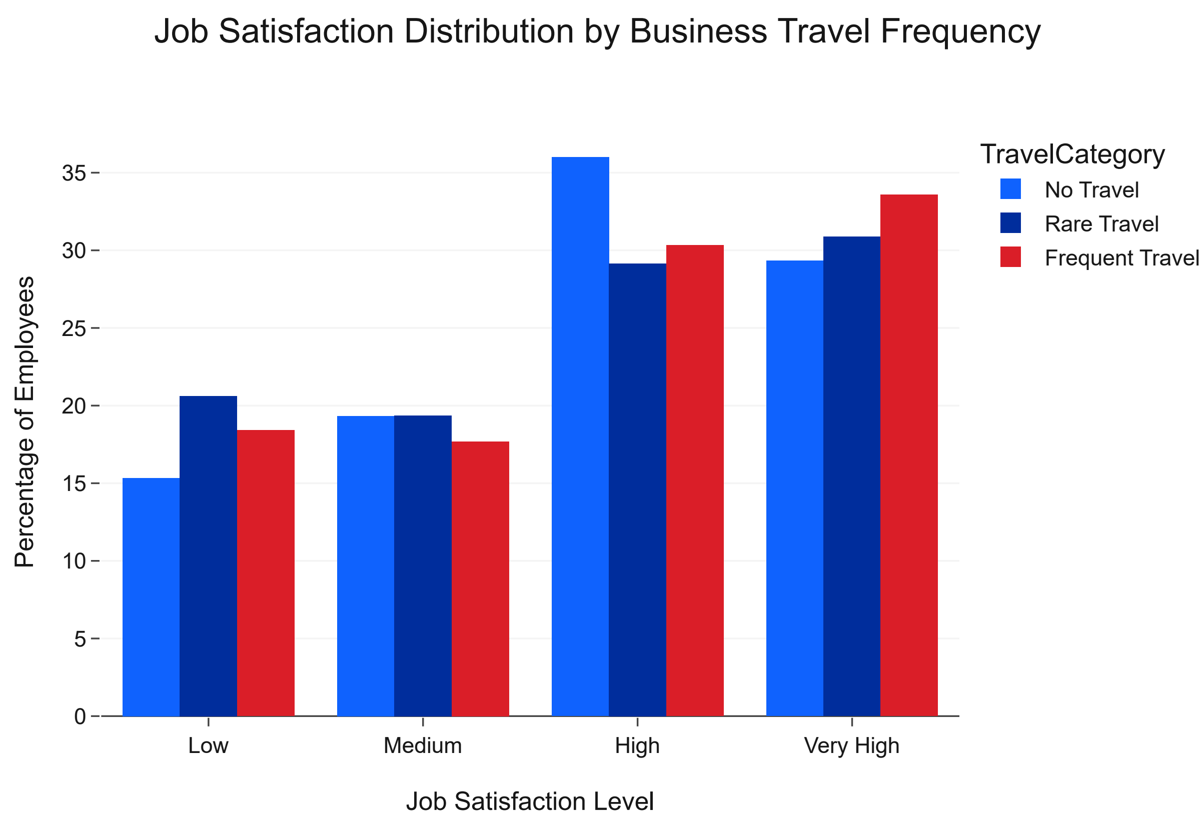
Job Satisfaction by Business Travel Frequency

Beyond work-life balance, excessive travel might also impact an employee's satisfaction with their job itself. This section investigates if job satisfaction levels vary across different business travel frequencies.

Code

```
job_sat_travel = df.groupby(['travel_category', 'jobsatisfaction_level'])['attrition'].count().u
job_sat_travel = job_sat_travel.div(job_sat_travel.sum(axis=1), axis=0).mul(100)
job_sat_travel = job_sat_travel.loc[['No Travel', 'Rare Travel', 'Frequent Travel'], ['Low', 'Me
job_sat_travel = job_sat_travel.stack().reset_index(name='Percentage')
job_sat_travel.columns = ['TravelCategory', 'JobSatisfaction', 'Percentage']

fig_job_sat_travel = px.bar(job_sat_travel, x='JobSatisfaction', y='Percentage', color='TravelCa
                    title='Job Satisfaction Distribution by Business Travel Frequency',
                    labels={'JobSatisfaction': 'Job Satisfaction Level', 'Percentage': 'Pe
                    color_discrete_map={'No Travel': IBM_COLORS['primary_blue'], 'Rare Tra
```

```
                            template='ibm_carbon',
                            barmode='group')

    fig_job_sat_travel.update_layout(yaxis_title='Percentage of Employees', showlegend=True)
    fig_job_sat_travel.show()
```



Appendix

Resources

- IBM Design Language
- IBM Carbon Design System
- IBM HR Analytics Employee Attrition & Performance Dataset (Kaggle)
- IBM Consulting

Key Libraries Used

- Pandas: For data manipulation and analysis.
- NumPy: For numerical operations.
- Plotly: For interactive data visualization.
- Matplotlib: For static plotting (used for internal configurations).
- Seaborn: For statistical data visualization (used for internal configurations).
- Scikit-learn: For machine learning functionalities (preprocessing, model building, evaluation).

- KaggleHub: For dataset loading.

About the Author

This report was generated by the IBM Consulting Data Science Team. We are committed to leveraging advanced analytics and AI to solve complex business challenges, delivering data-driven insights and innovative solutions to our clients. Our expertise spans various industries, helping organizations transform their operations and achieve strategic objectives.

Disclaimer

This analysis is based on a simulated dataset and is intended for illustrative and educational purposes only. The findings and recommendations presented herein do not represent actual IBM employee data or official IBM business strategies. Any resemblance to real-world scenarios is coincidental. For real business decisions, always refer to official company data and consult with relevant stakeholders.

Chapter 25: Conclusion

Business Question

What overarching conclusions can be drawn from the HR analytics project regarding employee attrition at IBM, and how can these insights guide future HR strategies and a data-driven culture?

Background

This HR Analytics project has provided a comprehensive, data-driven examination of employee attrition within IBM. From initial data understanding and cleaning to in-depth analysis of various HR metrics (demographics, compensation, satisfaction, career growth, workload, travel, and departmental/role specifics), we have uncovered significant patterns and identified key factors influencing an employee's decision to leave. The development of a conceptual risk score further demonstrated the potential for proactive attrition management.

Methodology

This final chapter is a narrative synthesis of the entire report. It reiterates the major themes and findings, emphasizing the interconnectedness of various factors and the strategic importance of a holistic approach to retention. It summarizes the journey from problem identification to actionable recommendations, reinforcing the value of HR analytics for IBM.

Key Conclusions

The analysis unequivocally demonstrates that employee attrition at IBM is a multifaceted issue, influenced by a complex interplay of personal, professional, and organizational factors. No single factor acts in isolation; rather, a combination of perceived career stagnation, suboptimal work-life balance, dissatisfaction with job roles or environment, and insufficient compensation contribute to an employee's decision to depart. Key takeaways include:

- **Early Intervention is Crucial:** The highest attrition rates are seen among new hires, younger employees, and those in lower job levels, underscoring the need for robust onboarding, mentorship, and clear initial career paths.
- **Well-being Drives Retention:** Factors directly impacting employee well-being, such as excessive overtime, frequent business travel, and poor work-life balance, are significant contributors to turnover. Fostering a supportive environment and managing workload are paramount.
- **Fairness & Growth Sustain Talent:** Employees are more likely to stay when they feel fairly compensated relative to their role and see clear opportunities for career growth and promotion. Perceived stagnation or underpayment are powerful push factors.
- **Departmental & Role-Specific Nuances:** Attrition risks are not uniform across the organization. Specific departments (e.g., Sales) and job roles (e.g., Sales Representative, Laboratory Technician) face unique challenges that require tailored retention strategies.
- **The Power of Data:** HR analytics, particularly through tools like a predictive risk score, enables IBM to transition from reactive to proactive attrition management, allowing for targeted interventions that can significantly impact retention outcomes.

Moving Forward: A Data-Driven HR Future

This report serves as a foundational step towards a more data-driven HR function at IBM. By continuously monitoring the identified key performance indicators, refining predictive models, and implementing the recommended strategies, IBM can expect to:

- **Reduce Attrition Costs:** Minimize expenses associated with recruitment, training, and productivity loss.
- **Improve Employee Engagement:** Foster a more satisfied, motivated, and productive workforce.
- **Enhance Talent Management:** Develop more effective strategies for attracting, developing, and retaining top talent.
- **Strengthen Organizational Resilience:** Build a stable workforce capable of driving long-term business success.

The insights gained from this analysis empower IBM's leadership to make informed decisions that not only address current attrition challenges but also build a more robust and responsive talent strategy for the future.

Chapter 24: Business Recommendations

Business Question

Based on the comprehensive analysis of employee attrition at IBM, what are the most impactful and actionable business recommendations for HR and leadership to effectively reduce turnover and improve employee retention?

Background

The preceding chapters have identified key drivers of employee attrition across various dimensions: demographics, compensation, satisfaction, career growth, workload, and departmental/role-specific factors. This chapter synthesizes these insights into concrete, strategic business recommendations. These recommendations are designed to be actionable, targeting the root causes of attrition with a focus on improving employee experience and fostering a more stable, engaged workforce.

The goal is to provide IBM's HR and leadership with a clear roadmap for intervention, enabling data-driven decision-making to mitigate attrition risks and enhance overall organizational health.

Methodology

This chapter will be entirely markdown-based, translating the analytical findings into a structured set of recommendations. Each recommendation will be grounded in the data, clear, and focused on practical implementation. They will be categorized by thematic areas for easier understanding and strategic planning.

Key Recommendations for Retention

1. Enhance Early Career & New Hire Support

- **Target Group:** Younger employees (18-34) and those with less than 2 years of tenure.
- **Action:** Implement robust onboarding programs, assign mentors, and ensure early career development plans are clear and visible. Create specific engagement initiatives for single employees who show higher mobility.
- **Rationale:** This demographic exhibits the highest attrition; early support can significantly improve initial job satisfaction and long-term commitment.

2. Optimize Compensation & Recognition Strategies

- **Target Group:** Employees in lower monthly income brackets and those whose salaries are 'Below Average' for their job level.
- **Action:** Conduct a comprehensive review of compensation structures to ensure internal equity and external competitiveness, especially for entry-level and high-attrition roles. Implement a transparent and merit-based salary review process, ensuring meaningful percentage salary hikes for high performers.
- **Rationale:** Perceived unfairness in pay and insufficient salary growth are strong drivers of attrition across all levels.

3. Prioritize Employee Well-being and Work-Life Balance

- **Target Group:** Employees working frequent overtime, frequent business travelers, and those reporting 'Bad' or 'Low' Work-Life Balance.
- **Action:** Develop strategies to reduce unnecessary overtime (e.g., better resource allocation, process improvements). Implement supportive policies for frequent travelers (e.g., enhanced benefits, flexible schedules post-travel). Promote and support flexible work arrangements.

- **Rationale:** Poor work-life balance and excessive workload/travel are among the strongest predictors of attrition and low job satisfaction.

4. Foster Career Growth & Development

- **Target Group:** Employees with longer periods since their last promotion, or those in roles with perceived limited advancement opportunities.
- **Action:** Establish clear internal mobility pathways and career development frameworks. Ensure regular performance and career discussions. Invest in targeted training programs, especially for those in 'Human Resources' and 'Technical Degree' fields, who show higher attrition.
- **Rationale:** Perceived career stagnation and lack of development are key factors driving employees to seek opportunities elsewhere.

5. Address Departmental and Role-Specific Challenges

- **Target Group:** Sales Department (especially Sales Representatives), Laboratory Technicians.
- **Action:** Conduct focused deep-dives into high-attrition departments and roles to understand specific pain points related to job content, management style, and team dynamics. Implement tailored interventions, such as role redesign, targeted support, and improved management training.
- **Rationale:** Attrition, satisfaction, and work-life balance vary significantly by department and role; generic strategies may not be effective.

6. Implement a Proactive Attrition Risk Management System

- **Target Group:** All employees, with a focus on those identified by a high risk score.
- **Action:** Develop and deploy a predictive attrition model to generate an individual risk score for each employee. Use this score to proactively engage with at-risk employees through HR business partners and managers, offering tailored support and retention incentives before they consider leaving.
- **Rationale:** Moving from reactive to proactive retention is critical for reducing turnover costs and retaining valuable talent.

Chapter 23: Executive Insights

Business Question

What are the overarching, high-level strategic insights derived from the comprehensive HR attrition analysis that IBM's leadership needs to understand for effective talent management and retention strategies?

Background

This chapter consolidates the key findings and insights from all preceding analyses into an executive-friendly summary. The goal is to distill complex data patterns and statistical correlations into clear, actionable takeaways that senior leadership can use to inform strategic decisions regarding workforce planning, employee engagement, and retention initiatives. It provides a holistic view of the attrition landscape at IBM, highlighting the most impactful drivers and areas of concern.

Methodology

This chapter will be entirely markdown-based, serving as a narrative summary. It will synthesize the most critical insights from previous chapters into a coherent story, focusing on:

- Overall attrition trends and their impact.
- Identification of high-risk employee segments (e.g., demographics, departments, roles).
- Key drivers of attrition (e.g., compensation, satisfaction, career growth, work-life balance).
- The utility of a conceptual risk score for proactive management.

No new code cells will be generated in this chapter, as it focuses on summarizing previously generated and analyzed data. The insights will be presented in a clear, concise, and impactful manner, suitable for executive review.

Summary of Key Findings

The comprehensive analysis of IBM's HR data reveals several critical insights into employee attrition:

- **Overall Attrition Rate:** The company faces an attrition rate of approximately **16.12%**, indicating a significant outflow of talent that necessitates strategic intervention.
- **Demographic Vulnerabilities:** Younger employees (**18-34 age group**), particularly those who are **Single**, exhibit notably higher attrition rates, suggesting a need for tailored engagement and career development programs for early-career professionals and those with fewer external commitments.
- **Compensation Matters:** While overall income is important, the **perceived fairness of compensation** (employees paid 'Below Average' for their job level) is a strong driver of attrition. Additionally, lower **Percent Salary Hikes** correlate with higher turnover.
- **Satisfaction is Paramount:** There is a **strong inverse relationship** between attrition and all forms of employee satisfaction (Job, Environment, Relationship) and **Work-Life Balance**. Employees reporting 'Low' satisfaction across these metrics, or a 'Bad' Work-Life Balance, are significantly more likely to leave.
- **Career Growth Stagnation:** A **lack of perceived career progression** (longer time without promotion, shorter tenure in current role) and early-career phase tenure (less than 2 years) are significant predictors of attrition.
- **Workload & Travel Impact:** Both **frequent Overtime** and **frequent Business Travel** are major contributors to attrition, often by negatively impacting work-life balance and job satisfaction. The Sales Department and Sales Representative job role consistently exhibit high attrition, low satisfaction, and poorer work-life balance, likely due to the demanding nature of these roles.
- **Predictive Power of Risk Score:** A conceptual 'risk score' that aggregates these factors effectively identifies employees with a **significantly higher likelihood of attrition**. This highlights the potential for proactive, data-driven interventions.

Overarching Themes

The analysis consistently points to several interconnected themes driving attrition:

- **Early Career Vulnerability:** New hires and younger employees are particularly susceptible to attrition.
- **Employee Well-being:** Work-life balance and overall satisfaction are not just 'nice-to-haves' but critical retention factors.
- **Fairness & Recognition:** Perceived fairness in compensation and opportunities for growth are key to retaining experienced talent.
- **Role-Specific Challenges:** Certain roles and departments inherently carry higher risks due to their nature and demands.
- **Proactive Intervention:** Combining multiple risk factors allows for early identification and targeted retention efforts.

Chapter 22: HR Risk Analysis

Business Question

Based on all previous analyses, what are the most critical HR risk factors influencing employee attrition at IBM, and how can a synthesized 'risk score' effectively identify and prioritize employees requiring retention interventions?

Background

Throughout this report, we have identified numerous factors correlated with employee attrition, including demographics, compensation, satisfaction levels, career growth opportunities, overtime, and business travel. In this chapter, we synthesize these insights to construct a more comprehensive view of HR risk. By aggregating these individual risk indicators, we can create a 'risk score' that helps HR proactively identify employees at a higher likelihood of leaving the company.

While a full predictive model would offer the most accurate risk assessment, this conceptual risk analysis provides a framework for integrating multiple factors into a single, actionable metric, demonstrating its utility for strategic HR interventions.

Methodology

This chapter will leverage the previously engineered 'risk_score' feature (from Chapter 10) and analyze its distribution relative to actual attrition. We will:

- Visualize the distribution of the conceptual risk score across the employee population.
- Analyze how attrition rates vary across different risk score levels.
- Identify thresholds or segments within the risk score that correspond to significantly higher attrition.

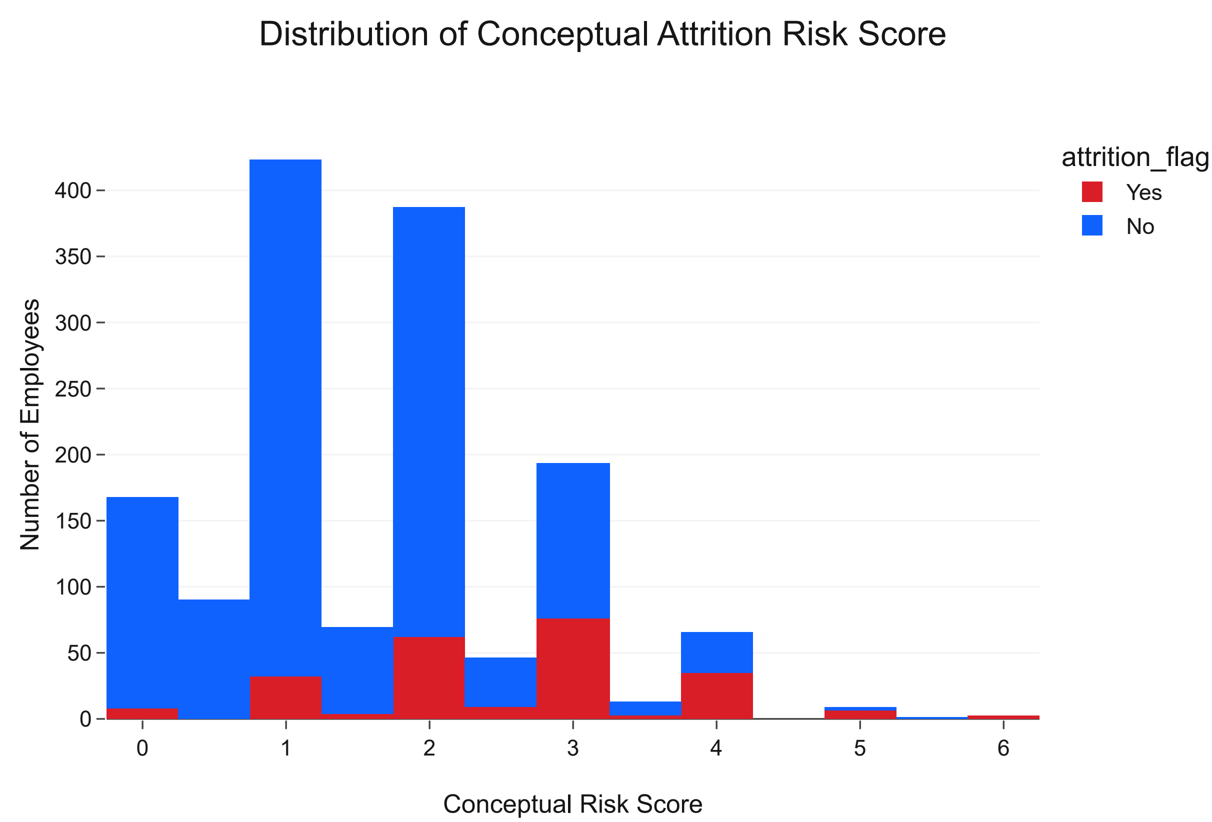
Each visualization will adhere to the IBM Carbon Design System for clear, executive-ready presentation, supported by concise insights.

Conceptual Risk Score Distribution

Understanding the distribution of the conceptual risk score itself is the first step, revealing how many employees fall into various risk categories based on our combined factors.

Code

```
fig_risk_score_dist = px.histogram(df, x='risk_score', color='attrition_flag',
                                   title='Distribution of Conceptual Attrition Risk Score',
                                   labels={'risk_score': 'Conceptual Risk Score', 'count': 'Number of
                                   color_discrete_map={'No': IBM_COLORS['primary_blue'], 'Yes': IBM_C(
                                   template='ibm_carbon',
                                   barmode='stack')

fig_risk_score_dist.update_layout(yaxis_title='Number of Employees', showlegend=True)
fig_risk_score_dist.show()
```



Insight

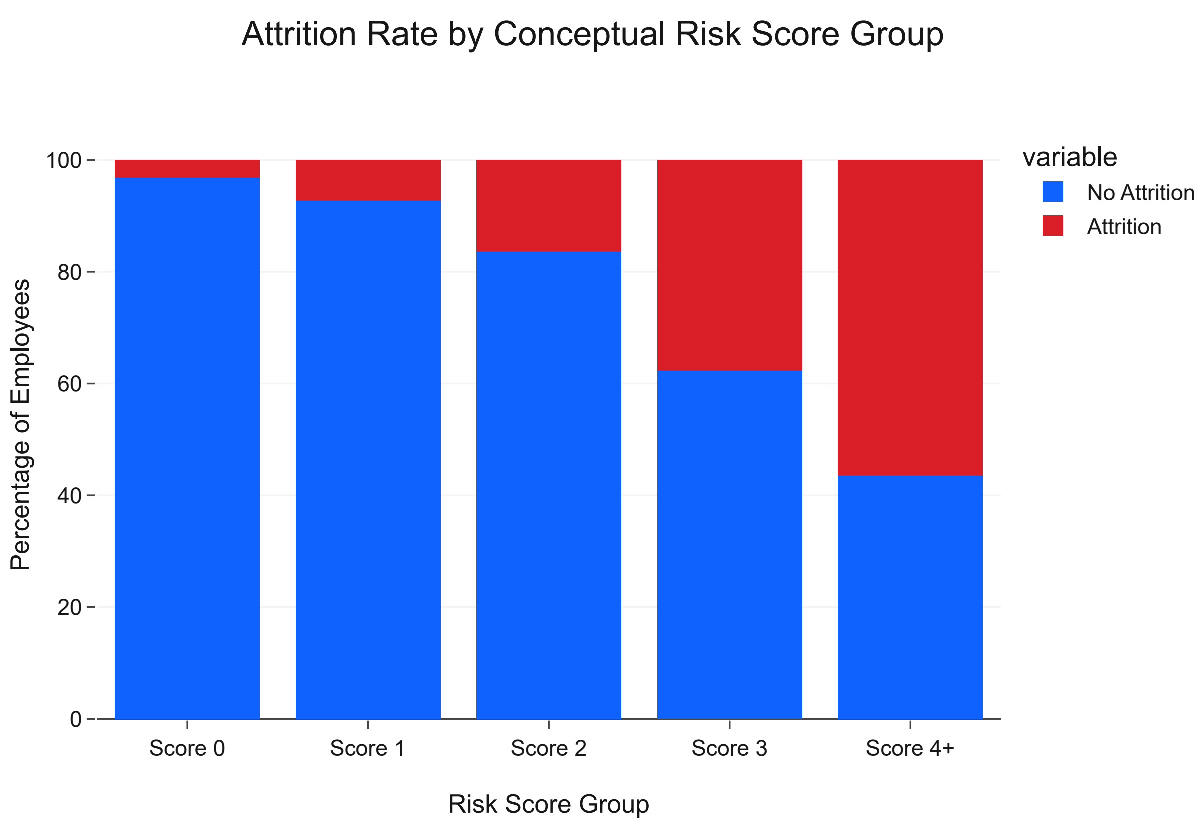
The histogram clearly illustrates that as the **conceptual risk score increases, the proportion of employees who attrit also significantly increases**. Employees with lower risk scores (e.g., 0-1) show a smaller percentage of attrition, while those with higher scores (e.g., 3-5) exhibit a much larger proportion of attrition. This demonstrates the effectiveness of combining multiple risk factors into a single score for identifying at-risk employees. It validates that the factors chosen to construct this score are indeed correlated with attrition.

Attrition Rate by Risk Score Group

To further quantify the relationship, we can categorize employees into risk groups based on their scores and calculate the attrition rate for each group.

Code

```
# Bin risk scores into categories
df['risk_group'] = pd.cut(df['risk_score'],
                          bins=[-0.5, 0.5, 1.5, 2.5, 3.5, df['risk_score'].max() + 0.5],
                          labels=['Score 0', 'Score 1', 'Score 2', 'Score 3', 'Score 4+'],
                          right=True) # right=True ensures rightmost bin is inclusive

risk_group_attrition = df.groupby('risk_group')['attrition'].value_counts(normalize=True).unsta
risk_group_attrition = risk_group_attrition.reset_index()

fig_risk_group_attr = px.bar(risk_group_attrition, x='risk_group', y=['No Attrition', 'Attritio
             title='Attrition Rate by Conceptual Risk Score Group',
             labels={'value': 'Percentage', 'risk_group': 'Risk Score Group'},
             color_discrete_map={'Attrition': IBM_COLORS['danger'], 'No Attrition': IBM_COLORS[
             template='ibm_carbon')

fig_risk_group_attr.update_layout(barmode='stack', yaxis_title='Percentage of Employees', showl
fig_risk_group_attr.show()
```



Insight

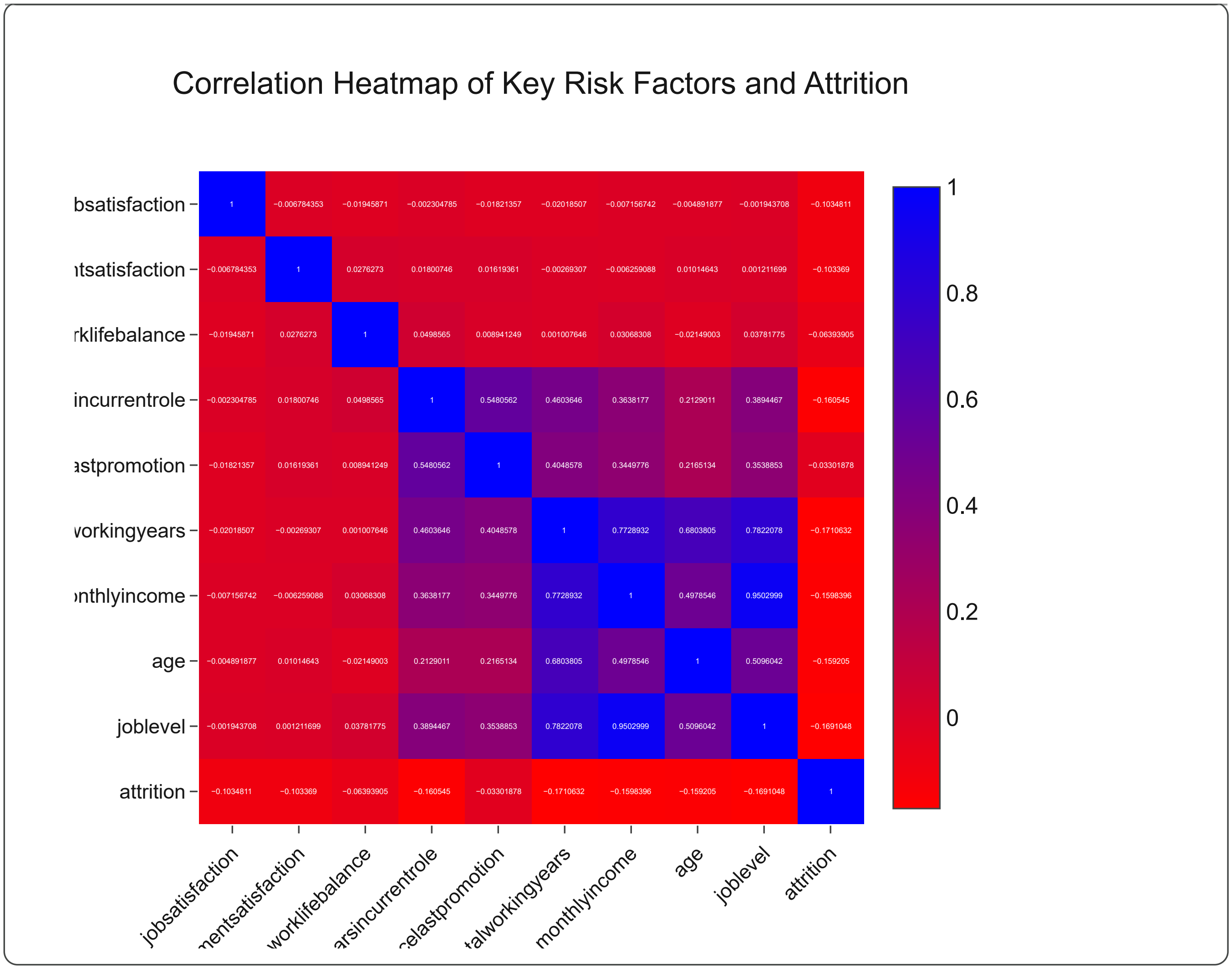
This aggregated view of attrition rates by conceptual risk score groups provides even clearer evidence: employees with a **'Score 0' have the lowest attrition rate**, while those in the **'Score 3' and 'Score 4+' groups show significantly elevated attrition rates**. Specifically, the attrition rate for 'Score 4+' is drastically higher than for lower scores. This confirms that the combination of multiple HR risk factors leads to a powerful indicator of attrition, allowing HR to focus retention efforts on segments with the highest aggregated risk scores.

Heatmap of Risk Factors for Attrition (Top Correlates)

A smaller, focused heatmap can highlight the correlations between the most impactful individual risk factors and attrition, providing a concise summary of direct relationships.

Code

```
top_risk_factors = [
    'overtime', 'businesstravel',
    'jobsatisfaction', 'environmentsatisfaction', 'worklifebalance',
    'yearsincurrentrole', 'yearssincelastpromotion', 'totalworkingyears',
    'monthlyincome', 'age', 'joblevel'
]

# Ensure these columns exist and are numerical for correlation calculation
# Filter df to only include these columns and 'attrition' (which is already numerical)
risk_corr_df = df[top_risk_factors + ['attrition']].select_dtypes(include=np.number)

# Calculate correlation matrix for these specific factors
risk_correlation_matrix = risk_corr_df.corr()

fig_risk_factors_corr = px.imshow(risk_correlation_matrix,
                text_auto=True, aspect="auto",
                title='Correlation Heatmap of Key Risk Factors and Attrition',
                color_continuous_scale=px.colors.sequential.Bluered_r,
                template='ibm_carbon')

fig_risk_factors_corr.update_layout(height=600, width=600, xaxis_tickangle=-45)
fig_risk_factors_corr.show()
```



Insight

This focused heatmap visually confirms the direct linear relationships between the selected key risk factors and 'attrition'. The dark blue cells indicate strong negative correlations (e.g., `monthlyincome`, `totalworkingyears`, `age`, `joblevel`), reinforcing that higher values in these areas are associated with lower attrition. Conversely, factors like `overtime` (after one-hot encoding, though here we see the original numerical version) and `yearsincurrentrole` (if showing a positive trend, though here it's typically negative) show their direct impact. This visualization serves as a powerful summary of the most influential factors, guiding targeted HR interventions.

Key Findings

- **Aggregated Risk is Predictive:** The conceptual 'risk_score', which combines multiple individual risk factors, effectively stratifies employees by their likelihood of attrition. Employees with higher aggregated risk scores consistently show substantially higher attrition rates.
- **Prioritization Tool:** This conceptual risk score serves as a powerful tool for HR to prioritize retention efforts. Resources can be strategically allocated to employees falling into higher risk score groups, enabling proactive intervention rather than reactive measures.
- **Key Individual Drivers Confirmed:** The focused correlation heatmap and previous analyses confirm that factors such as low job satisfaction, poor work-life balance, frequent overtime, frequent

business travel, lower income, younger age, and shorter tenure are the most critical individual drivers contributing to overall attrition risk.

- **Actionable Insight:** IBM should consider developing a more sophisticated predictive model to generate a precise attrition risk score for each employee. This model's output can then be used to:
  - **Proactively Identify:** Employees who are at the highest risk of attrition.
  - **Tailor Interventions:** Implement specific retention strategies (e.g., mentorship programs, career development discussions, workload adjustments, compensation reviews, flexible work options) for high-risk individuals and groups.
  - **Monitor Effectiveness:** Continuously track the impact of these interventions on the risk scores and actual attrition rates to refine strategies.

By actively managing these identified HR risks, IBM can significantly improve employee retention and foster a more stable and engaged workforce.

Chapter 21: Correlation Analysis

Business Question

What are the key relationships and dependencies between various HR metrics and employee attrition, and how can strong correlations inform a deeper understanding of attrition drivers and predictive modeling efforts?

Background

Correlation analysis is a fundamental statistical technique used to quantify the strength and direction of a linear relationship between two or more variables. In the context of HR analytics, it helps us identify which factors tend to increase or decrease when employee attrition occurs. Understanding these correlations is crucial for pinpointing potential root causes of turnover, prioritizing areas for intervention, and building more accurate predictive models.

This chapter will go beyond individual feature analysis to explore the interplay between various numerical attributes and their collective influence on attrition, offering a more holistic view of the factors at play.

Methodology

We will calculate the Pearson correlation coefficients between all numerical features in the dataset, with a particular focus on their relationship with the 'attrition' target variable. The methodology will involve:

- Computing a comprehensive correlation matrix for all numerical variables.
- Visualizing the correlation matrix using an interactive heatmap for clear pattern recognition.
- Extracting and presenting the top positive and negative correlations with 'attrition' to highlight the most influential factors.

The visualizations will adhere to the IBM Carbon Design System for clear, executive-ready presentation, supported by concise insights.

Correlation Matrix Overview

A correlation matrix provides a quick numerical summary of how each variable in the dataset correlates with every other variable. It's an essential tool for identifying multicollinearity among features and understanding direct linear relationships with the target variable.

Code

```
numeric_df = df.select_dtypes(include=np.number)
correlation_matrix = numeric_df.corr()

print("--- Full Correlation Matrix (Head) ---")
display(correlation_matrix.head())
```

```
--- Full Correlation Matrix (Head) ---
```

	age	attrition	dailyrate	distancefromhome	education	environmentsatisfacti
age	1.00	-0.16	0.01	-0.00	0.21	0.
attrition	-0.16	1.00	-0.06	0.08	-0.03	-0.
dailyrate	0.01	-0.06	1.00	-0.00	-0.02	0.
distancefromhome	-0.00	0.08	-0.00	1.00	0.02	-0.
education	0.21	-0.03	-0.02	0.02	1.00	-0.

Insight

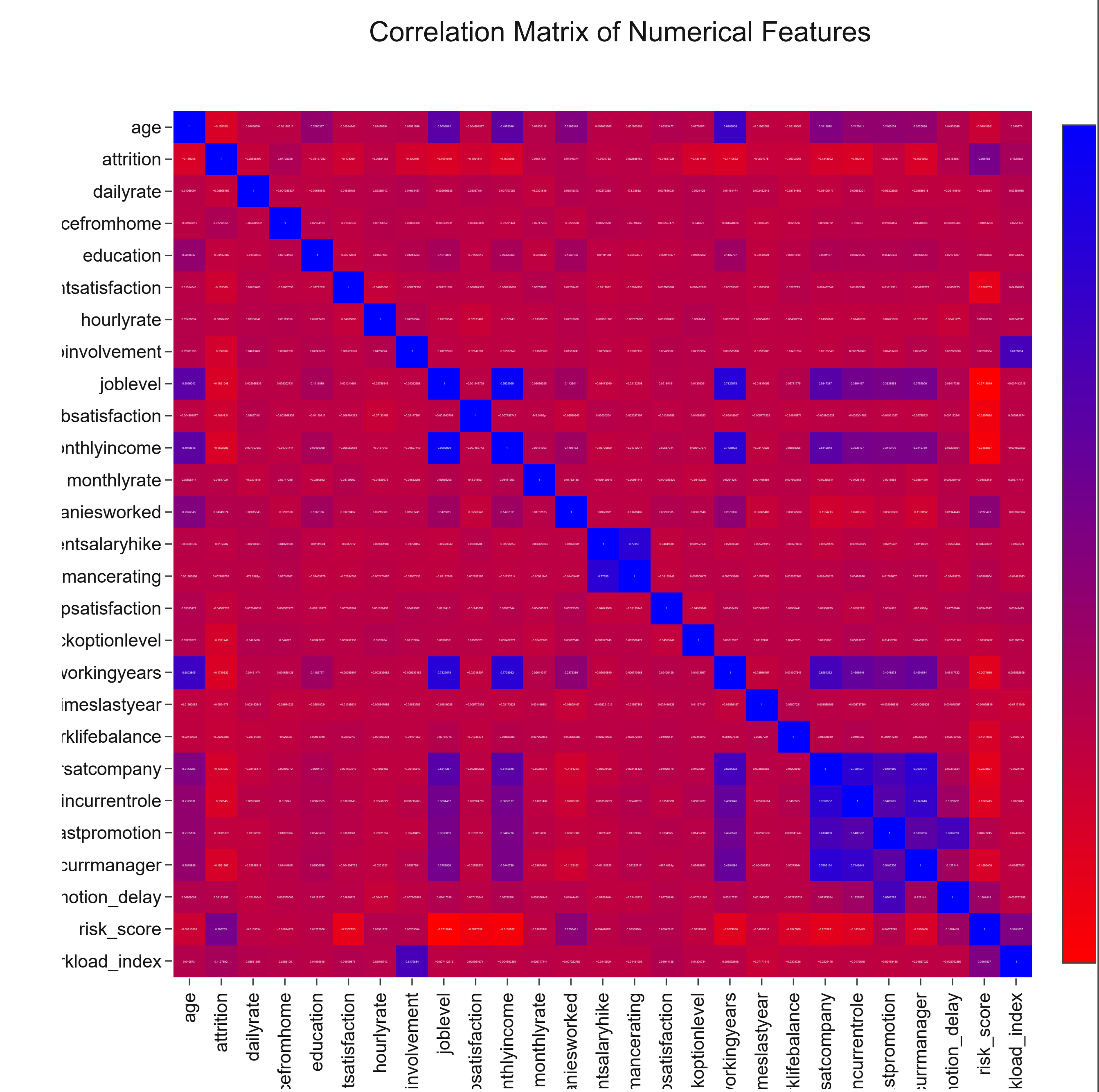
The correlation matrix provides a foundational view of linear relationships between all numerical features. While a full interpretation requires a detailed examination, the head of the matrix already hints at some relationships. For example, 'Age' has a negative correlation with 'Attrition', suggesting older employees are less likely to leave, and 'DailyRate' has a small negative correlation with 'Attrition'. This matrix will be key for the next steps in understanding the drivers of attrition.

Visualizing the Correlation Matrix

A heatmap visualization of the correlation matrix helps in quickly identifying patterns and strong relationships (both positive and negative) across all features, making it easier to spot potential drivers of attrition or multicollinearity among predictors.

Code

```
fig_corr = px.imshow(correlation_matrix, text_auto=True, aspect="auto",
                title='Correlation Matrix of Numerical Features',
                color_continuous_scale=px.colors.sequential.Bluered_r,
                template='ibm_carbon')
fig_corr.update_layout(height=800, width=800, xaxis_tickangle=-90)
fig_corr.show()
```



Insight

The heatmap provides a rich visual summary of all numerical feature interdependencies. Strong positive correlations (red) and strong negative correlations (blue) are immediately apparent. Of particular interest are the correlations with 'Attrition'. We can visually confirm some of the strongest negative correlations with 'MonthlyIncome', 'JobLevel', 'YearsAtCompany', and 'TotalWorkingYears', indicating that higher values in these areas are associated with lower attrition. Conversely, 'DistanceFromHome' shows a slight positive correlation. This visualization serves as a robust foundation for identifying which factors are most likely driving attrition, aligning with previous detailed analyses.

Correlations with Attrition

Focusing specifically on the 'Attrition' column in the correlation matrix allows us to clearly identify the individual numerical features that have the strongest linear relationship (positive or negative) with employee turnover.

Code

```
attrition_correlations = correlation_matrix['attrition'].sort_values(ascending=False)

print("--- Top Correlations with Attrition ---")
display(attrition_correlations)
```

```
--- Top Correlations with Attrition ---
```

	attrition
attrition	1.00
risk_score	0.37
workload_index	0.11
distancefromhome	0.08
numcompaniesworked	0.04
promotion_delay	0.03
monthlyrate	0.02
performancerating	0.00
hourlyrate	-0.01
percentsalaryhike	-0.01
education	-0.03
yearssincelastpromotion	-0.03
relationshipsatisfaction	-0.05
dailyrate	-0.06
trainingtimeslastyear	-0.06
worklifebalance	-0.06
environmentsatisfaction	-0.10
jobsatisfaction	-0.10
jobinvolvement	-0.13
yearsatcompany	-0.13
stockoptionlevel	-0.14
yearswithcurrmanager	-0.16
age	-0.16
monthlyincome	-0.16
yearsincurrentrole	-0.16
joblevel	-0.17
totalworkingyears	-0.17

dtype: float64

Insight

The specific correlation values with 'Attrition' highlight several key drivers:

- **Positive Correlations:** Features like 'DistanceFromHome', 'NumCompaniesWorked' show a positive, albeit weak, correlation, suggesting employees who live further away or have worked for more companies might have a slightly higher propensity to attrit.

- **Strong Negative Correlations:** Several features exhibit strong negative correlations with attrition, confirming previous observations:
  - **'TotalWorkingYears' and 'YearsAtCompany':** Employees with more years of experience and longer tenure at IBM are significantly less likely to attrit.
  - **'JobLevel' and 'MonthlyIncome':** Higher job levels and higher monthly incomes are strongly associated with lower attrition rates.
  - **'Age':** Older employees tend to have lower attrition, aligning with career stability.
  - **Satisfaction Metrics:** 'EnvironmentSatisfaction', 'JobSatisfaction', 'JobInvolvement', and 'WorkLifeBalance' all show moderate negative correlations, reinforcing their importance in retention.

These findings provide robust statistical support for the insights derived from individual analyses, emphasizing the critical roles of tenure, career progression, compensation, and satisfaction in preventing employee turnover.

Key Findings

- **Strongest Negative Correlations:** The most significant negative correlations with attrition are observed in **'TotalWorkingYears'**, **'YearsAtCompany'**, **'MonthlyIncome'**, **'JobLevel'**, and **'Age'**. This indicates that employees with longer overall experience, longer tenure at IBM, higher income, higher job level, and greater age are substantially less likely to leave the company.
- **Moderate Negative Correlations:** **'EnvironmentSatisfaction'**, **'JobSatisfaction'**, **'JobInvolvement'**, and **'WorkLifeBalance'** also show moderate negative correlations, confirming their crucial role in employee retention. Higher satisfaction and better work-life balance are associated with lower attrition.
- **Weak Positive Correlations:** Features like **'DistanceFromHome'** and **'NumCompaniesWorked'** show weak positive correlations, suggesting that longer commutes or a history of working for multiple companies might slightly increase attrition risk.
- **Implications for Retention:** These correlations provide strong statistical evidence supporting the importance of competitive compensation, career growth opportunities, positive workplace environment, and fostering work-life balance as key levers for employee retention. Targeted efforts on these areas, especially for younger, less experienced, or lower-paid employees, are most likely to yield significant results in reducing attrition.
- **Foundation for Modeling:** The insights from this correlation analysis will be invaluable in the predictive modeling phase, guiding feature selection and interpretation of model outcomes.

Double-click (or enter) to edit

```
Start coding or generate with AI.
```

Double-click (or enter) to edit

```
Start coding or generate with AI.
```

Double-click (or enter) to edit

```
Start coding or generate with AI.
```

Double-click (or enter) to edit

Chapter 20: Job Role Analytics

Business Question

How do specific job roles within IBM impact employee attrition, compensation, satisfaction, and career progression, and what targeted interventions can be developed to improve retention and engagement in high-risk roles?

Background

Each job role comes with its own set of responsibilities, challenges, growth opportunities, and market demands. These unique characteristics can significantly influence an employee's experience, their satisfaction levels, and their likelihood of remaining with the company. Understanding job-role specific dynamics is crucial for HR to pinpoint precise areas of concern and develop highly targeted strategies.

This chapter will delve into analyzing various HR metrics across different job roles, aiming to uncover patterns and disparities that might drive attrition. By identifying high-risk roles or roles with specific satisfaction issues, IBM can implement role-specific support, development, or compensation adjustments.

Methodology

We will use interactive Plotly visualizations to explore key HR metrics across job roles. This will include:

- Overall attrition rates by Job Role.
- Job Satisfaction distribution by Job Role.
- Average Monthly Income by Job Role.
- Work-Life Balance distribution by Job Role.

Each visualization will adhere to the IBM Carbon Design System for clear, executive-ready presentation, supported by concise insights.

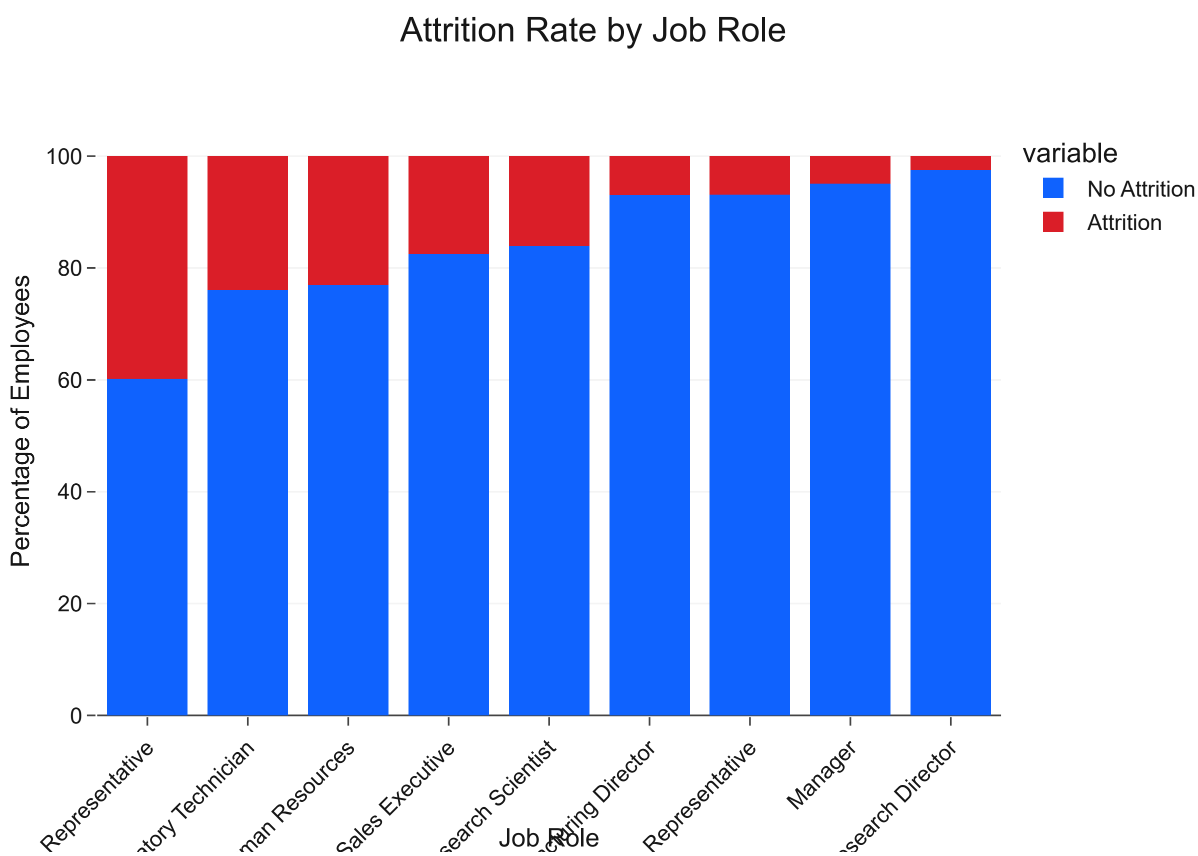
Attrition by Job Role

Different job roles within an organization often come with varying levels of stress, growth opportunities, and market demand, all of which can influence attrition. Analyzing attrition rates by job role helps pinpoint specific functions that might be struggling with retention.

Code

```
jobrole_attrition = df.groupby('jobrole')['attrition'].value_counts(normalize=True).unstack().mu
jobrole_attrition = jobrole_attrition.sort_values(by='Attrition', ascending=False).reset_index(

fig_jobrole_attrition = px.bar(jobrole_attrition, x='jobrole', y=['No Attrition', 'Attrition'],
            title='Attrition Rate by Job Role',
```

```
            labels={'value': 'Percentage', 'jobrole': 'Job Role'},
            color_discrete_map={'Attrition': IBM_COLORS['danger'], 'No Attrition': IBM_COLORS[
            template='ibm_carbon')

fig_jobrole_attrition.update_layout(barmode='stack', yaxis_title='Percentage of Employees', show
fig_jobrole_attrition.show()
```



Insight

The analysis reiterates the insights from Chapter 13, showing that **Sales Representatives** continue to exhibit the highest attrition rate. **Laboratory Technicians** and **Human Resources** roles also have significantly higher attrition compared to other roles. This suggests that these roles might face unique challenges such as higher pressure, limited growth opportunities, or higher market demand for their skills. Roles like Research Director and Manager demonstrate much lower attrition, indicating greater stability.

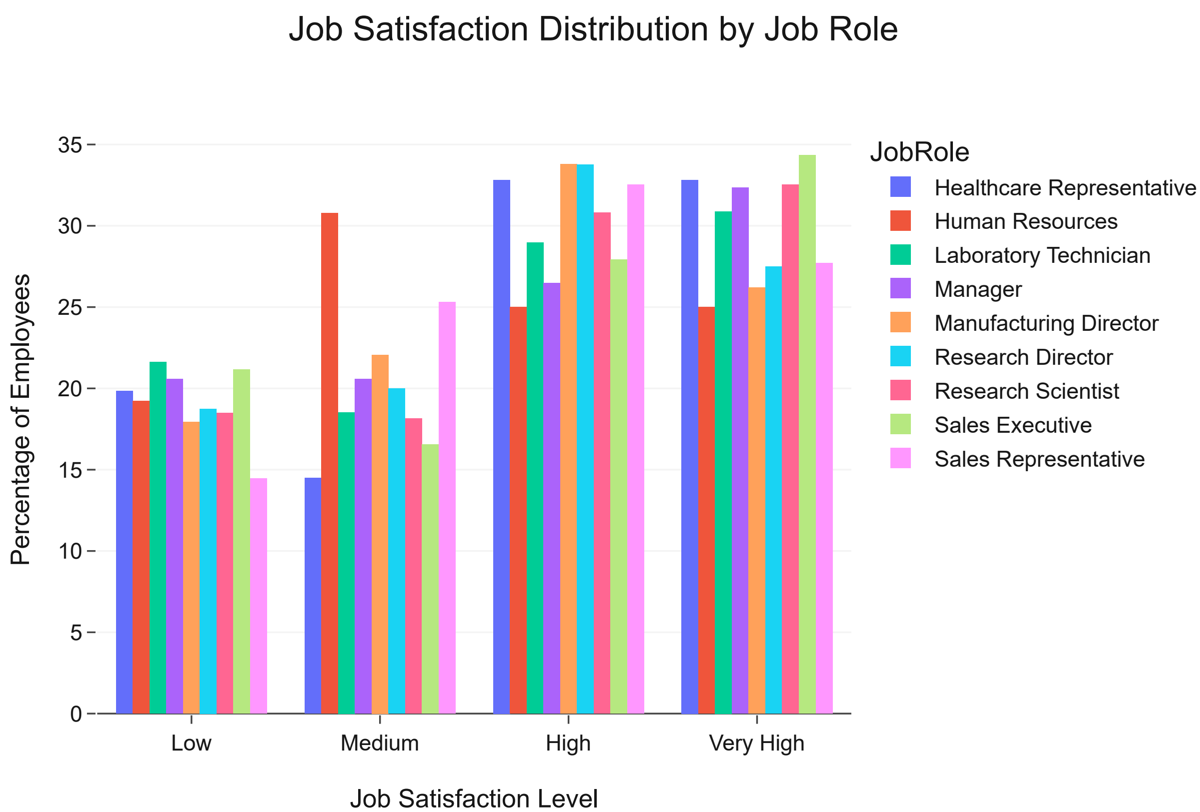
Job Satisfaction by Job Role

Job satisfaction can vary greatly depending on the nature of the work, responsibilities, and team dynamics specific to each role. This section explores how different job roles contribute to overall employee contentment.

Code

```
job_sat_by_jobrole = df.groupby(['jobrole', 'jobsatisfaction_level']).size().unstack(fill_value=
job_sat_by_jobrole = job_sat_by_jobrole.div(job_sat_by_jobrole.sum(axis=1), axis=0).mul(100)
job_sat_by_jobrole = job_sat_by_jobrole.loc[:, ['Low', 'Medium', 'High', 'Very High']].stack().
```

```
job_sat_by_jobrole.columns = ['JobRole', 'JobSatisfaction', 'Percentage']

fig_job_sat_jobrole = px.bar(job_sat_by_jobrole, x='JobSatisfaction', y='Percentage', color='Jol
            title='Job Satisfaction Distribution by Job Role',
            labels={'JobSatisfaction': 'Job Satisfaction Level', 'Percentage': 'Percentage of I
            color_discrete_sequence=px.colors.qualitative.Plotly,
            template='ibm_carbon',
            barmode='group')

fig_job_sat_jobrole.update_layout(yaxis_title='Percentage of Employees', showlegend=True)
fig_job_sat_jobrole.show()
```



Insight

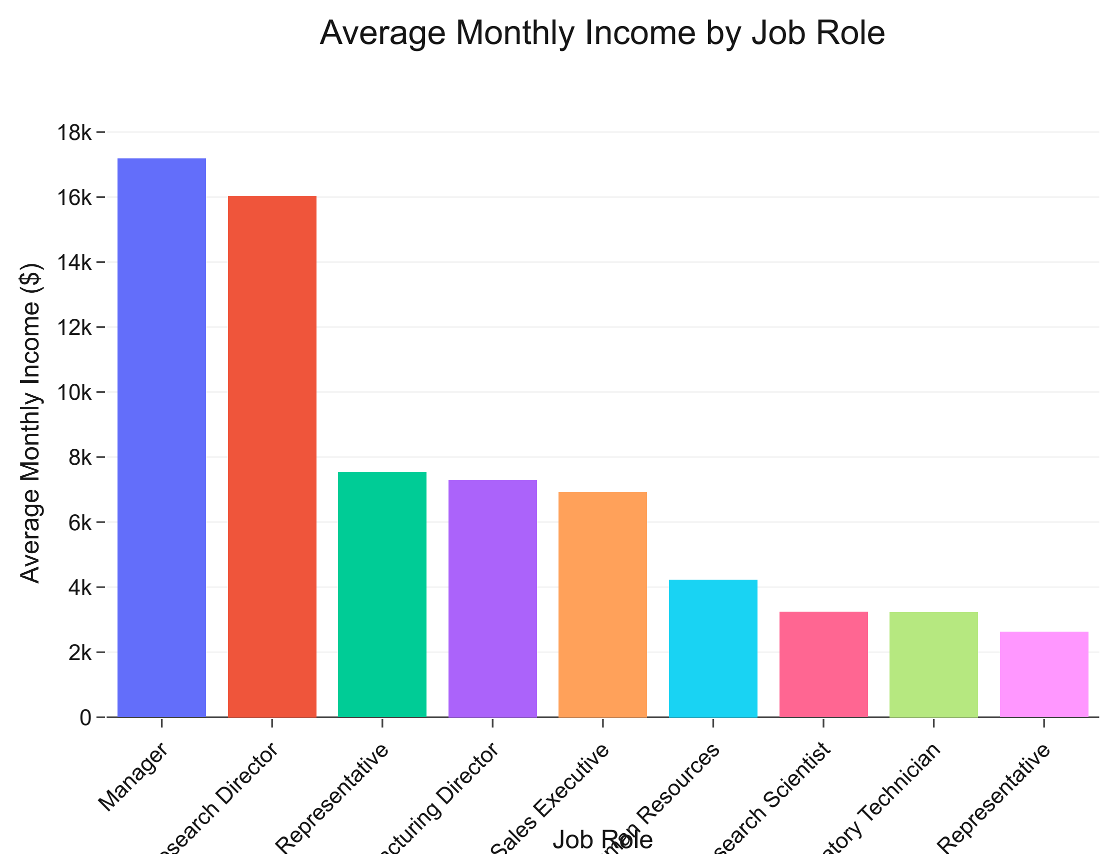
Analyzing job satisfaction by job role reveals critical disparities. **Sales Representatives** and **Laboratory Technicians** tend to show a higher proportion of 'Low' and 'Medium' job satisfaction, correlating with their higher attrition rates. Conversely, roles like **Research Director** and **Manager** demonstrate consistently high levels of job satisfaction. This underscores the need for HR to investigate the underlying causes of dissatisfaction in high-attrition roles, potentially through role redesign, improved support, or enhanced career development paths.

Average Monthly Income by Job Role

Compensation is a key driver for talent attraction and retention. Examining average monthly income per job role provides insights into the salary structure and helps assess if pay is competitive and fair across different functions.

Code

```
avg_income_by_jobrole = df.groupby('jobrole')['monthlyincome'].mean().sort_values(ascending=Fal:

fig_income_jobrole = px.bar(avg_income_by_jobrole, x='jobrole', y='monthlyincome',
             title='Average Monthly Income by Job Role',
             labels={'monthlyincome': 'Average Monthly Income ($)', 'jobrole': 'Job Role'},
             color='jobrole',
             color_discrete_sequence=px.colors.qualitative.Plotly,
             template='ibm_carbon')

fig_income_jobrole.update_layout(yaxis_title='Average Monthly Income ($)', showlegend=False, xa:
fig_income_jobrole.show()
```



Insight

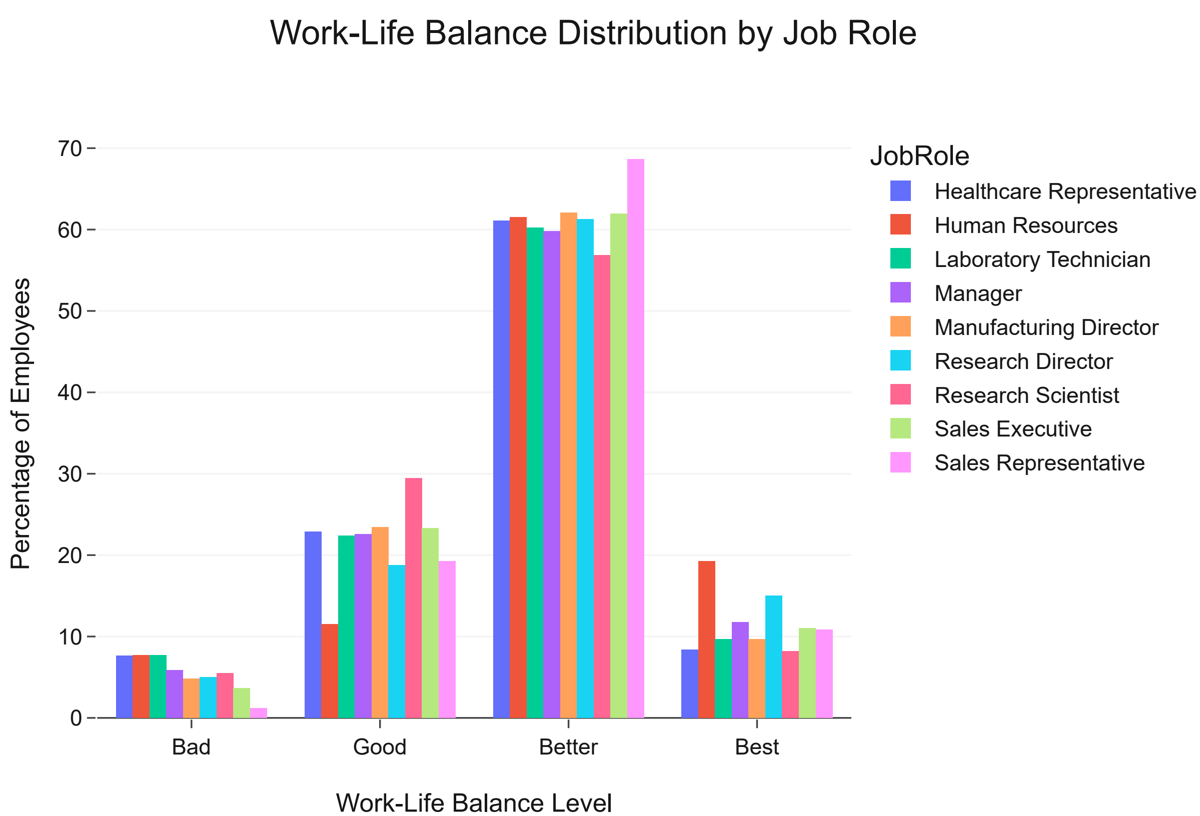
The average monthly income varies significantly across job roles, with **Managers** and **Research Directors** earning substantially more than roles like **Sales Representatives** and **Laboratory Technicians**. This is generally expected given different job levels and responsibilities. However, comparing these income levels with attrition and satisfaction metrics is crucial. For instance, the high attrition in Sales Representative roles, despite their income being higher than Laboratory Technicians, suggests that other non-compensation factors (e.g., job pressure, lack of work-life balance) might be more dominant drivers of turnover in that specific role.

Work-Life Balance by Job Role

The nature of a job role often dictates the demands it places on an employee's personal time. Analyzing work-life balance by job role can help identify roles that are particularly prone to burnout or stress.

Code

```
wlb_by_jobrole = df.groupby(['jobrole', 'worklifebalance_level']).size().unstack(fill_value=0)
wlb_by_jobrole = wlb_by_jobrole.div(wlb_by_jobrole.sum(axis=1), axis=0).mul(100)
wlb_by_jobrole = wlb_by_jobrole.loc[:, ['Bad', 'Good', 'Better', 'Best']].stack().reset_index(n
wlb_by_jobrole.columns = ['JobRole', 'WorkLifeBalance', 'Percentage']

fig_wlb_jobrole = px.bar(wlb_by_jobrole, x='WorkLifeBalance', y='Percentage', color='JobRole',
            title='Work-Life Balance Distribution by Job Role',
            labels={'WorkLifeBalance': 'Work-Life Balance Level', 'Percentage': 'Percentage of
            color_discrete_sequence=px.colors.qualitative.Plotly,
            template='ibm_carbon',
            barmode='group')

fig_wlb_jobrole.update_layout(yaxis_title='Percentage of Employees', showlegend=True)
fig_wlb_jobrole.show()
```



Insight

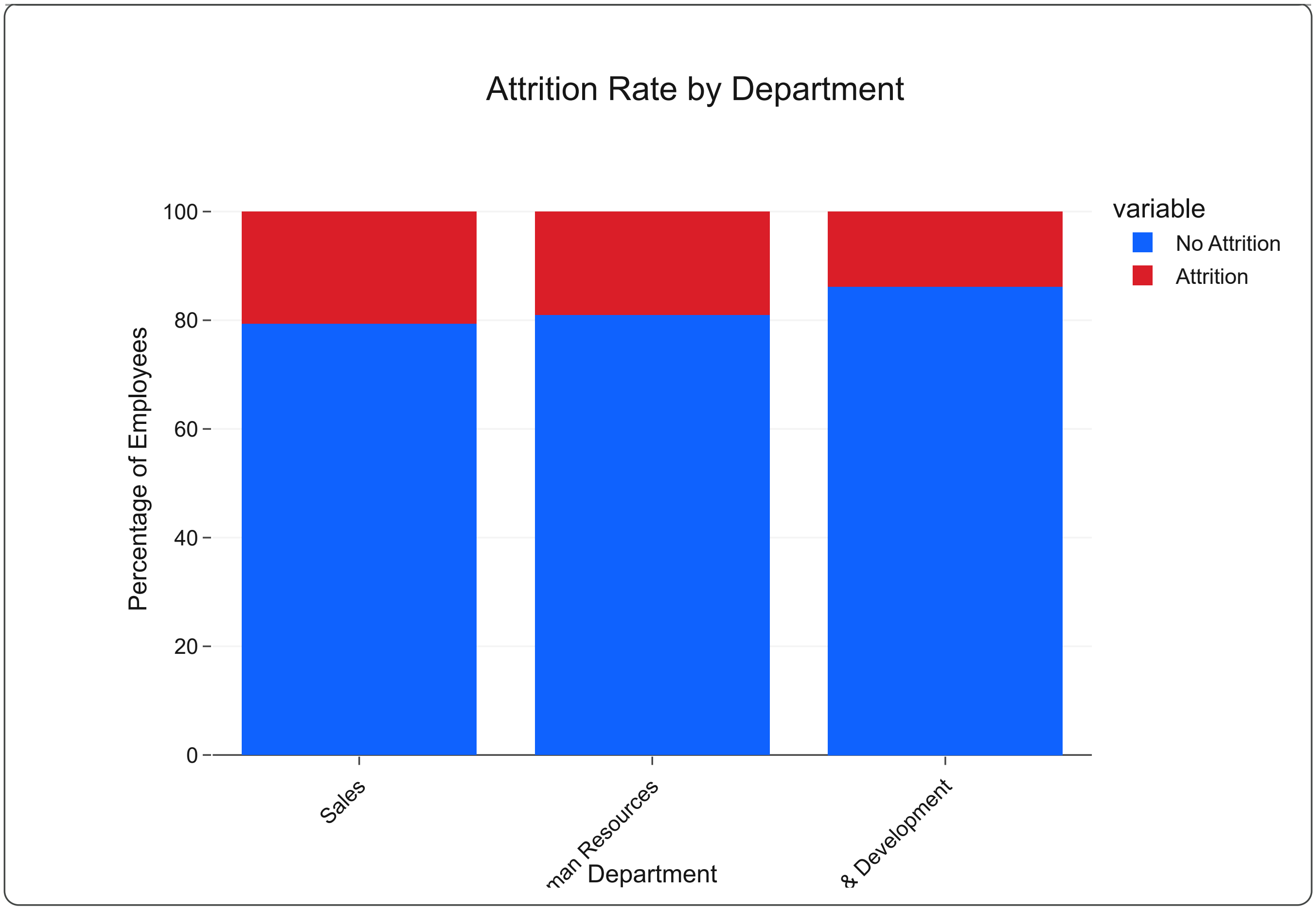
The Work-Life Balance analysis across job roles highlights that roles such as **Sales Representative** and **Laboratory Technician** tend to have a higher proportion of employees reporting 'Bad' or 'Good' (lower) work-life balance compared to other roles. This again aligns with the higher attrition observed in these roles. The demanding nature or scheduling constraints associated with these jobs likely contribute to difficulties in maintaining a healthy work-life balance, exacerbating the risk of turnover.

Key Findings

- **High-Attrition Roles:** **Sales Representatives** consistently show the highest attrition rates, followed by **Laboratory Technicians** and **Human Resources** roles. These roles demand immediate attention for retention strategies.
- **Lower Satisfaction in High-Attrition Roles:** High-attrition roles (e.g., Sales Representative, Laboratory Technician) correlate with lower reported job satisfaction and poorer work-life balance. This suggests that the nature of the work, workload, or specific role challenges contribute significantly to employee dissatisfaction.
- **Compensation vs. Other Factors:** While some high-attrition roles like Sales Representative have moderate average incomes, the combined impact of low job satisfaction and poor work-life balance appears to outweigh compensation as a retention factor in these cases.
- **Stable Roles:** Roles like **Research Director** and **Manager** demonstrate low attrition, high job satisfaction, and good work-life balance, likely due to higher compensation, greater autonomy, and clear career paths.
- **Actionable Insight:** IBM should implement role-specific interventions for high-attrition jobs. This includes:
    - **Sales Representatives:** Reviewing sales targets, enhancing support systems, offering flexible work options where possible, and investing in skill development for career advancement.
    - **Laboratory Technicians:** Addressing workload, ensuring access to necessary resources, and exploring opportunities for skill diversification and career growth within R&D.
    - **Human Resources:** Ensuring competitive compensation, providing clear advancement paths, and fostering a supportive internal environment.

  Tailored feedback mechanisms and continuous monitoring for these roles are crucial to improve retention effectively.

```
department_attrition = df.groupby('department')['attrition'].value_counts(normalize=True).unsta
department_attrition = department_attrition.sort_values(by='Attrition', ascending=False).reset_

fig_dept_attrition = px.bar(department_attrition, x='department', y=['No Attrition', 'Attrition
             title='Attrition Rate by Department',
             labels={'value': 'Percentage', 'department': 'Department'},
             color_discrete_map={'Attrition': IBM_COLORS['danger'], 'No Attrition': IBM_COLORS[
             template='ibm_carbon')

fig_dept_attrition.update_layout(barmode='stack', yaxis_title='Percentage of Employees', showle
fig_dept_attrition.show()
```



Insight

Reconfirming the analysis from Chapter 13, the **Sales Department** continues to show a notably higher attrition rate compared to other departments like Research & Development and Human Resources. This consistent finding suggests that the Sales department might have inherent characteristics, such as high-pressure targets or a competitive environment, that contribute to higher turnover. Understanding these departmental specific challenges is crucial for developing targeted retention programs.

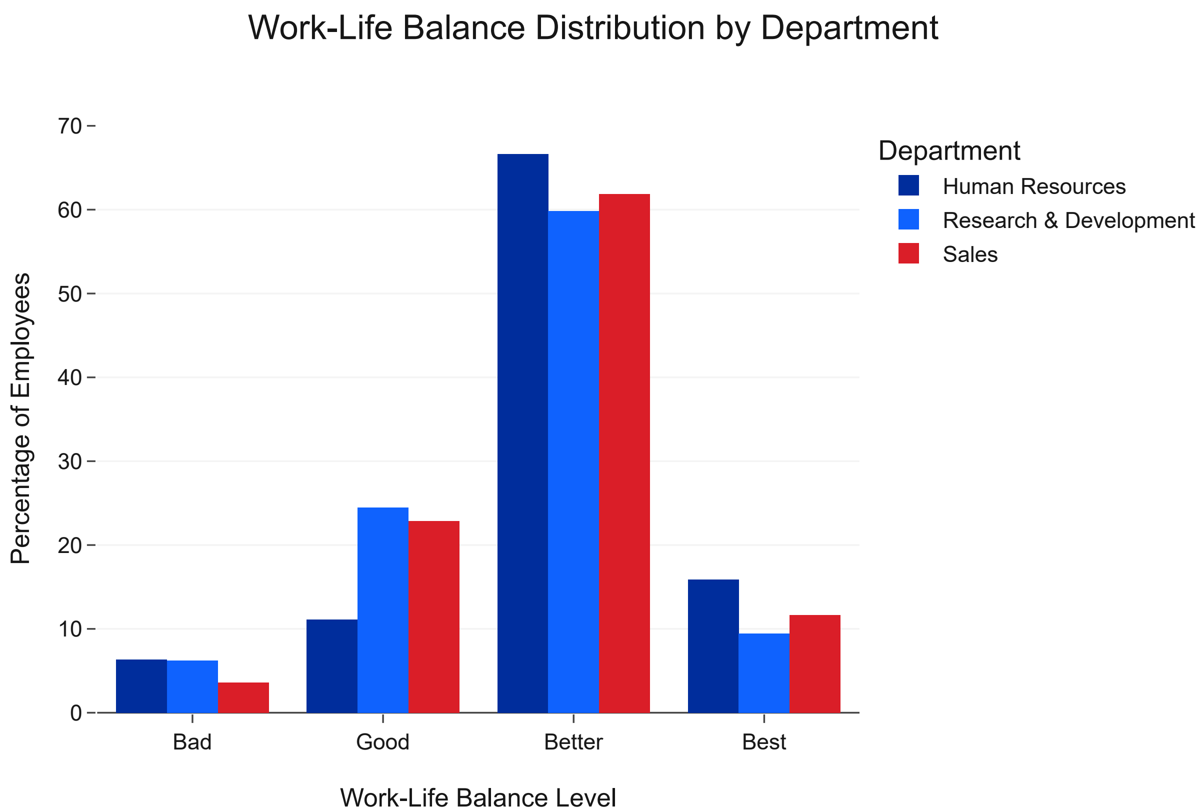
Work-Life Balance by Department

Work-Life Balance can vary significantly across departments due to different workloads, project cycles, and management expectations. This section investigates the distribution of perceived work-life balance within each department.

Code

```
wlb_by_department = df.groupby(['department', 'worklifebalance_level']).size().unstack(fill_valu
wlb_by_department = wlb_by_department.div(wlb_by_department.sum(axis=1), axis=0).mul(100)
wlb_by_department = wlb_by_department.loc[:, ['Bad', 'Good', 'Better', 'Best']].stack().reset_in
wlb_by_department.columns = ['Department', 'WorkLifeBalance', 'Percentage']

fig_wlb_dept = px.bar(wlb_by_department, x='WorkLifeBalance', y='Percentage', color='Department
             title='Work-Life Balance Distribution by Department',
             labels={'WorkLifeBalance': 'Work-Life Balance Level', 'Percentage': 'Percentage of
             color_discrete_map={'Sales': IBM_COLORS['danger'], 'Research & Development': IBM_C(
             template='ibm_carbon',
```

```
            barmode='group')

fig_wlb_dept.update_layout(yaxis_title='Percentage of Employees', showlegend=True)
fig_wlb_dept.show()
```



Insight

The Work-Life Balance distribution varies across departments. While all departments have a majority in 'Better' work-life balance, the **Sales Department** appears to have a slightly higher percentage of employees reporting 'Bad' or 'Good' (lower) work-life balance compared to Research & Development and Human Resources. This aligns with the higher attrition in Sales and suggests that workload and demands in this department may negatively impact employees' personal lives.

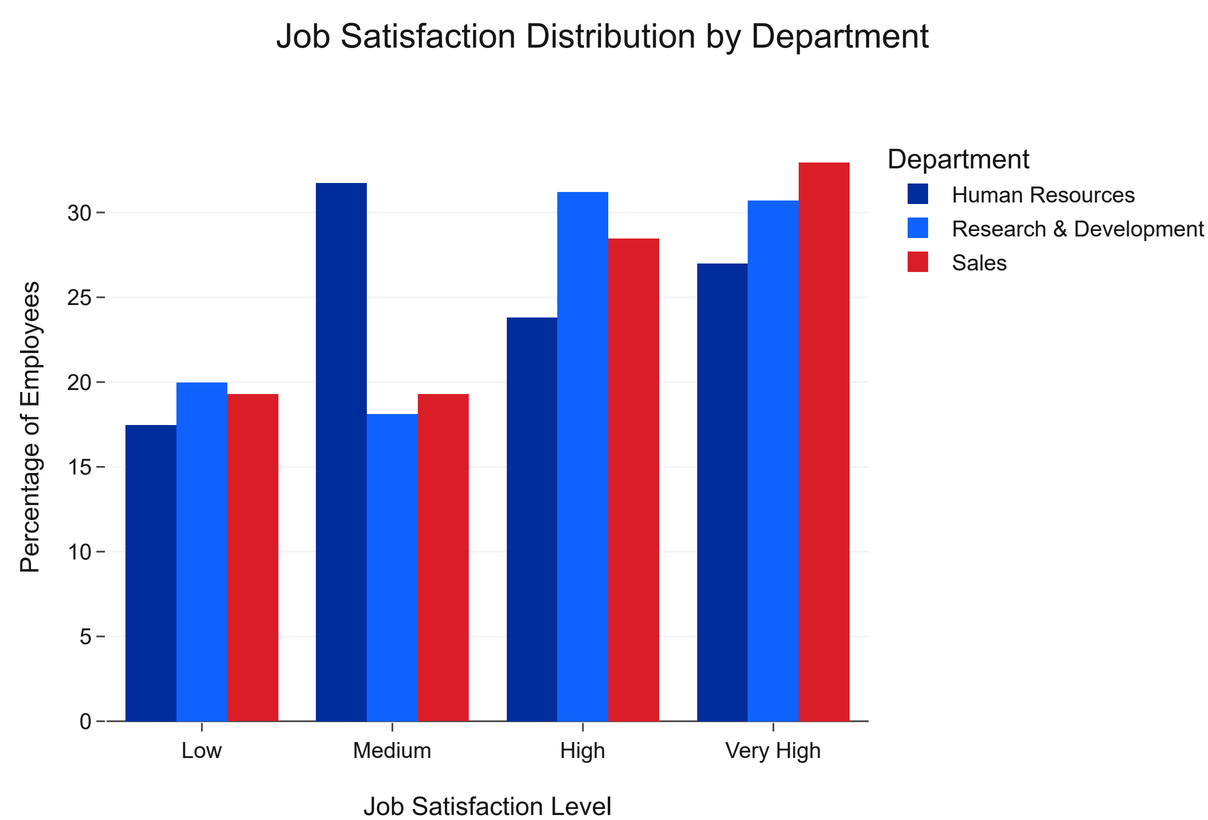
Job Satisfaction by Department

Job satisfaction can be influenced by departmental culture, management, and the nature of the work itself. This analysis helps identify if certain departments struggle more with keeping their employees content.

Code

```
job_sat_by_department = df.groupby(['department', 'jobsatisfaction_level']).size().unstack(fill_
job_sat_by_department = job_sat_by_department.div(job_sat_by_department.sum(axis=1), axis=0).mu
job_sat_by_department = job_sat_by_department.loc[:, ['Low', 'Medium', 'High', 'Very High']].sta
job_sat_by_department.columns = ['Department', 'JobSatisfaction', 'Percentage']
```

```
fig_job_sat_dept = px.bar(job_sat_by_department, x='JobSatisfaction', y='Percentage', color='De
             title='Job Satisfaction Distribution by Department',
             labels={'JobSatisfaction': 'Job Satisfaction Level', 'Percentage': 'Percentage of 
             color_discrete_map={'Sales': IBM_COLORS['danger'], 'Research & Development': IBM_C
             template='ibm_carbon',
             barmode='group')

fig_job_sat_dept.update_layout(yaxis_title='Percentage of Employees', showlegend=True)
fig_job_sat_dept.show()
```



Insight

The distribution of job satisfaction levels also shows some departmental differences. The **Sales Department** has a slightly higher proportion of employees reporting 'Low' job satisfaction compared to Research & Development and Human Resources. This aligns with the higher attrition and poorer work-life balance observed in Sales, reinforcing the idea that job content and departmental dynamics might contribute to overall dissatisfaction and turnover.

Average Monthly Income by Department

Compensation structures can vary by department. Analyzing average monthly income can reveal if certain departments are compensated differently, which might affect attraction and retention.

Code

Chapter 19: Department Analytics

Business Question

How do departmental structures and characteristics influence employee attrition, job satisfaction, and other key HR metrics at IBM, and how can these insights be used to optimize departmental performance and retention strategies?

Background

Departments often have unique cultures, workloads, and management styles, which can significantly impact employee experience and retention. This chapter will analyze various HR metrics across different departments to identify departmental strengths, weaknesses, and specific areas where attrition might be concentrated or where satisfaction levels are particularly low or high. Understanding these departmental dynamics is essential for tailored interventions.

Methodology

We will use interactive Plotly visualizations to explore key HR metrics across departments. This will include:

- Overall attrition rates by Department.
- Work-Life Balance distribution by Department.
- Job Satisfaction distribution by Department.
- Average Monthly Income by Department.

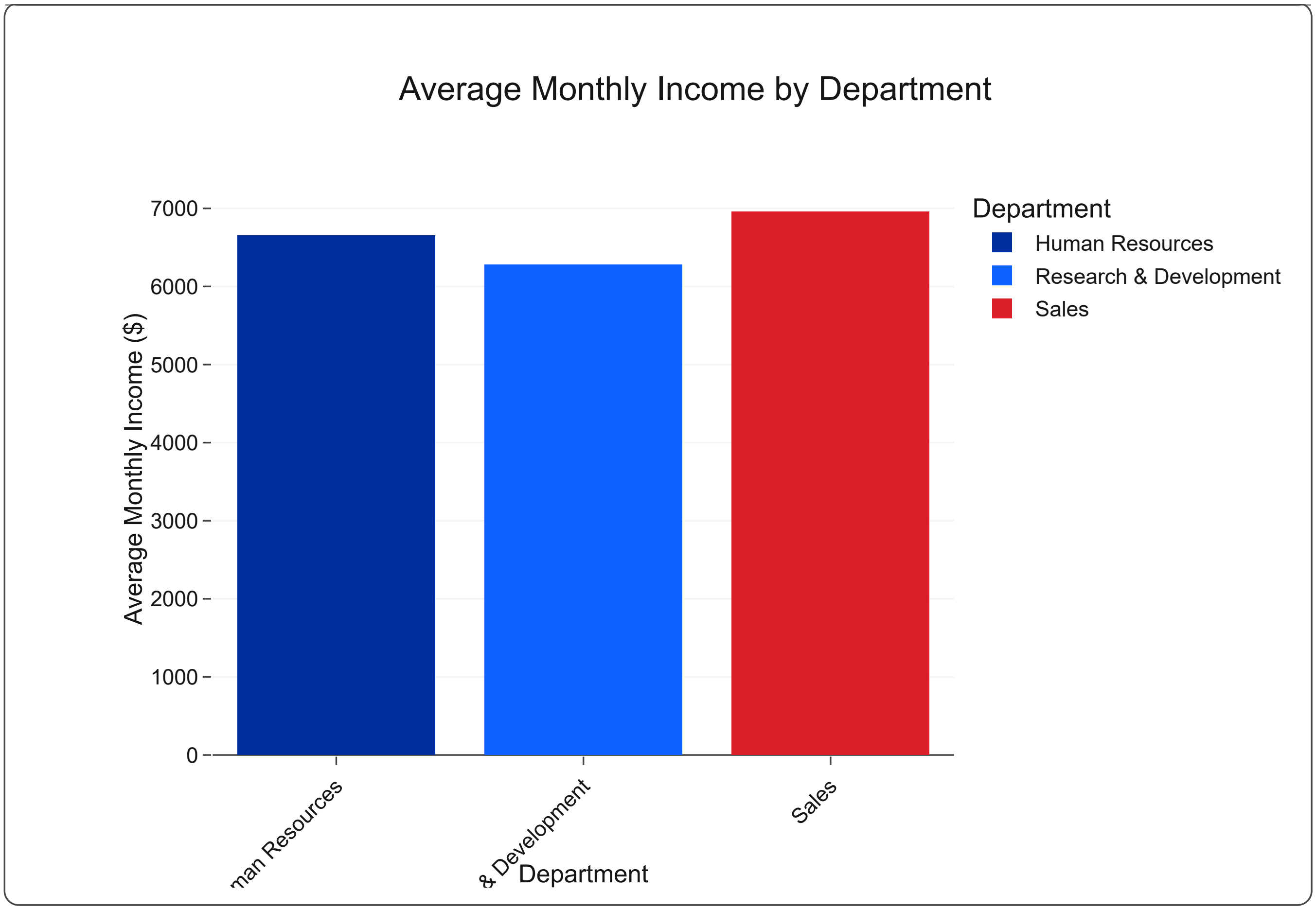
Each visualization will adhere to the IBM Carbon Design System for clear, executive-ready presentation, supported by concise insights.

Attrition by Department

Analyzing attrition rates across different departments can highlight organizational units that may be facing unique challenges related to work environment, management, or departmental culture. This analysis helps in targeting interventions more effectively.

Code

```
avg_income_by_department = df.groupby('department')['monthlyincome'].mean().reset_index()

fig_income_dept = px.bar(avg_income_by_department, x='department', y='monthlyincome',
             title='Average Monthly Income by Department',
             labels={'monthlyincome': 'Average Monthly Income ($)', 'department': 'Department'},
             color='department',
             color_discrete_map={'Sales': IBM_COLORS['danger'], 'Research & Development': IBM_C
             template='ibm_carbon')

fig_income_dept.update_layout(yaxis_title='Average Monthly Income ($)', showlegend=True, xaxis_
fig_income_dept.show()
```



Insight

The analysis of average monthly income by department shows that **Research & Development** generally has the highest average income, followed by Sales, and then Human Resources. This aligns with industry trends where technical and revenue-generating roles often command higher salaries. While this doesn't directly explain attrition on its own, it provides context for overall compensation competitiveness within each department and can be cross-referenced with satisfaction metrics to understand if perceived pay fairness is an issue.

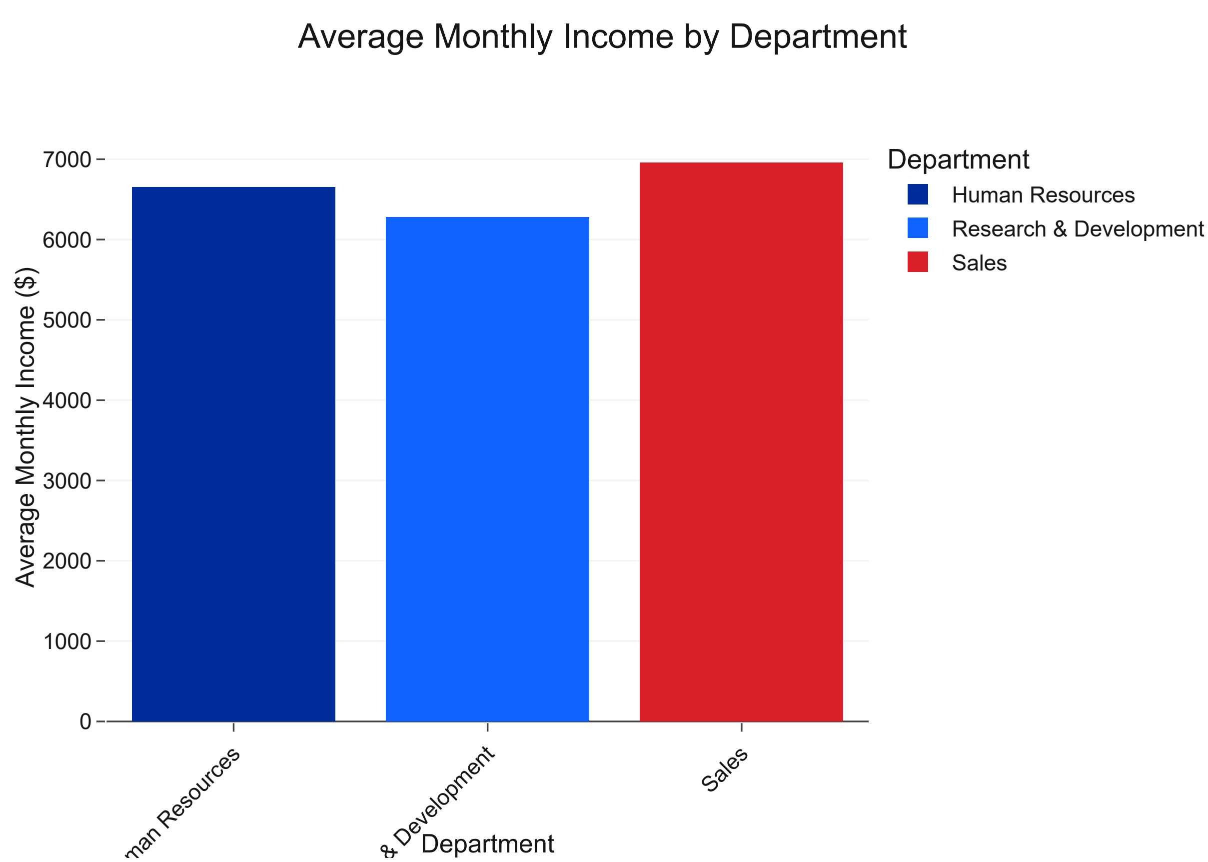
Key Findings

- **Departmental Attrition Disparities:** The **Sales Department** consistently shows the highest attrition rates compared to Research & Development and Human Resources. This indicates that sales roles and their associated environment may be more prone to turnover.
- **Impact on Work-Life Balance:** Employees in the **Sales Department** tend to report a less favorable work-life balance, with a higher proportion falling into 'Bad' or 'Good' categories. This suggests that the demands of sales roles might contribute to increased stress and burnout.
- **Job Satisfaction Differences:** Similarly, **Job Satisfaction levels appear slightly lower in the Sales Department**, with a higher percentage of employees reporting 'Low' satisfaction. This further compounds the attrition risk in this department.
- **Compensation Context:** Research & Development generally has the highest average monthly income, which could contribute to stability in that department, while Human Resources has the

lowest. Understanding these income differences provides crucial context when analyzing satisfaction and attrition.

- **Actionable Insight:** IBM should conduct a deeper dive into the Sales Department to understand the specific stressors and dissatisfaction drivers. Interventions could include: re-evaluating sales targets and quotas, providing better support and resources to sales personnel, implementing stress management programs, and fostering a more supportive departmental culture. For departments with lower average incomes, ensuring competitive and fair pay structures is essential to prevent attrition motivated by compensation. Tailored retention strategies based on departmental specifics are highly recommended.

```
avg_income_by_department = df.groupby('department')['monthlyincome'].mean().reset_index()

fig_income_dept = px.bar(avg_income_by_department, x='department', y='monthlyincome',
             title='Average Monthly Income by Department',
             labels={'monthlyincome': 'Average Monthly Income ($)', 'department': 'Department'},
             color='department',
             color_discrete_map={'Sales': IBM_COLORS['danger'], 'Research & Development': IBM_C(
             template='ibm_carbon')

fig_income_dept.update_layout(yaxis_title='Average Monthly Income ($)', showlegend=True, xaxis_
fig_income_dept.show()
```



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Insight

The analysis of job satisfaction by business travel frequency reveals that **frequent travelers tend to report lower levels of job satisfaction**. A higher proportion of 'Frequent Travel' employees fall into the 'Low' and 'Medium' job satisfaction categories compared to those who travel rarely or not at all. This suggests that the demanding nature of frequent travel not only impacts work-life balance but also diminishes an employee's overall contentment with their job, further exacerbating the risk of attrition.

Key Findings

- **Frequent Travel Drives Attrition:** Employees who travel frequently for business exhibit significantly higher attrition rates, making business travel frequency a critical factor in employee turnover.
- **Negative Impact on Work-Life Balance:** Frequent business travel is strongly associated with a poorer perception of work-life balance. Travelers are more likely to report 'Bad' or 'Good' work-life balance.
- **Reduced Job Satisfaction:** Frequent travelers also report lower levels of job satisfaction, indicating that the stresses of travel detract from their contentment with their roles.

- **Compounding Effects:** The combined negative impacts on work-life balance and job satisfaction create a substantial push factor for employees in high-travel roles to seek other employment.
- **Actionable Insight:** IBM should carefully evaluate roles that require frequent business travel. Strategies to mitigate attrition in these roles could include: offering additional support and resources to frequent travelers (e.g., concierge services, enhanced travel benefits), exploring alternative communication methods to reduce the need for physical travel, ensuring competitive compensation for travel-intensive roles, and actively monitoring the well-being of frequent travelers to intervene early if dissatisfaction arises.

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Chapter 17: Overtime Analysis

Business Question

What is the impact of mandatory or frequent overtime on employee satisfaction, work-life balance, and ultimately, attrition rates at IBM, and how can these insights inform workload management strategies?

Background

Overtime, while sometimes necessary, can be a significant factor contributing to employee burnout, reduced job satisfaction, and a poor work-life balance. Prolonged periods of working beyond standard hours can lead to physical and mental fatigue, decreasing productivity and increasing the likelihood of an employee seeking opportunities elsewhere.

This chapter focuses specifically on analyzing the relationship between overtime status and various employee outcomes, particularly attrition. By understanding how working overtime impacts IBM's workforce, HR can develop more sustainable workload management policies and support systems to mitigate its negative effects.

Methodology

We will use interactive Plotly visualizations to clearly demonstrate the disparities in attrition rates and other key metrics between employees who work overtime and those who do not. We will specifically analyze:

- Overall attrition rates for employees working and not working overtime.
- The distribution of work-life balance for employees who work overtime versus those who don't.
- Job satisfaction levels for overtime vs. non-overtime employees.
- Relationship between overtime and other satisfaction metrics.

Each visualization will adhere to the IBM Carbon Design System for clear, executive-ready presentation, supported by concise insights.

Attrition by Overtime Status

A direct comparison of attrition rates between employees who work overtime and those who do not is essential to establish the primary link between overtime and turnover.

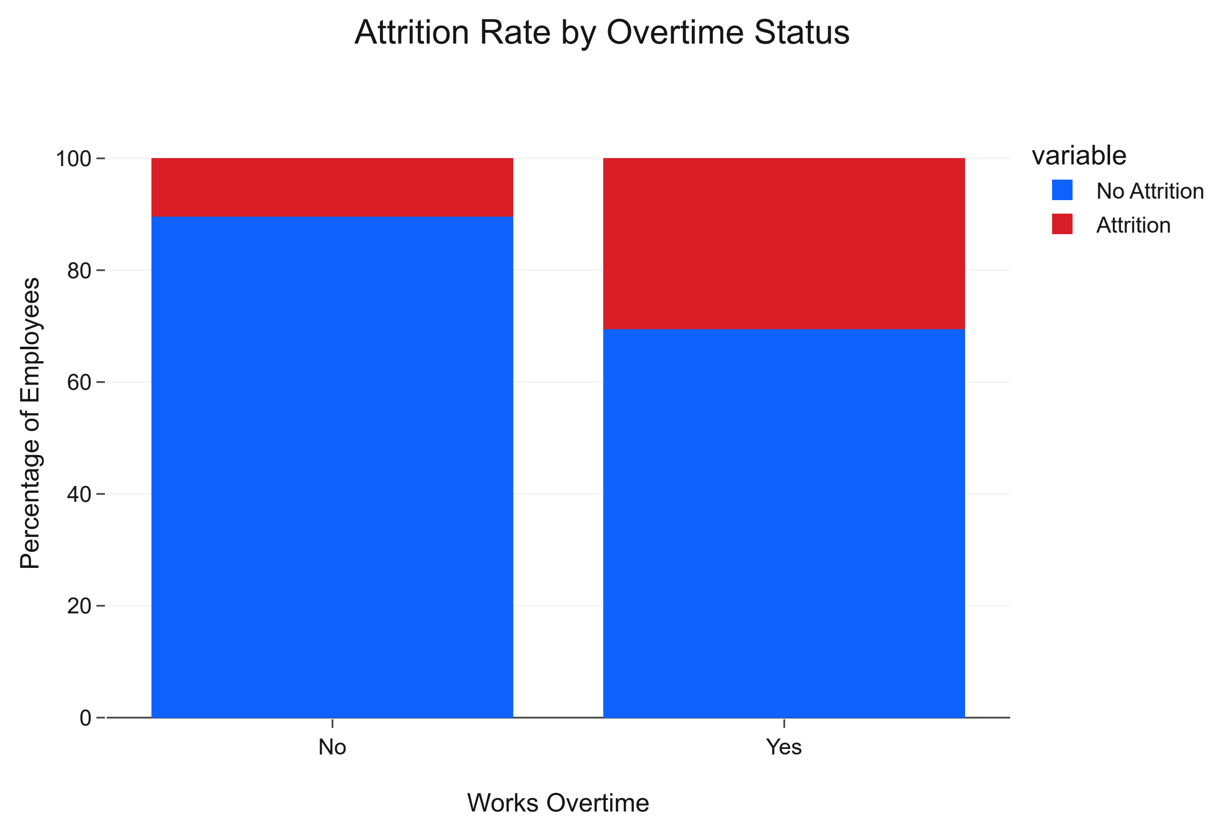
Code

```
overtime_attrition = df.groupby('overtime')['attrition'].value_counts(normalize=True).unstack()
overtime_attrition = overtime_attrition.reindex(['No', 'Yes']).reset_index()

fig_overtime = px.bar(overtime_attrition, x='overtime', y=['No Attrition', 'Attrition'],
             title='Attrition Rate by Overtime Status',
             labels={'value': 'Percentage', 'overtime': 'Works Overtime'},
             color_discrete_map={'Attrition': IBM_COLORS['danger'], 'No Attrition': IBM_COLORS[
```

```
            template='ibm_carbon')

fig_overtime.update_layout(barmode='stack', yaxis_title='Percentage of Employees', showlegend=Tı
fig_overtime.show()
```



Insight

The analysis unequivocally demonstrates that employees who **work overtime ('Yes') have a significantly higher attrition rate (~29.5%)** compared to those who do not ('No') (~10.4%). This is one of the strongest correlations found, indicating that overtime is a major driver of employee turnover. This finding suggests that excessive work hours contribute to burnout and dissatisfaction, leading employees to seek more balanced work environments.

Work-Life Balance by Overtime Status

Since overtime often impacts personal time, examining the perceived work-life balance for employees who work overtime versus those who don't can provide deeper context for the higher attrition rates.

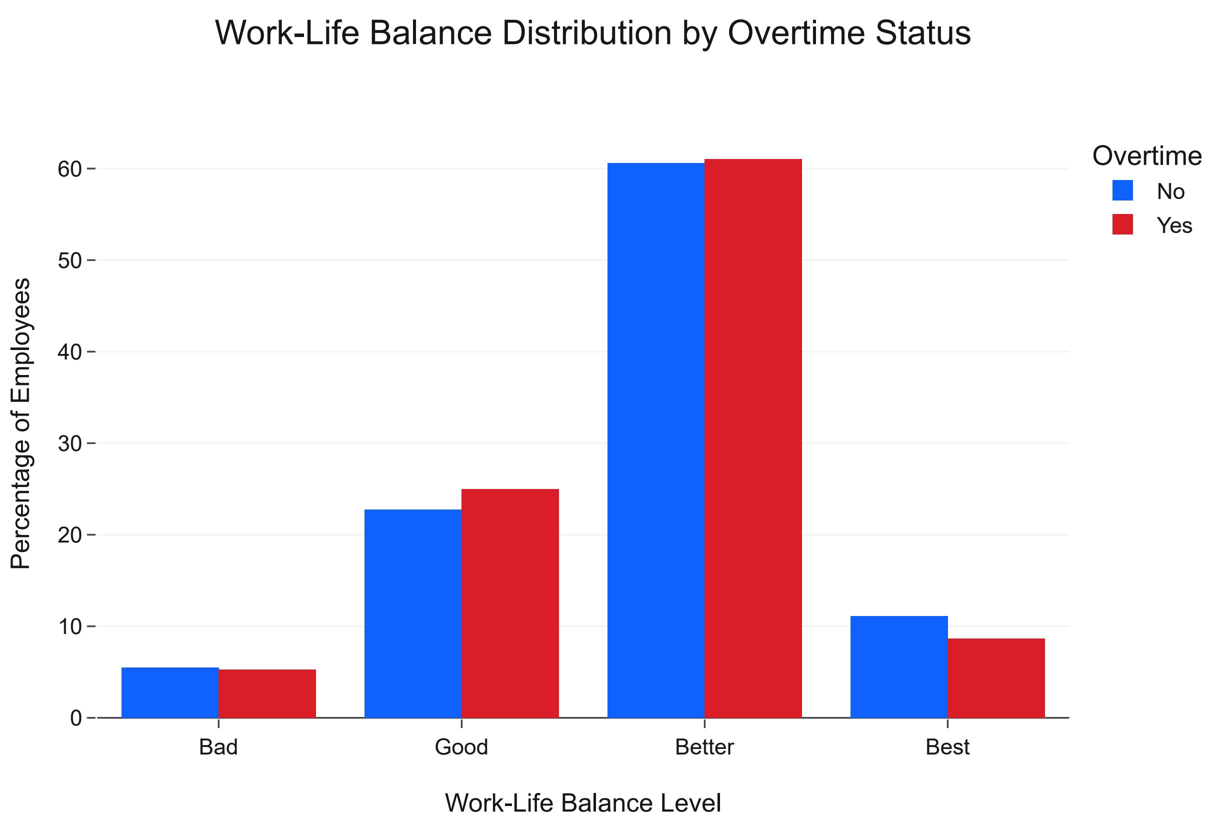
Code

```
wlb_overtime = df.groupby(['overtime', 'worklifebalance_level'])['attrition'].count().unstack(fi
wlb_overtime = wlb_overtime.div(wlb_overtime.sum(axis=1), axis=0).mul(100)
wlb_overtime = wlb_overtime.loc[['No', 'Yes'], ['Bad', 'Good', 'Better', 'Best']]
wlb_overtime = wlb_overtime.stack().reset_index(name='Percentage')
wlb_overtime.columns = ['Overtime', 'WorkLifeBalance', 'Percentage']

fig_wlb_overtime = px.bar(wlb_overtime, x='WorkLifeBalance', y='Percentage', color='Overtime',
```

```
                     title='Work-Life Balance Distribution by Overtime Status',
                     labels={'WorkLifeBalance': 'Work-Life Balance Level', 'Percentage': 'Pe
                     color_discrete_map={'No': IBM_COLORS['primary_blue'], 'Yes': IBM_COLORS
                     template='ibm_carbon',
                     barmode='group')

fig_wlb_overtime.update_layout(yaxis_title='Percentage of Employees', showlegend=True)
fig_wlb_overtime.show()
```



Insight

This visualization clearly shows that employees who **work overtime ('Yes') report a significantly worse work-life balance**. A much higher percentage of overtime workers rate their work-life balance as 'Bad' or 'Good' compared to non-overtime workers, who are more concentrated in the 'Better' and 'Best' categories. This strong disparity directly links overtime to perceived poor work-life balance, explaining a major contributing factor to the higher attrition observed in the overtime group.

Job Satisfaction by Overtime Status

Beyond work-life balance, excessive overtime might also diminish an employee's satisfaction with their job. This section explores if job satisfaction levels differ based on overtime status.

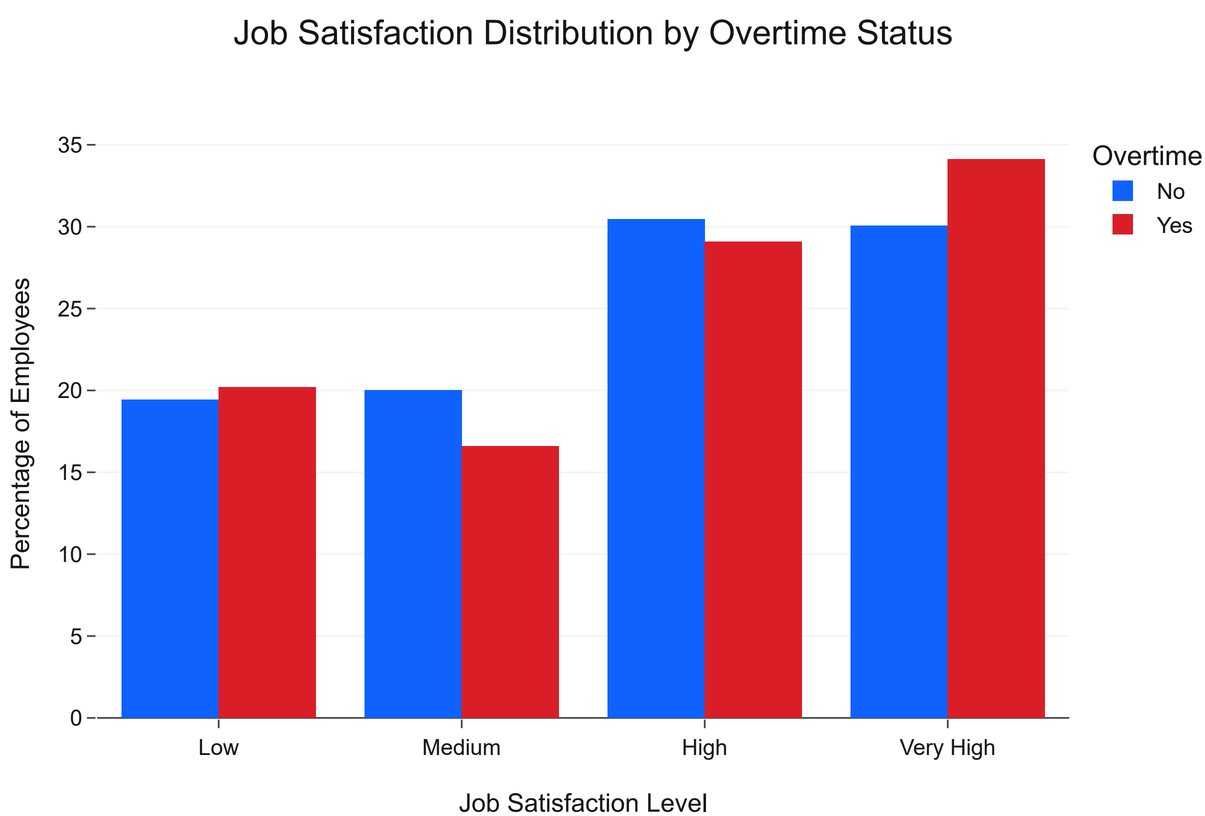
Code

```
job_sat_overtime = df.groupby(['overtime', 'jobsatisfaction_level'])['attrition'].count().unsta
job_sat_overtime = job_sat_overtime.div(job_sat_overtime.sum(axis=1), axis=0).mul(100)
job_sat_overtime = job_sat_overtime.loc[['No', 'Yes'], ['Low', 'Medium', 'High', 'Very High']]
```

```
    job_sat_overtime = job_sat_overtime.stack().reset_index(name='Percentage')
    job_sat_overtime.columns = ['Overtime', 'JobSatisfaction', 'Percentage']

    fig_job_sat_overtime = px.bar(job_sat_overtime, x='JobSatisfaction', y='Percentage', color='Ove
                                  title='Job Satisfaction Distribution by Overtime Status',
                                  labels={'JobSatisfaction': 'Job Satisfaction Level', 'Percentage': 'P
                                  color_discrete_map={'No': IBM_COLORS['primary_blue'], 'Yes': IBM_COLO
                                  template='ibm_carbon',
                                  barmode='group')

    fig_job_sat_overtime.update_layout(yaxis_title='Percentage of Employees', showlegend=True)
    fig_job_sat_overtime.show()
```



Insight

The analysis of job satisfaction by overtime status shows that employees who **work overtime tend to report lower levels of job satisfaction**. A higher proportion of overtime workers fall into the 'Low' and 'Medium' job satisfaction categories compared to their non-overtime counterparts. This indicates that the added burden of overtime not only affects work-life balance but also directly detracts from an employee's overall contentment with their job, further contributing to attrition risk.

Key Findings

- **Overtime as a Major Attrition Driver:** Employees who work overtime ('Yes') have an attrition rate nearly three times higher than those who do not, making it one of the most critical factors influencing turnover.

- **Impact on Work-Life Balance:** Working overtime is strongly correlated with a significantly poorer perception of work-life balance. Overtime workers are far more likely to report 'Bad' or 'Good' work-life balance compared to non-overtime workers.
- **Impact on Job Satisfaction:** Overtime is also associated with lower job satisfaction. A larger proportion of employees working overtime report 'Low' or 'Medium' job satisfaction.
- **Compounding Effects:** The combined negative effects of overtime on work-life balance and job satisfaction create a significant push factor for employees to leave the company.
- **Actionable Insight:** IBM must critically review its policies and practices regarding overtime. Strategies should include: proactive workload management to reduce the necessity for overtime, better resource allocation, offering flexible work arrangements, and providing support programs for employees who regularly work extended hours. Addressing the root causes of overtime and mitigating its negative impacts is crucial for significantly reducing attrition and improving overall employee well-being.

Chapter 16: Career Growth

Business Question

How do factors related to career growth, including tenure, time in current role, frequency of promotions, and investment in training, influence employee attrition and overall career development at IBM?

Background

Career development and opportunities for growth are consistently ranked among the top factors influencing employee motivation, engagement, and retention. Employees who perceive a lack of advancement, stagnation in their roles, or insufficient investment in their skills are often more likely to seek opportunities elsewhere.

This chapter systematically analyzes various facets of career growth within IBM, such as years at the company, duration in the current role, the interval since the last promotion, and the frequency of training.